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(12) **United States Patent**
He

(10) **Patent No.: US 12,408,798 B2**
(45) **Date of Patent: Sep. 9, 2025**

(54) **MICROWAVE COOKING SYSTEMS**

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(72) Inventor: **Zhengxu He**, Reno, NV (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 1026 days.

(21) Appl. No.: **17/383,729**

(22) Filed: **Jul. 23, 2021**

(65) **Prior Publication Data**

US 2022/0031123 A1 Feb. 3, 2022

Related U.S. Application Data

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(51) **Int. Cl.**

A47J 44/00	(2006.01)
A47J 36/32	(2006.01)
A47J 37/10	(2006.01)
A47J 47/01	(2006.01)
B25J 11/00	(2006.01)
B67D 7/02	(2010.01)
H05B 6/64	(2006.01)

(52) **U.S. Cl.**

CPC ***A47J 44/00*** (2013.01); ***A47J 36/32*** (2013.01); ***A47J 37/108*** (2013.01); ***A47J 47/01*** (2013.01); ***B25J 11/0045*** (2013.01); ***B67D 7/02*** (2013.01); ***H05B 6/6411*** (2013.01)

(58) **Field of Classification Search**

None

See application file for complete search history.

(56) **References Cited**

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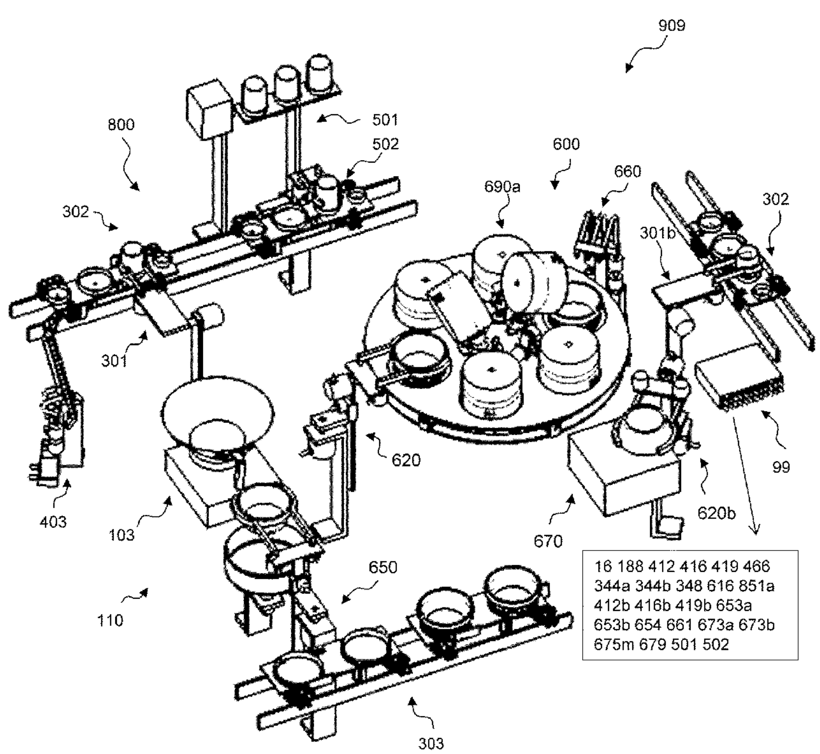
Primary Examiner — Steven W Crabb
Assistant Examiner — Theodore J Evangelista

(57) **ABSTRACT**

The present application firstly discloses a cooking system comprising two cooking apparatuses, each capable of producing a cooked food from food or food ingredient(s). The first cooking apparatus comprises a plurality of microwave ovens which can be moved by an intermittent motion mechanism or a transport system. The second cooking apparatus comprises a cookware and a motion mechanism to move the cookware to dispense a cooked food to a food container. A semi-cooked food is produced by a microwave oven of the first cooking apparatus and is used as an ingredient in the second cooking apparatus to produce cooked food.

20 Claims, 49 Drawing Sheets



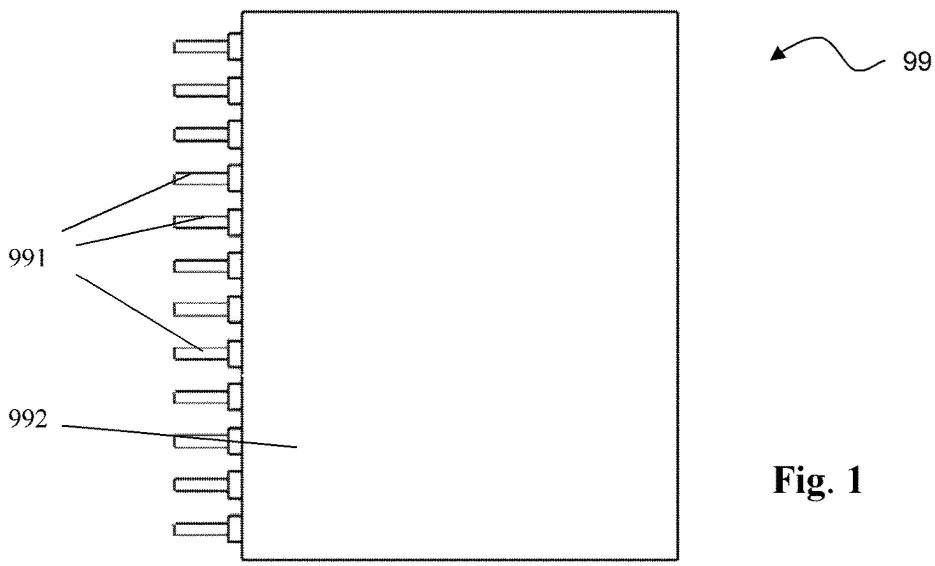


Fig. 1

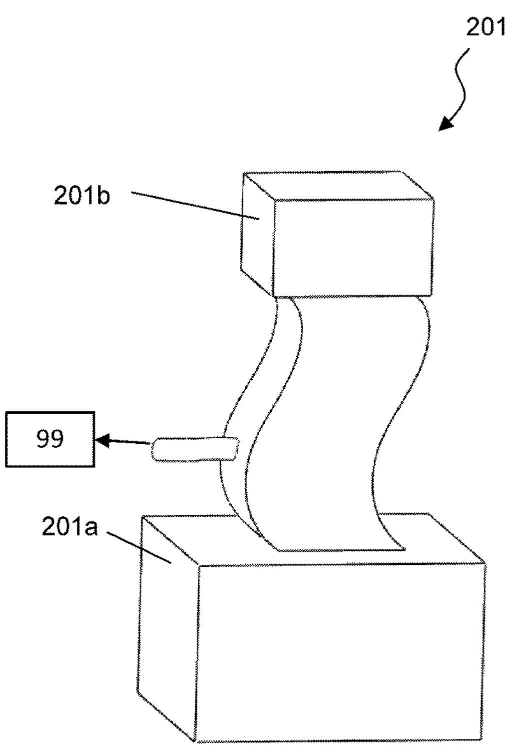


Fig. 2A

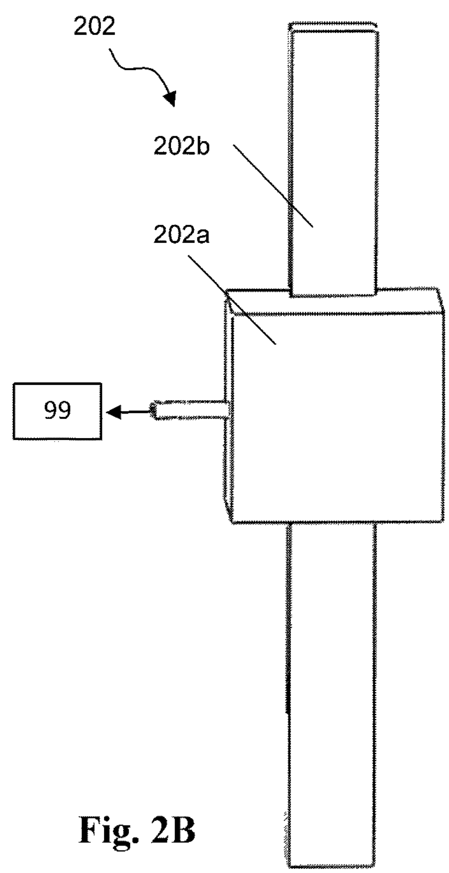


Fig. 2B

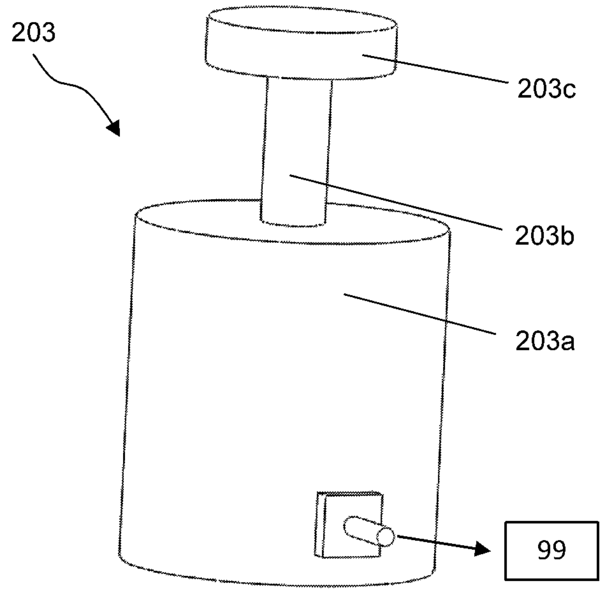

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Fig. 2C

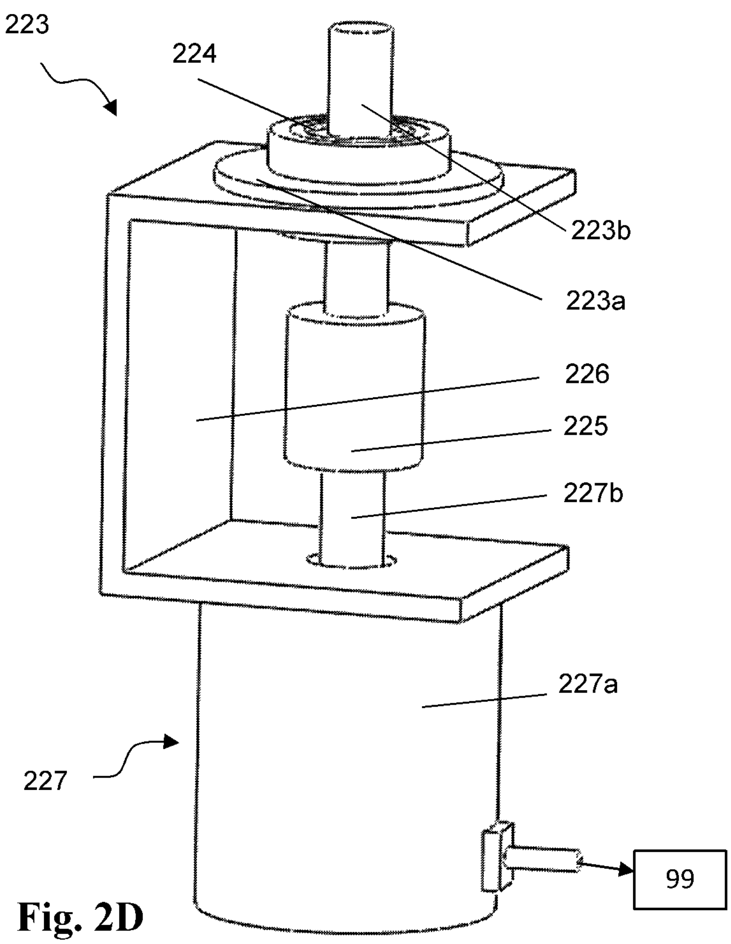
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Fig. 2D

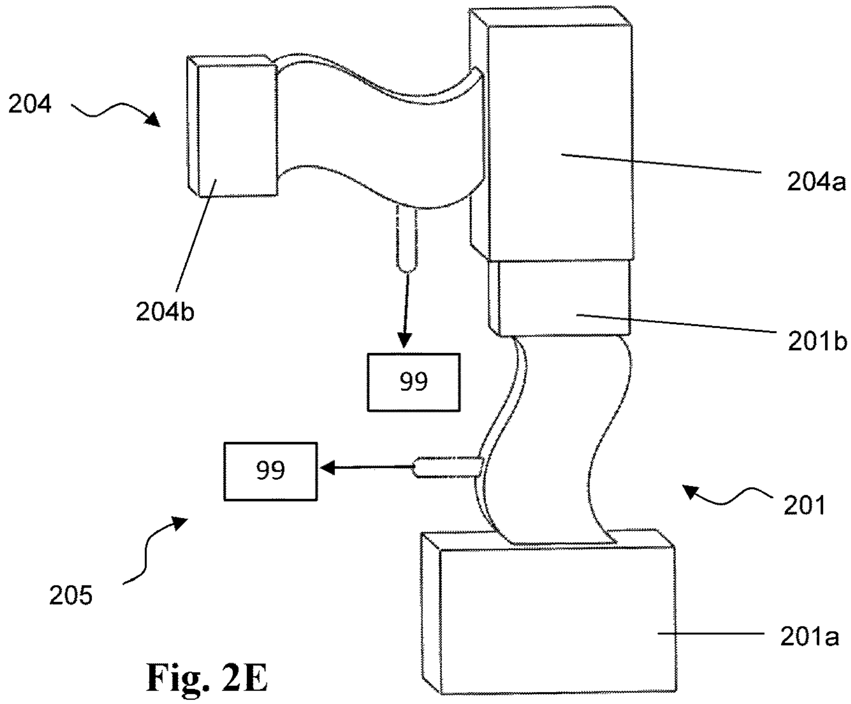
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Fig. 2E

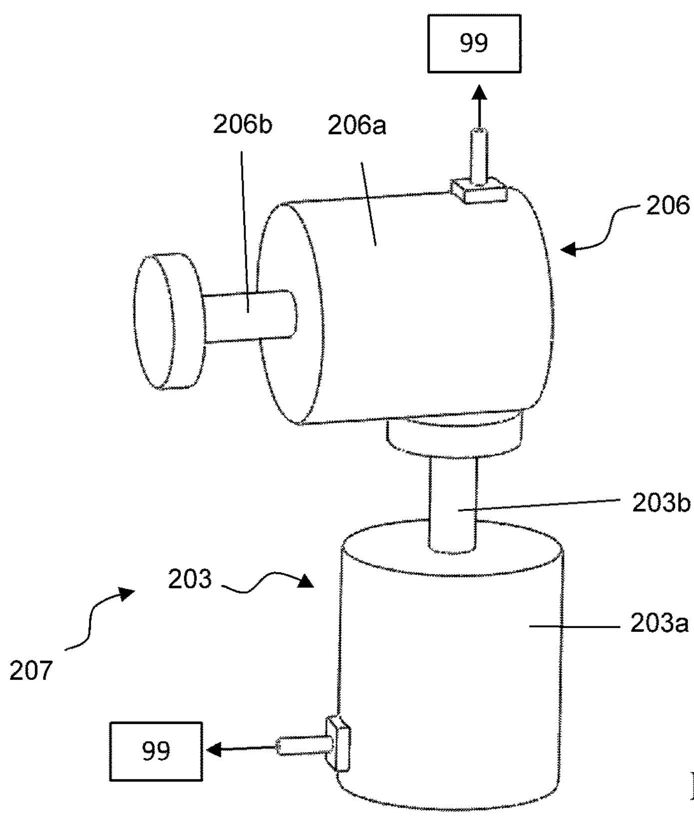
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Fig. 2F

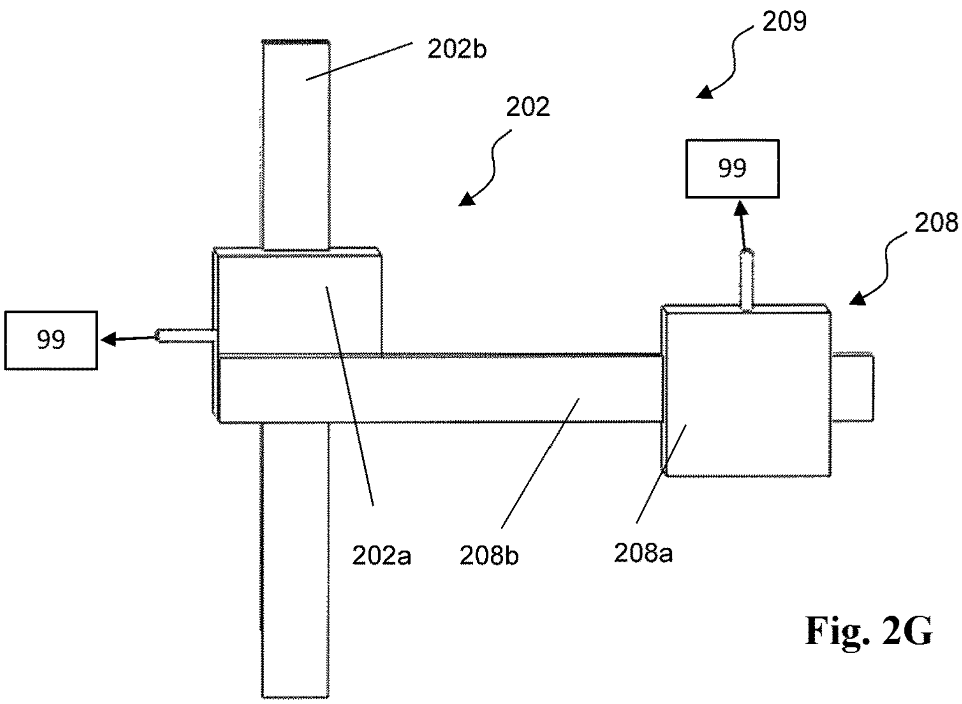
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Fig. 2G

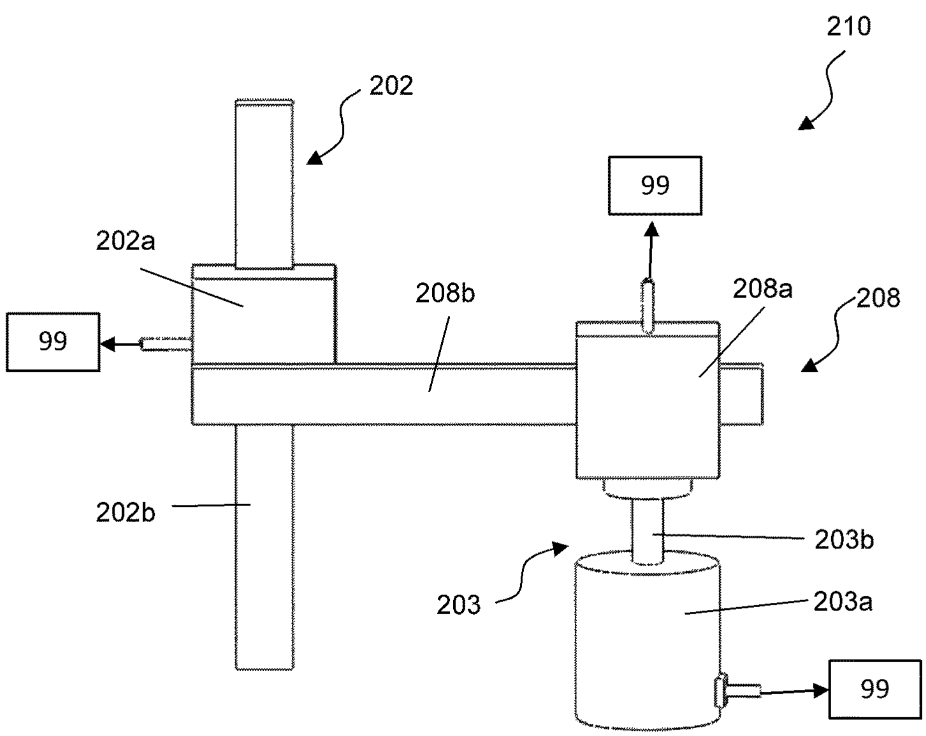
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Fig. 2H

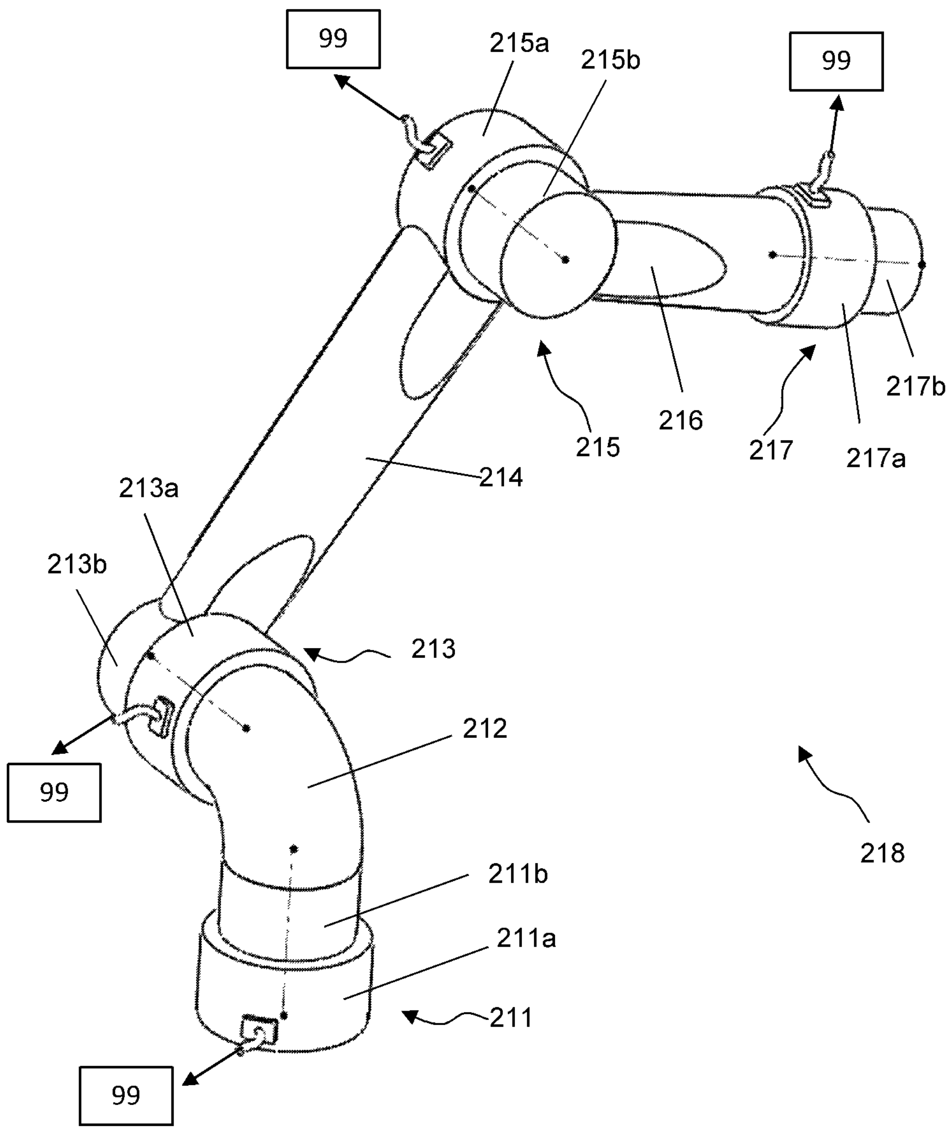
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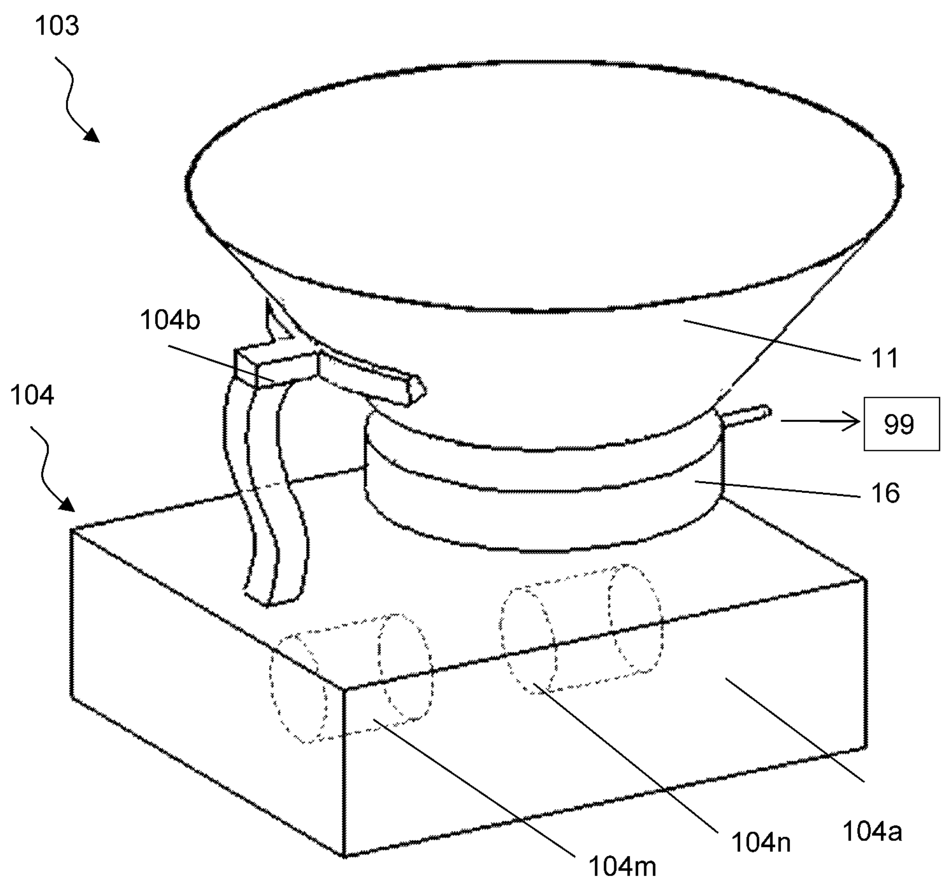
Fig. 2I



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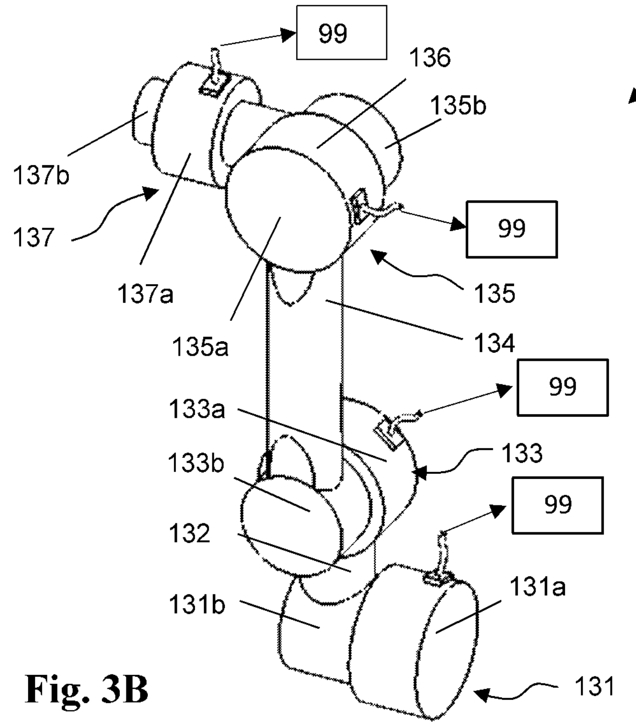
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131a
131b
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Fig. 3B

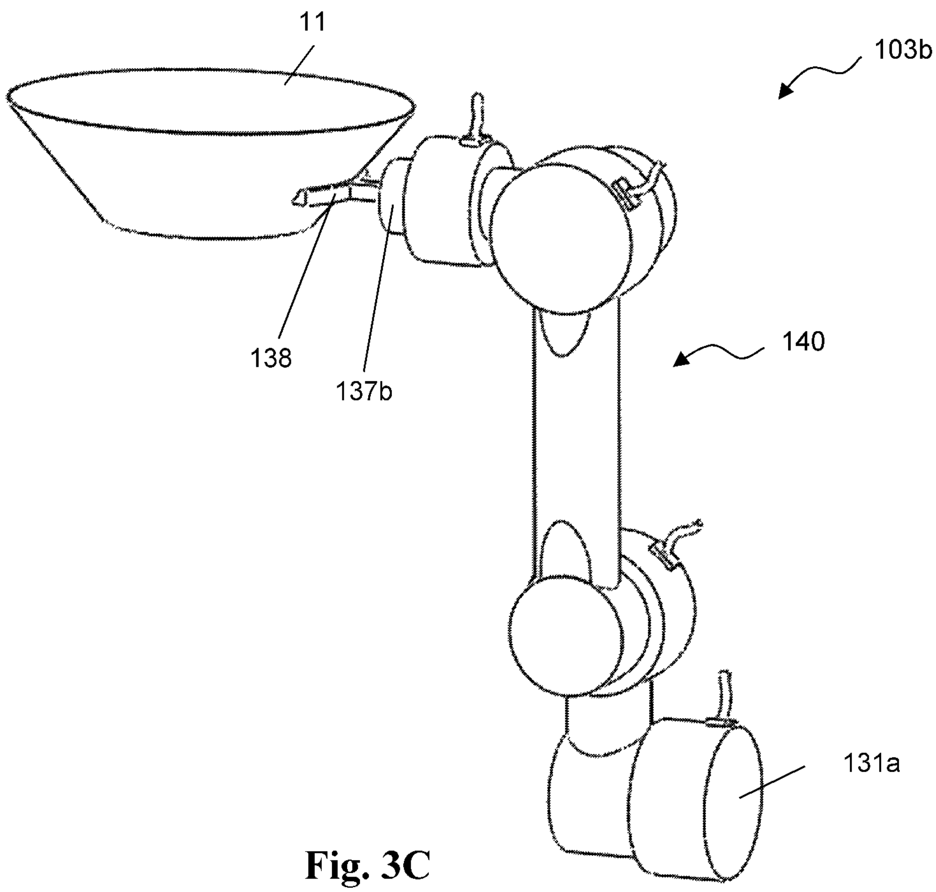
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Fig. 3C

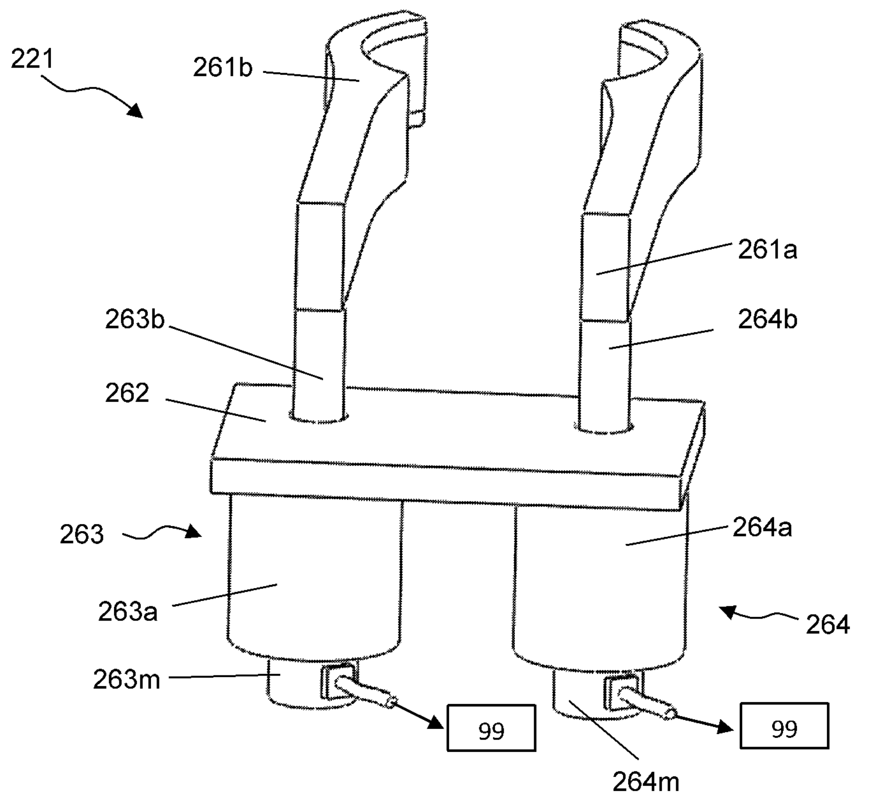
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Fig. 4A

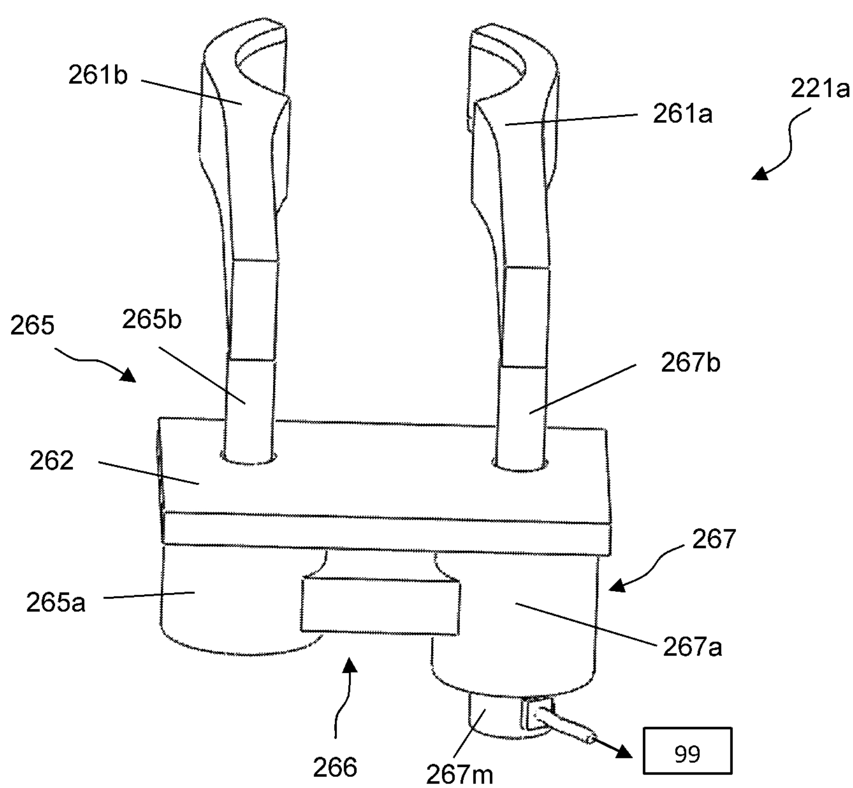
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Fig. 4B

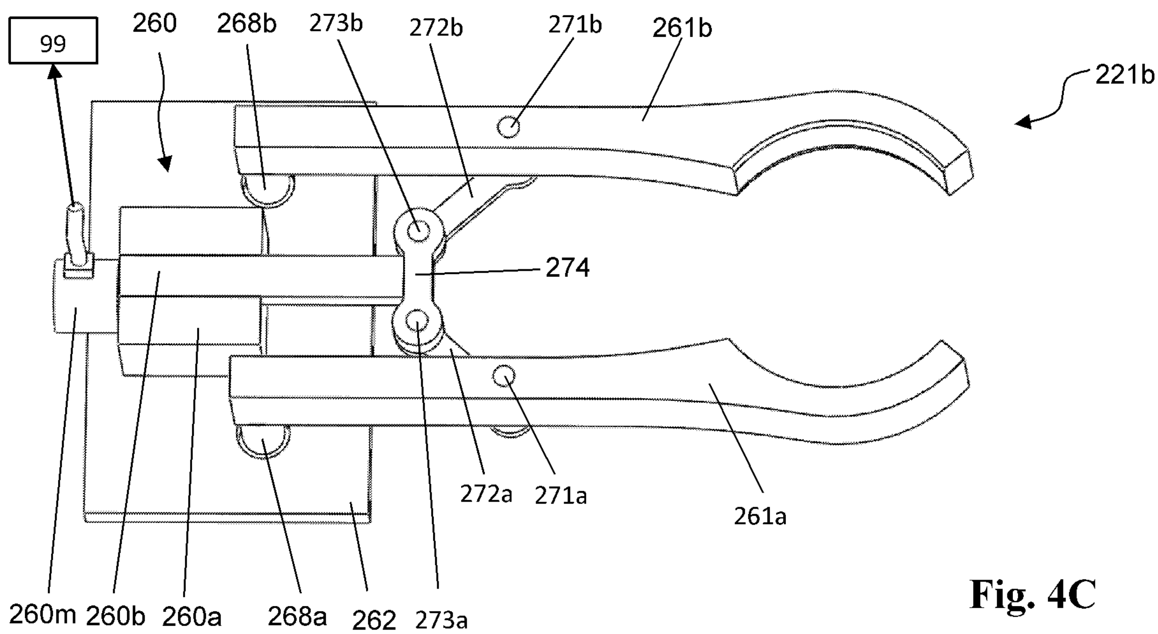
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Fig. 4C

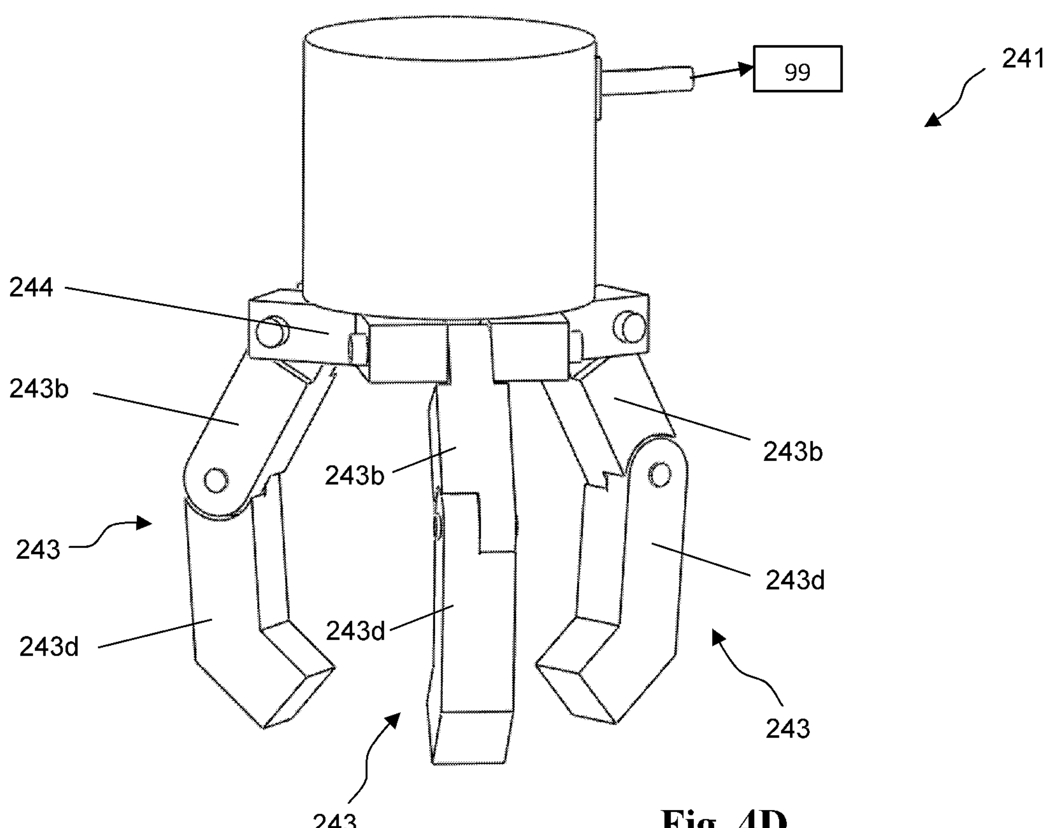
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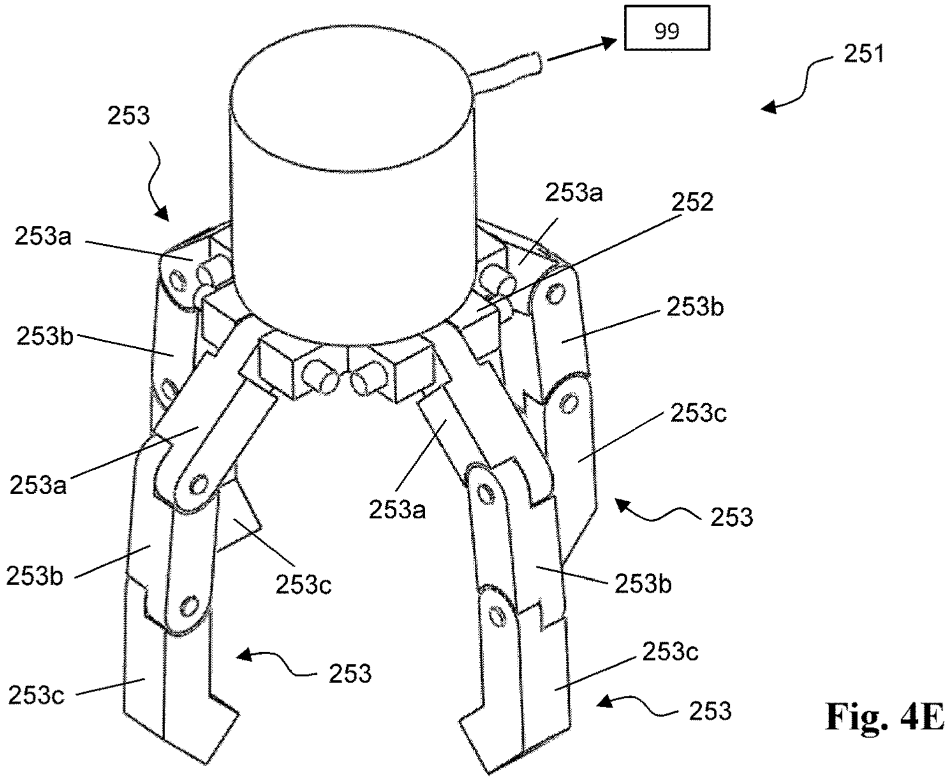
Fig. 4D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253c
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Fig. 4E

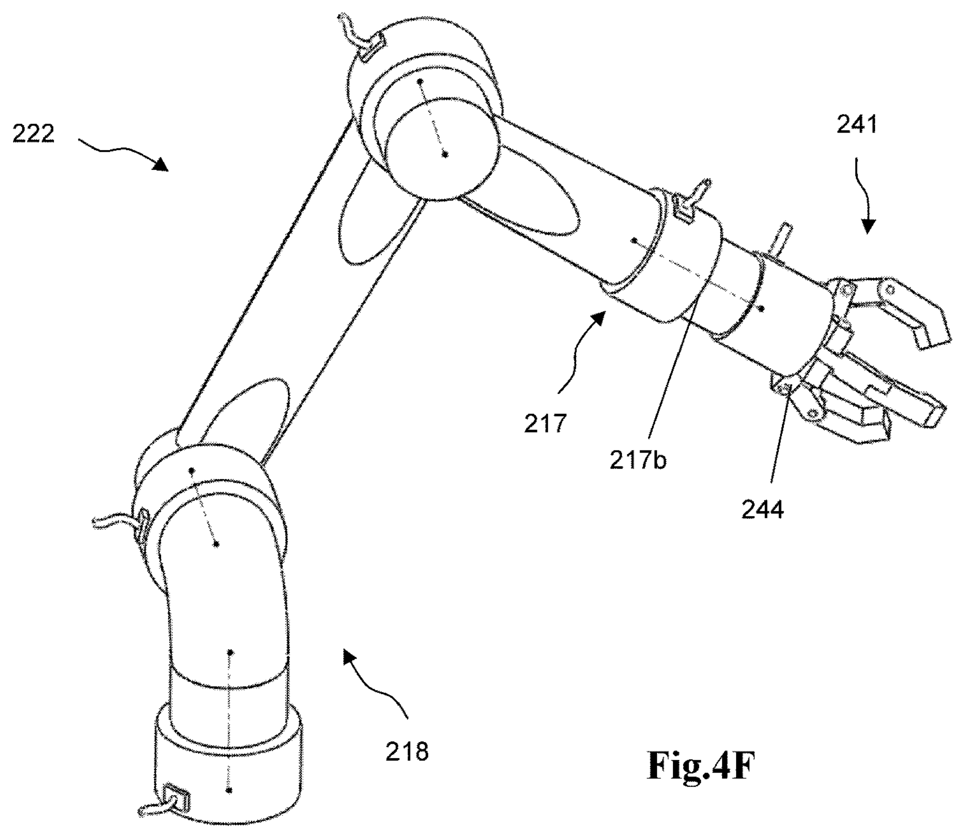
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Fig.4F

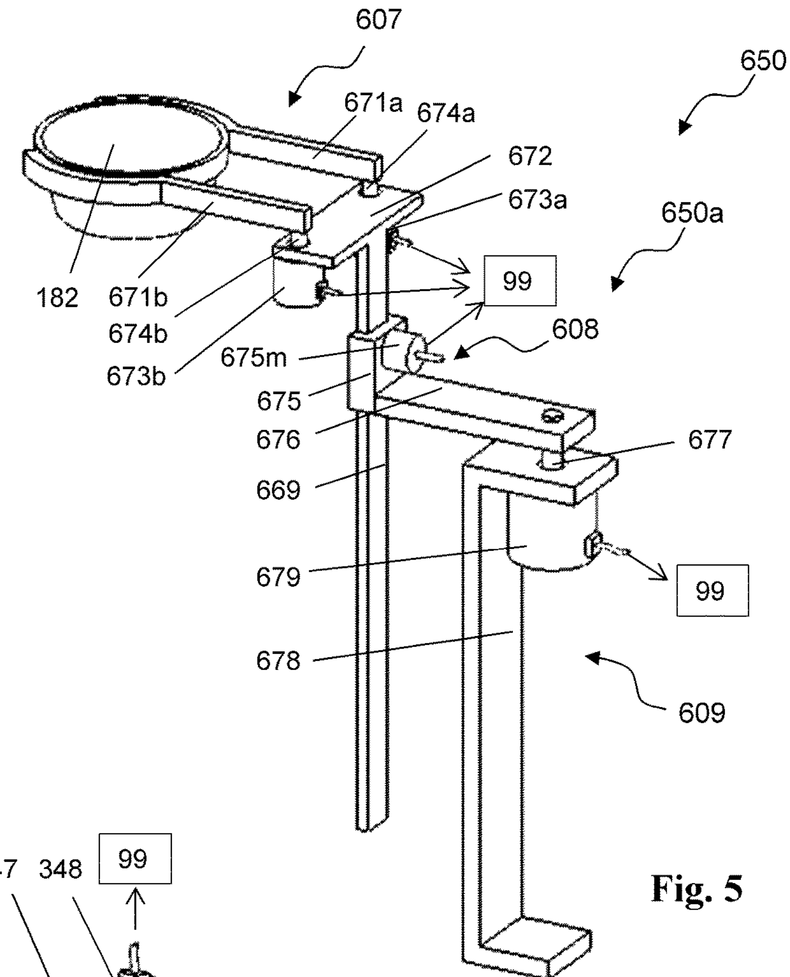
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Fig. 5

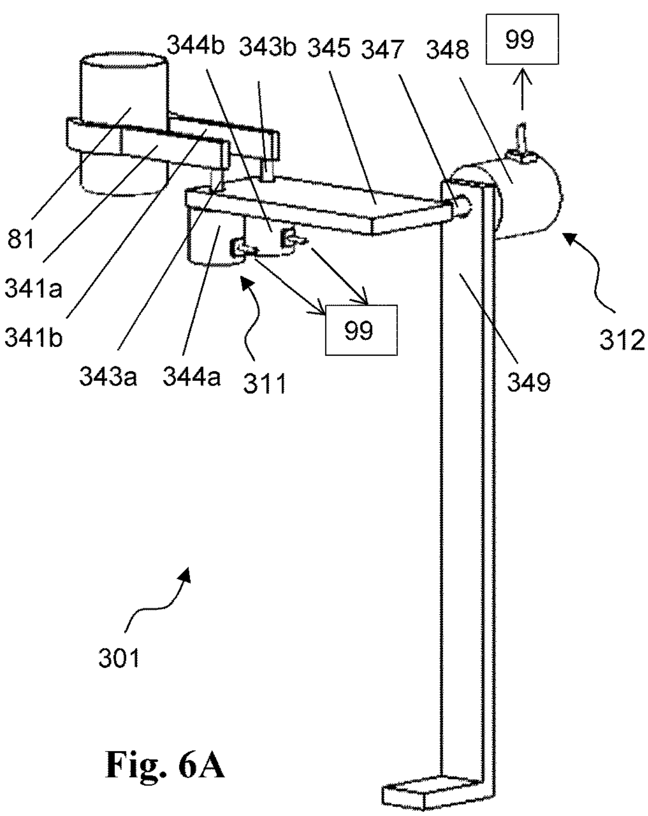
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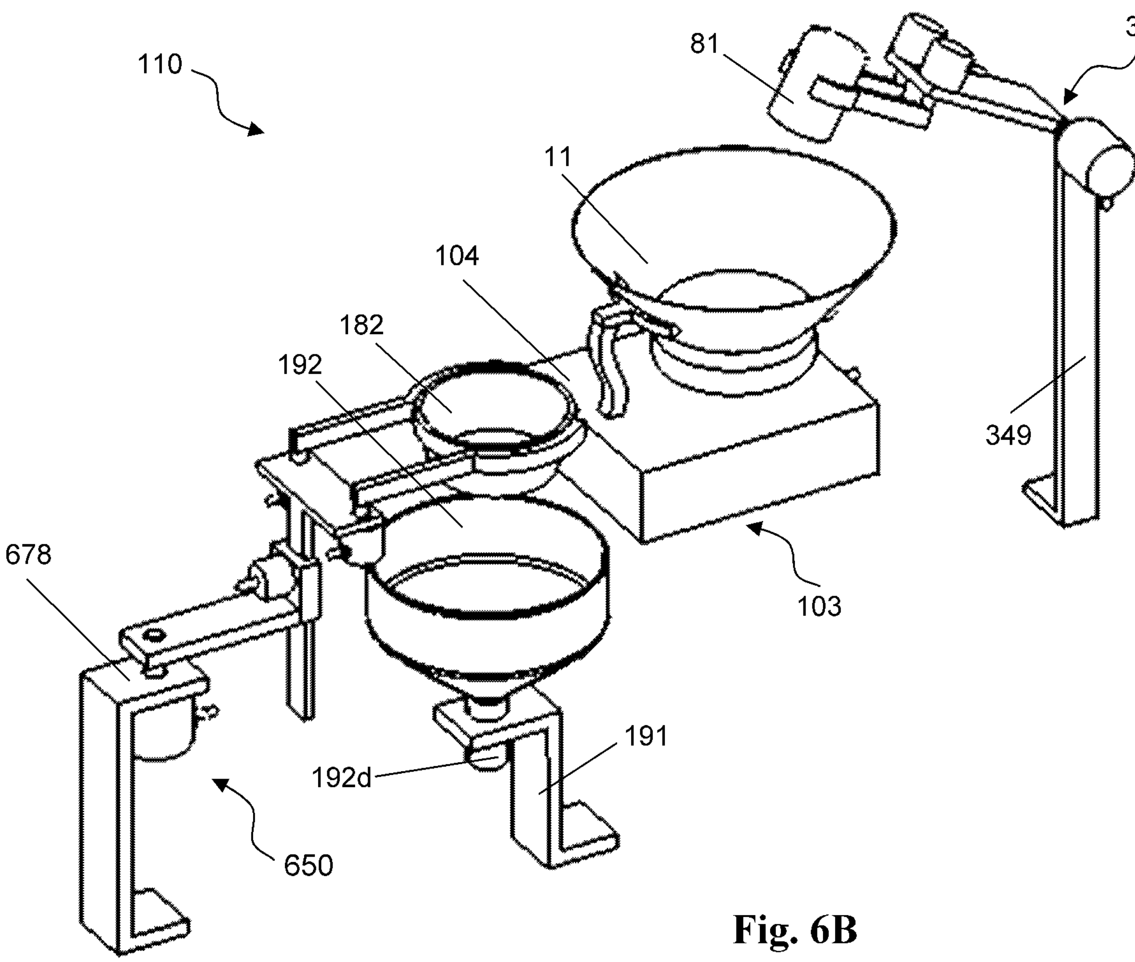
Fig. 6A



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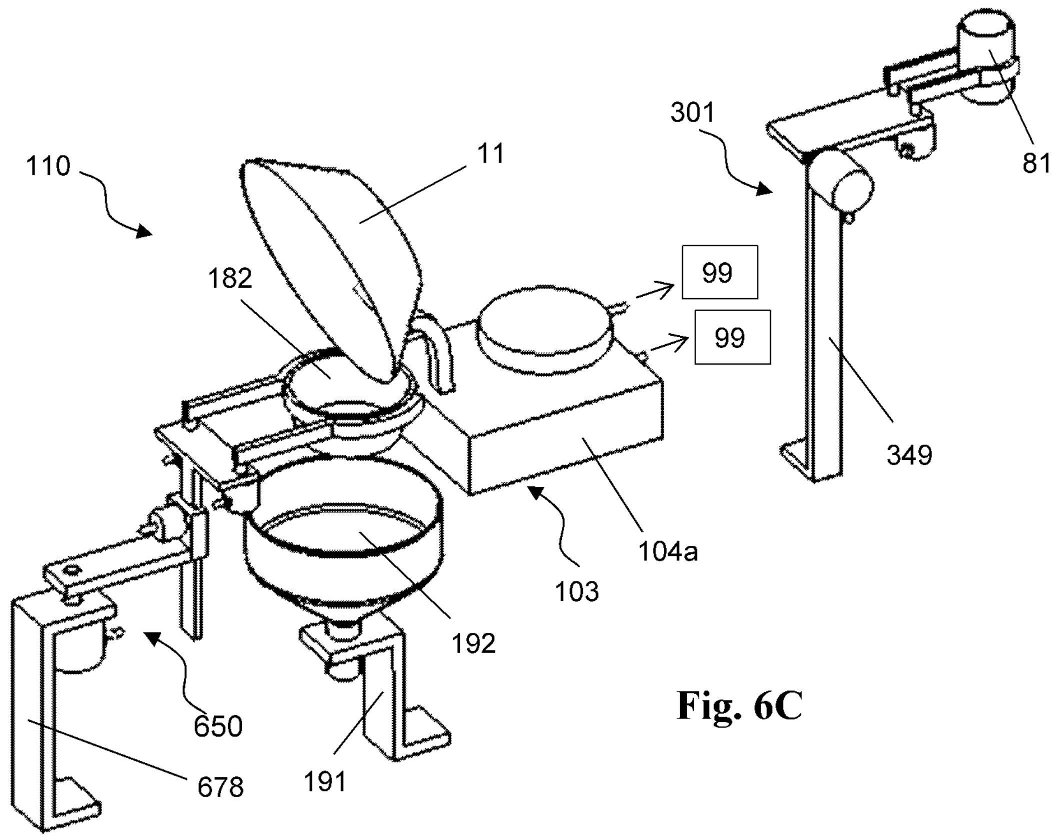
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Fig. 6C

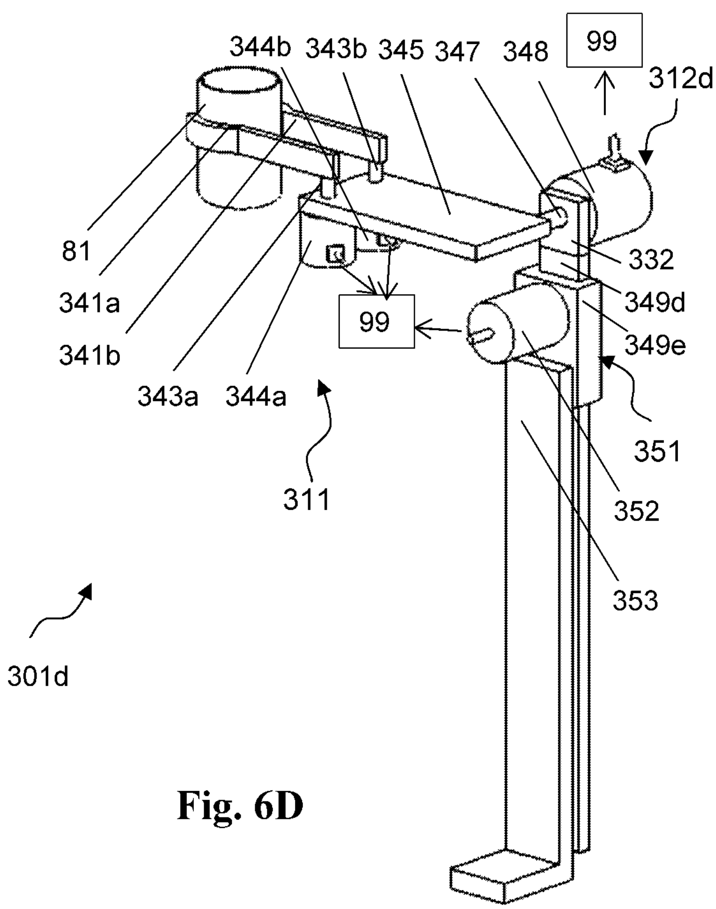
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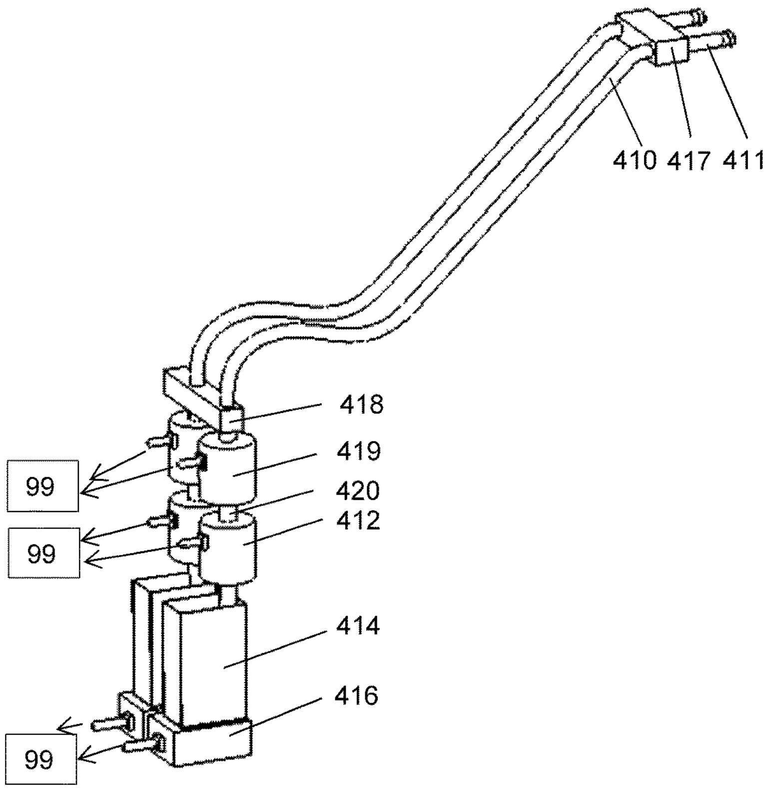
Fig. 6D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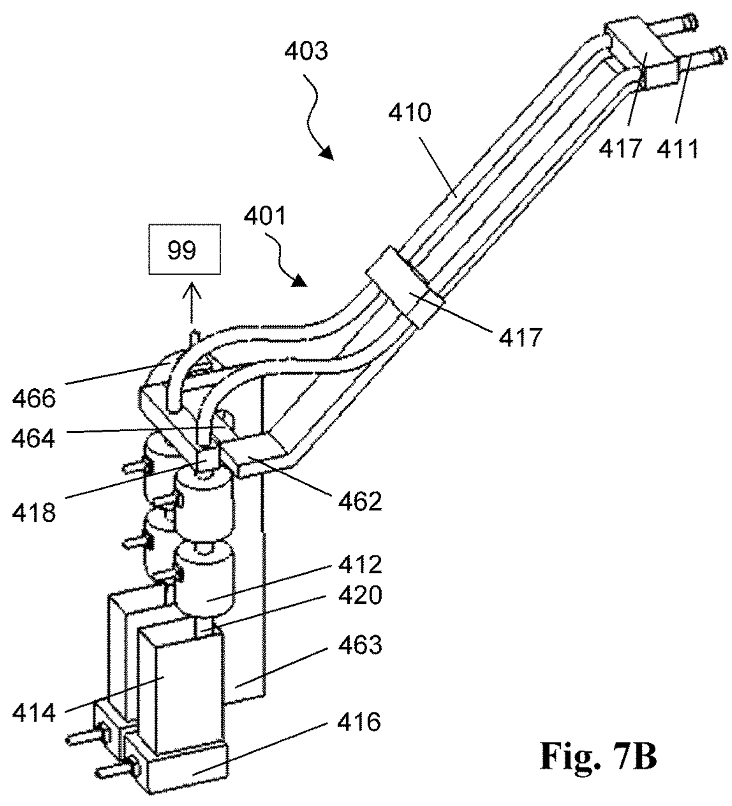
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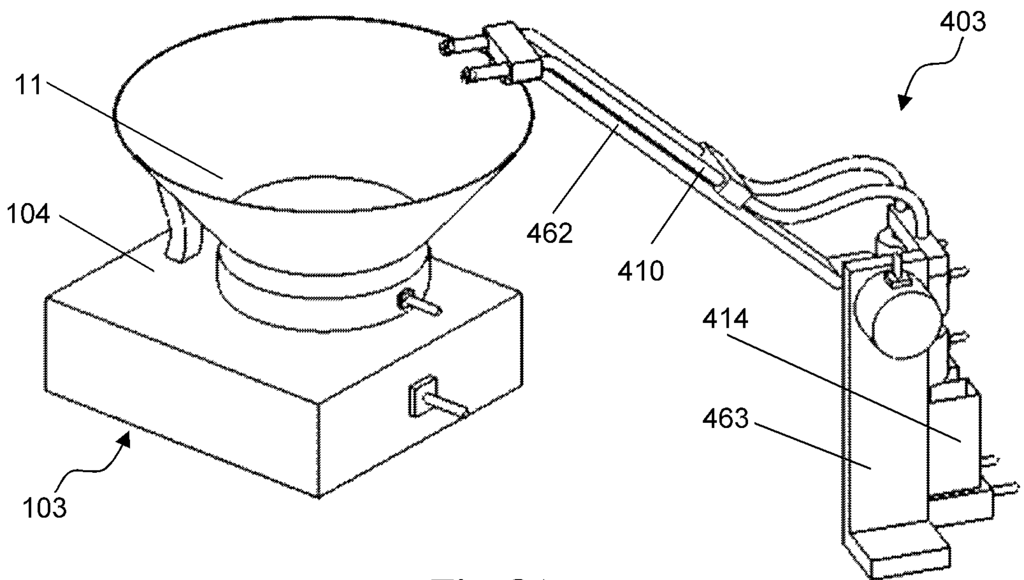

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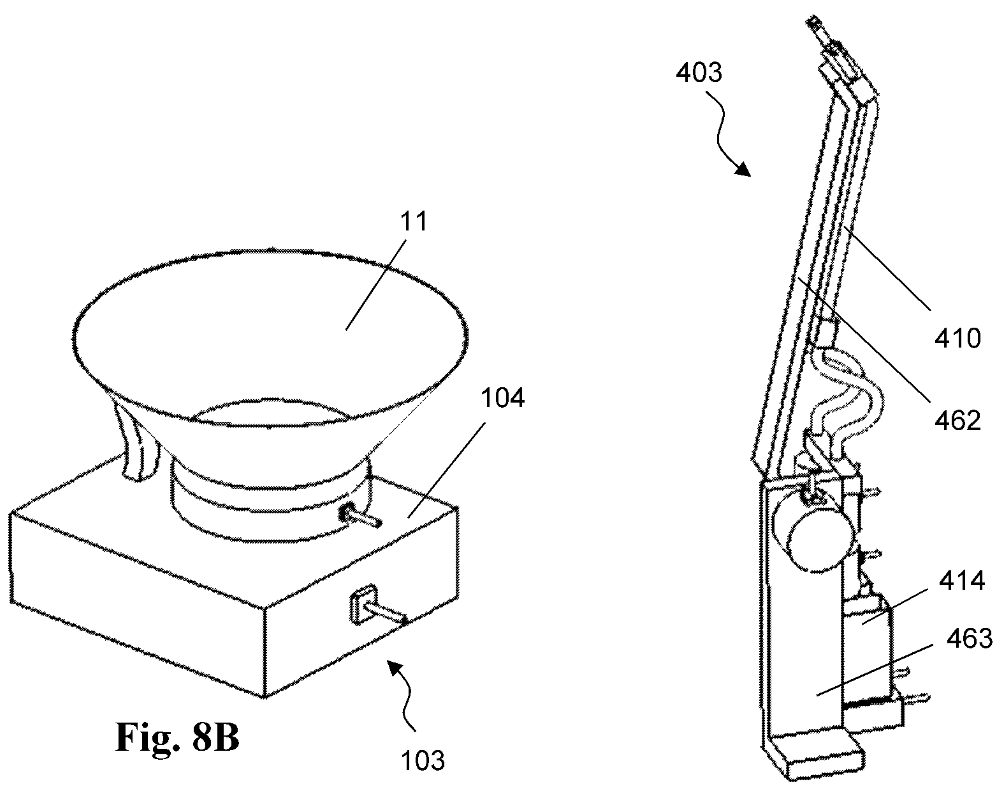

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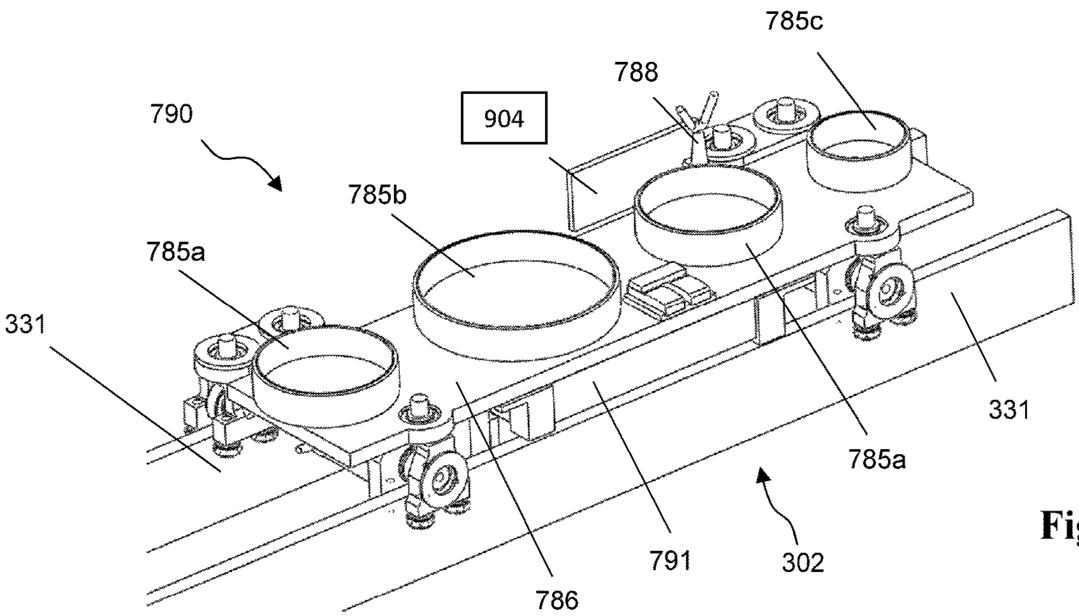
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Fig. 9A

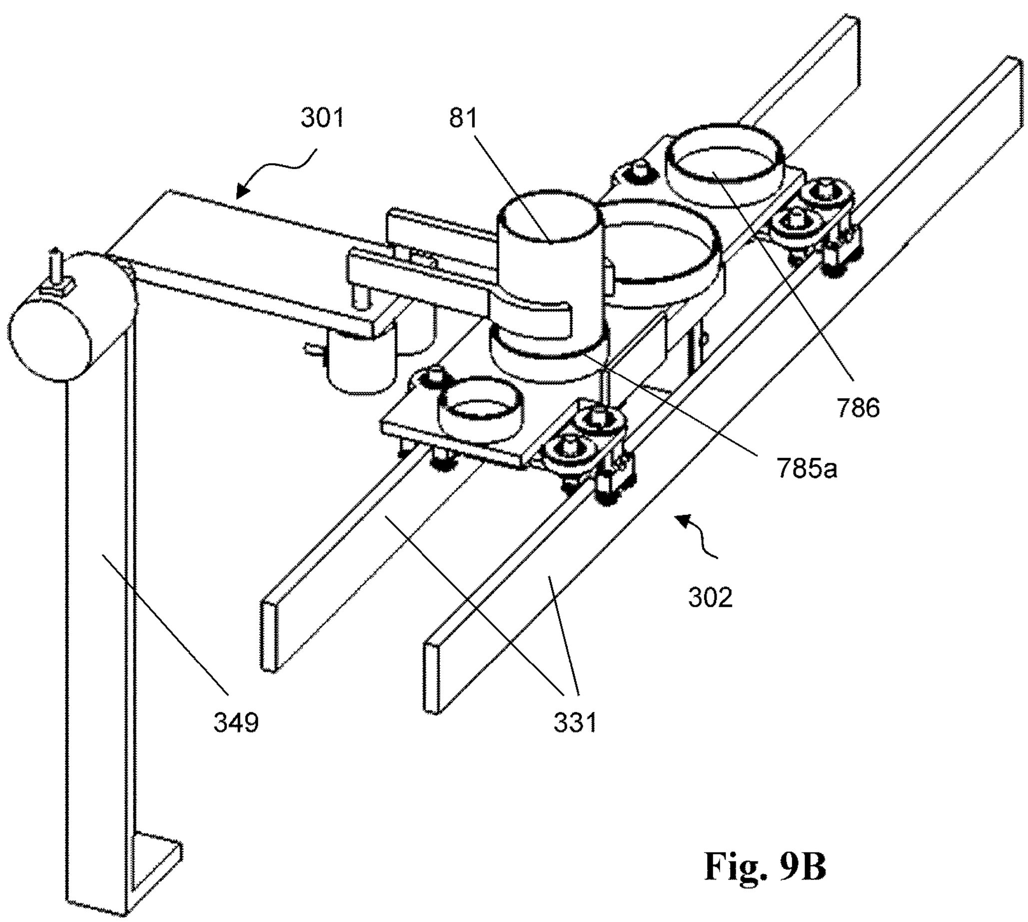
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Fig. 9B

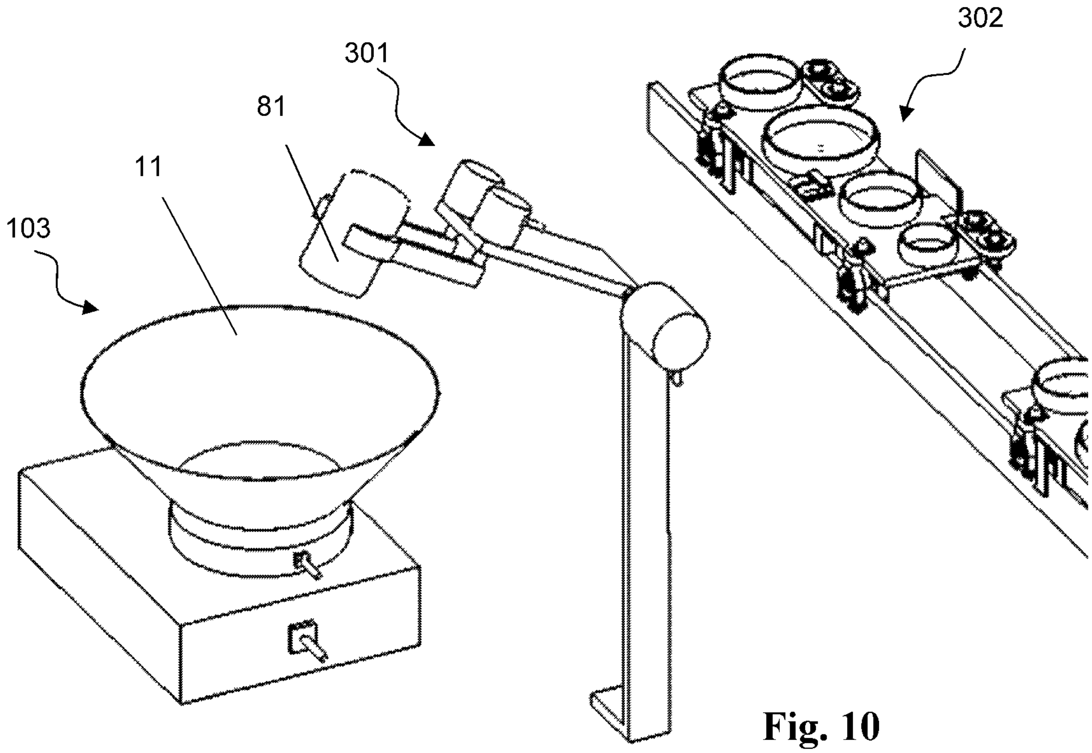
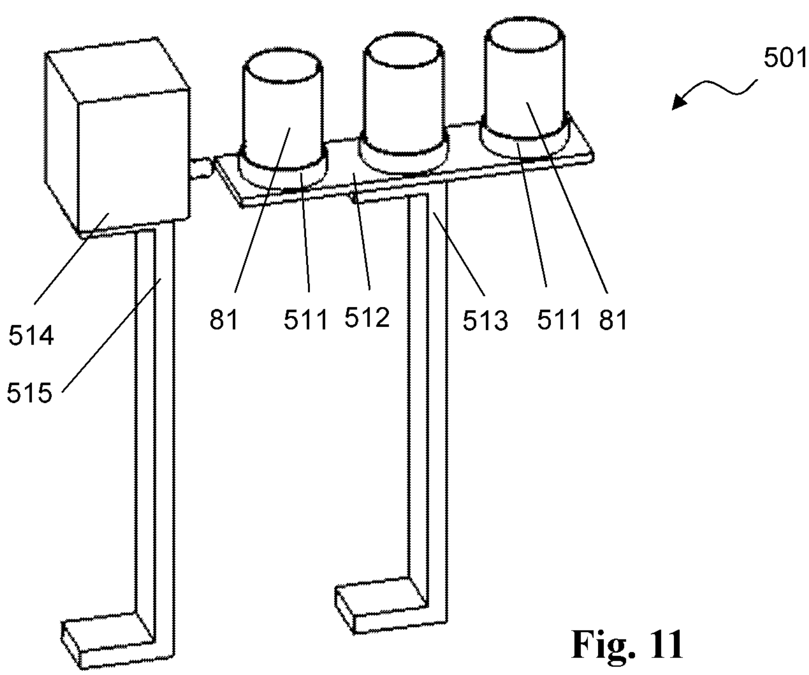


Fig. 10



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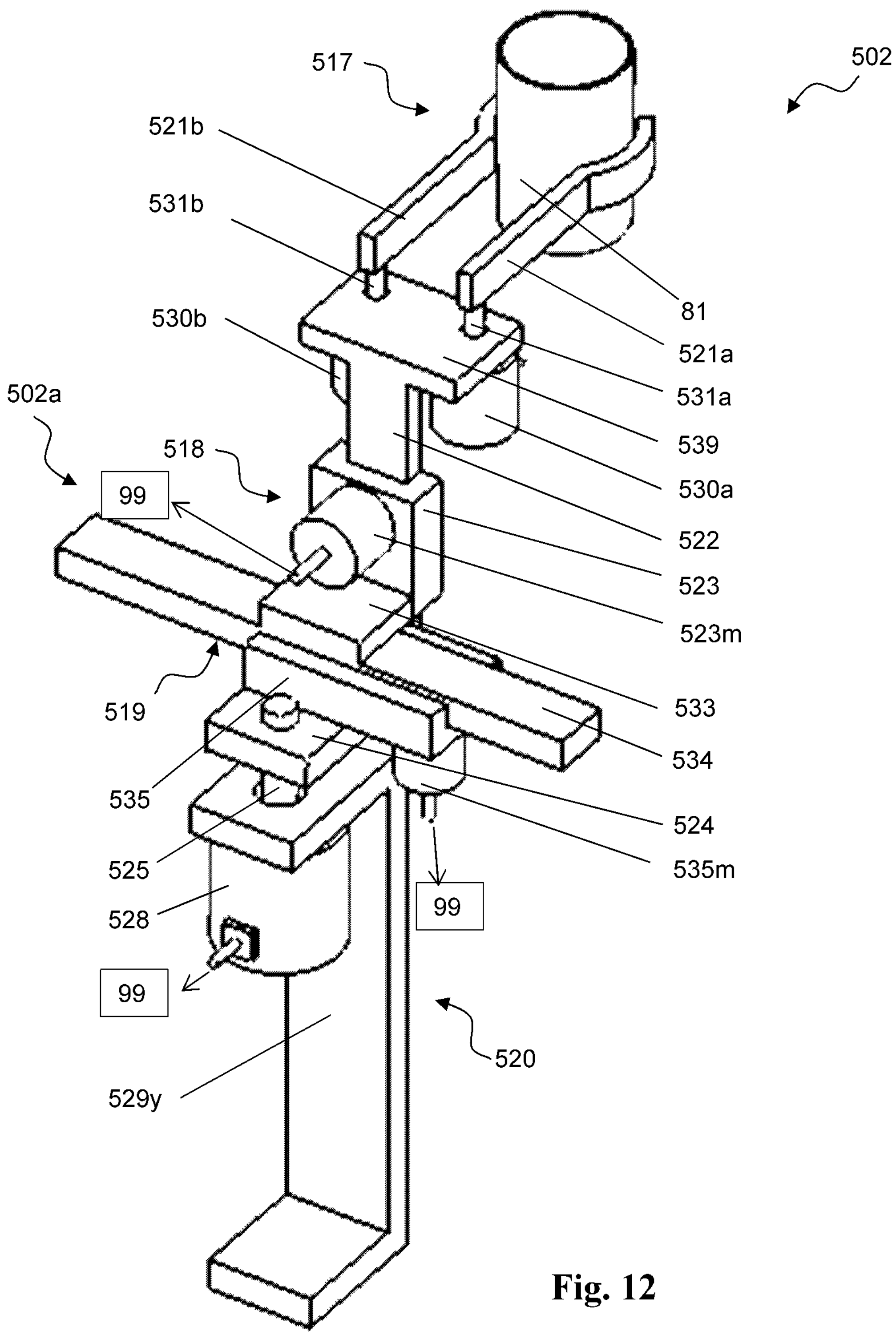

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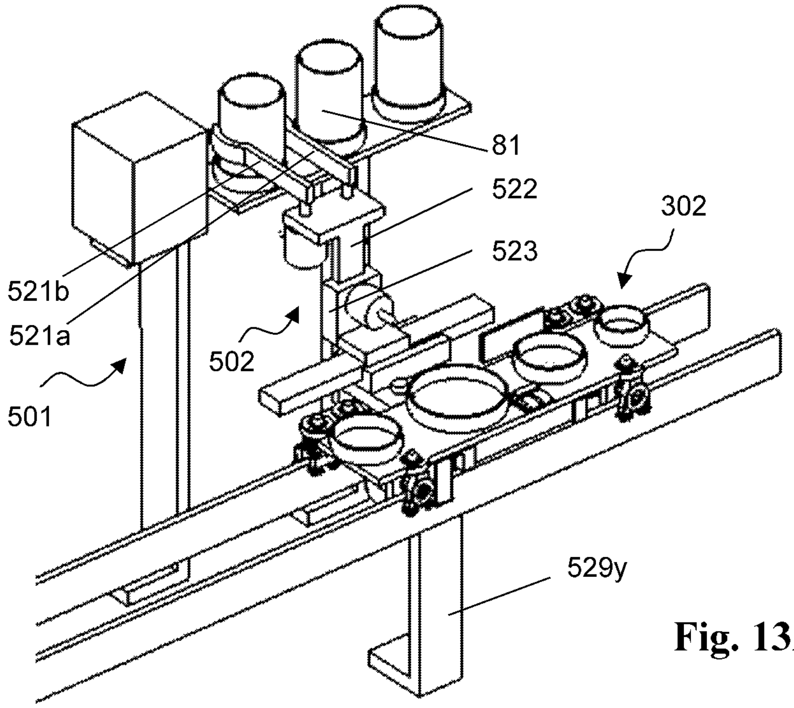
Fig. 12



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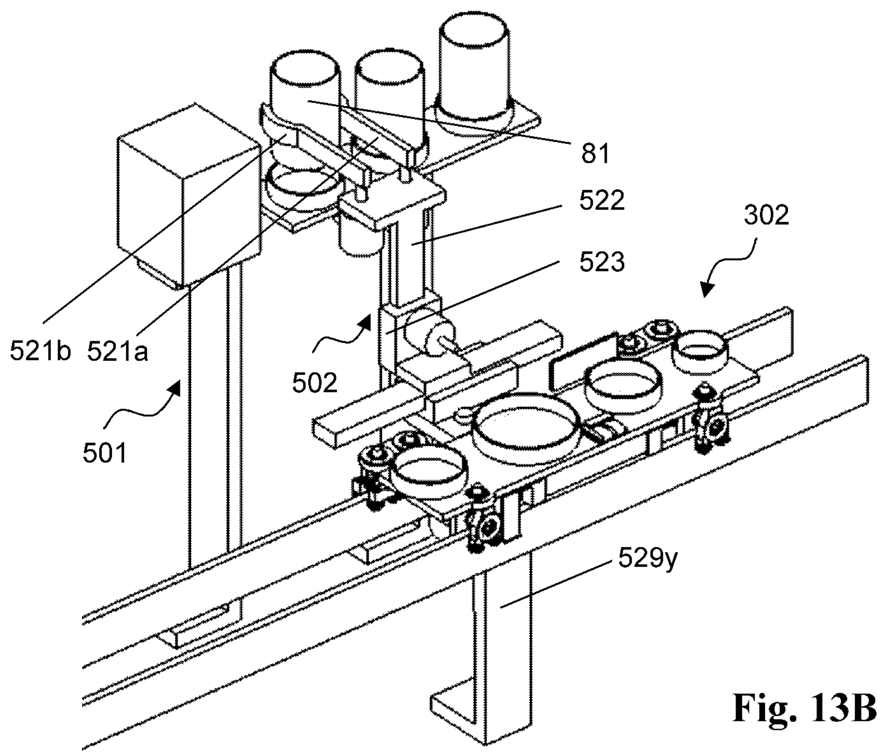

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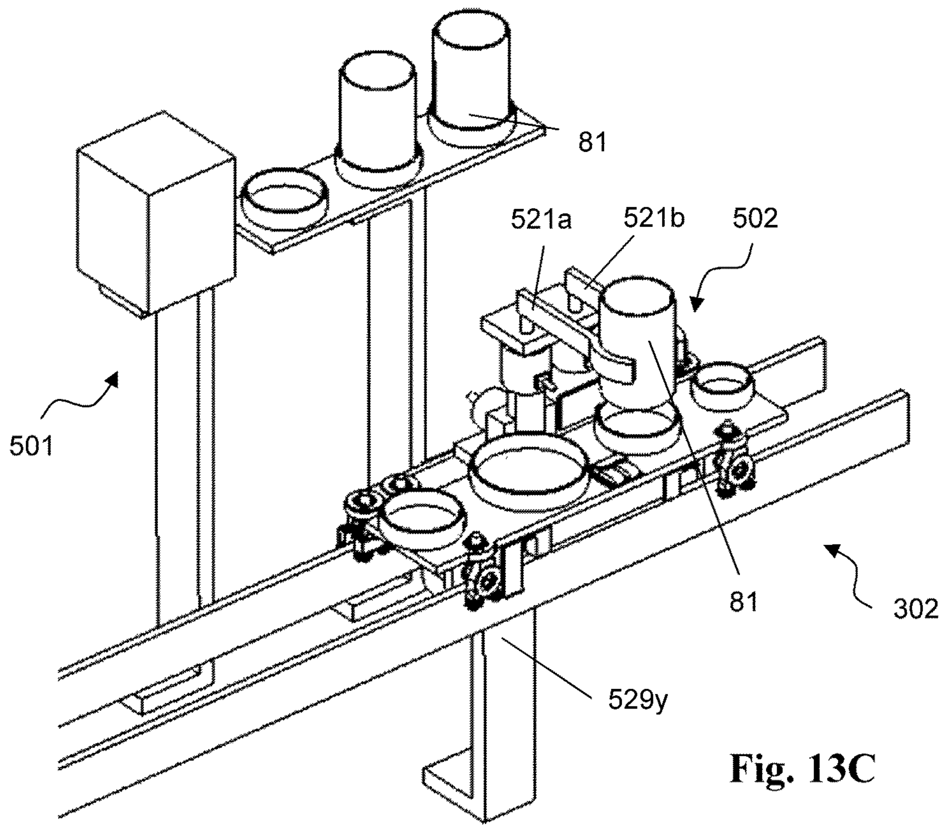

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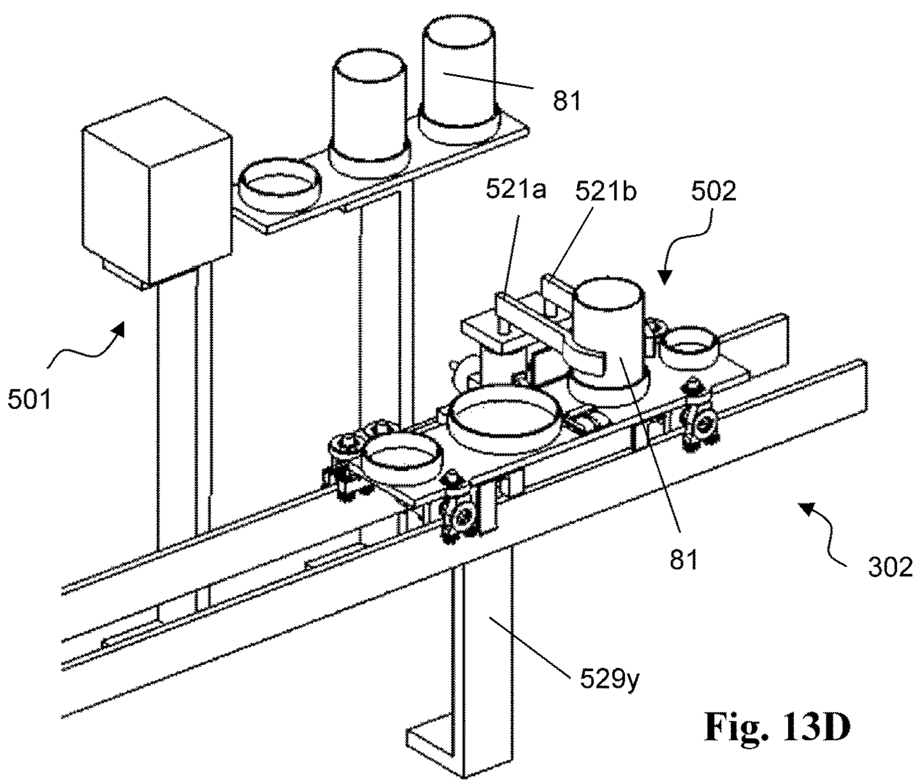

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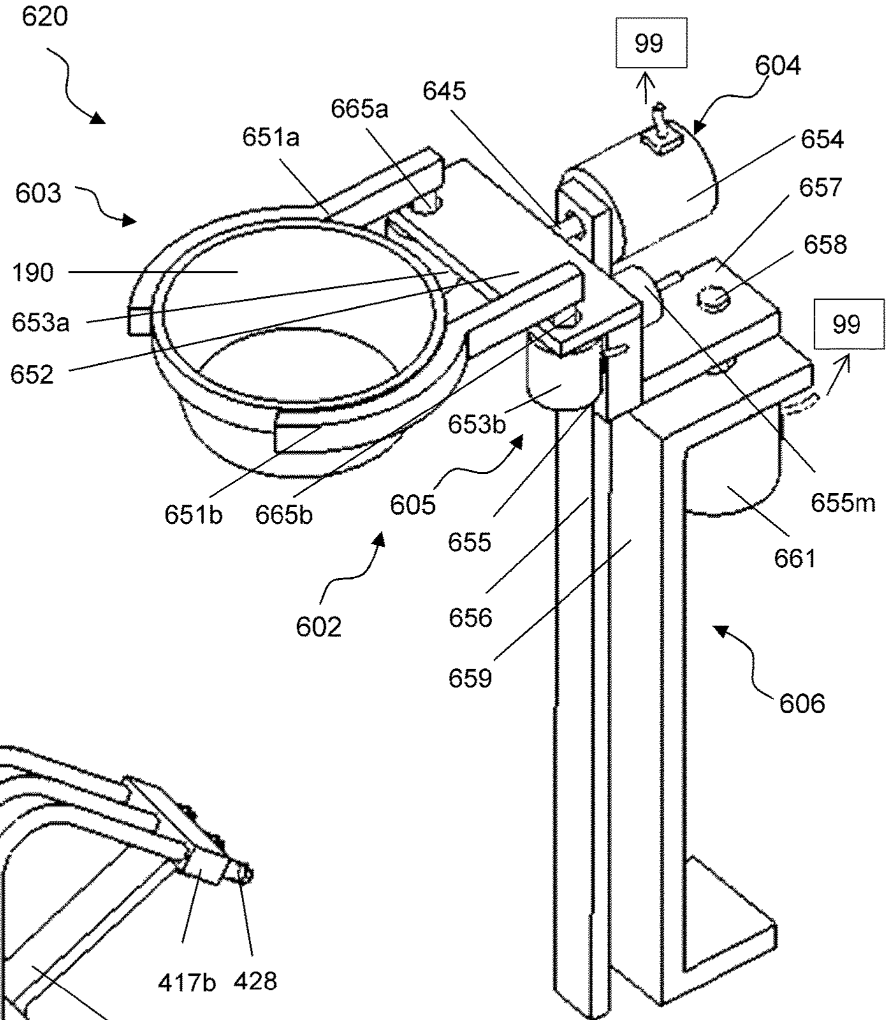
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Fig. 14

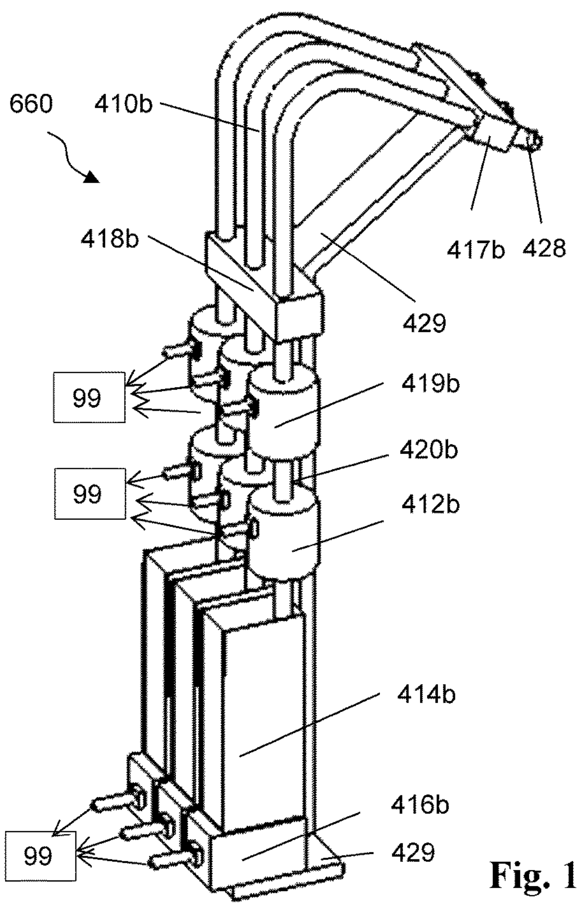
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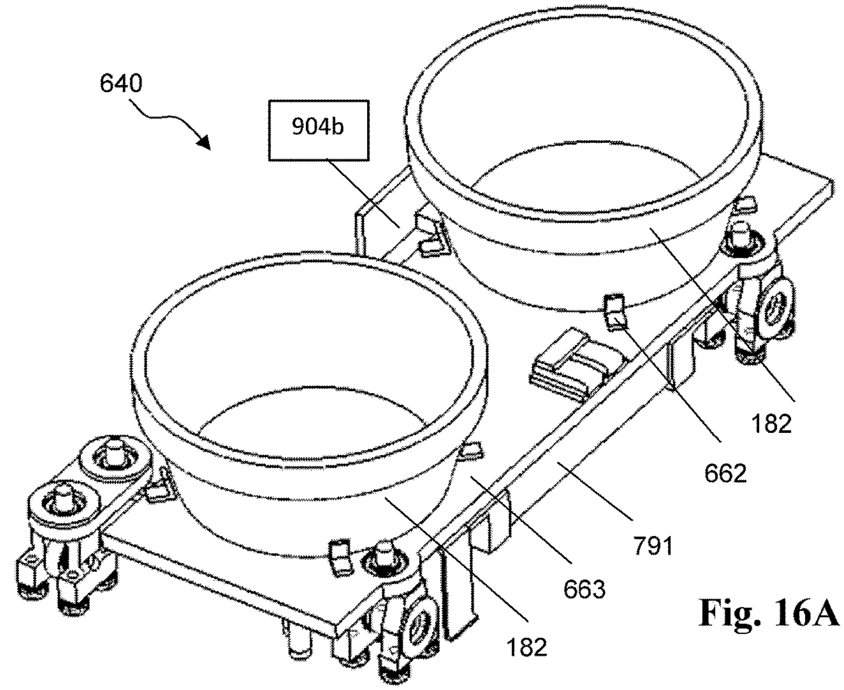
Fig. 15



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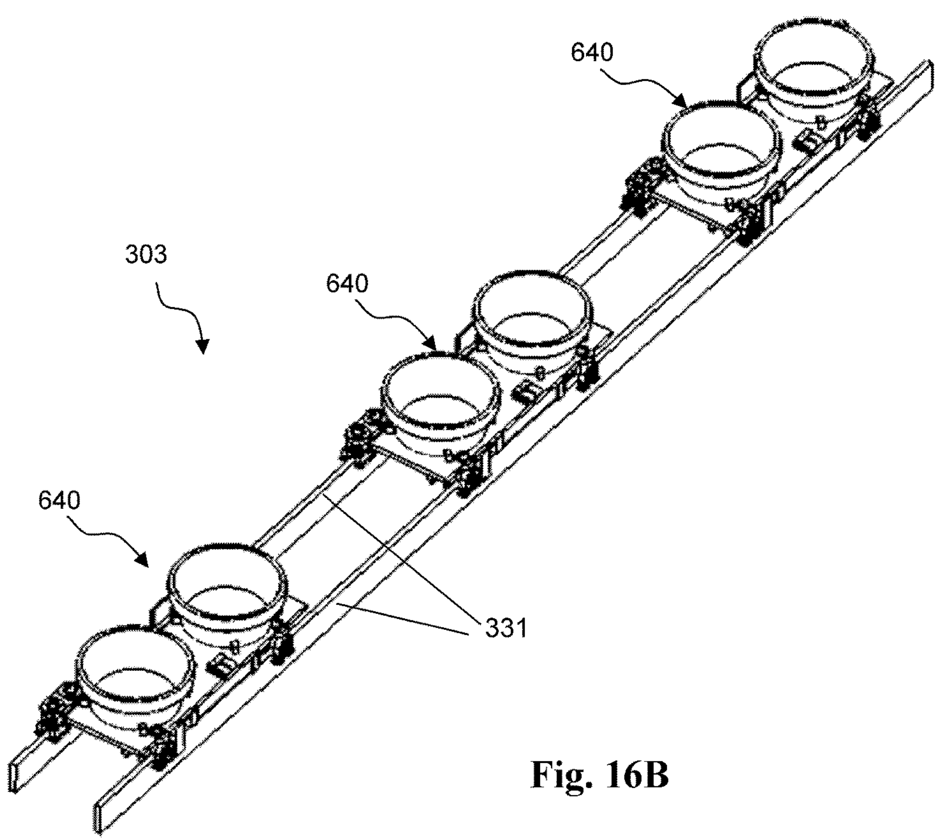

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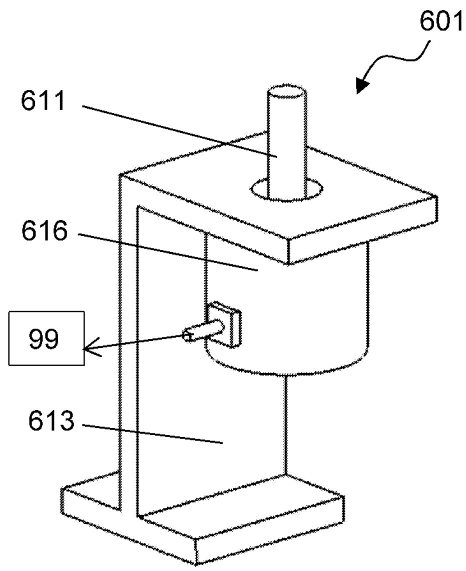
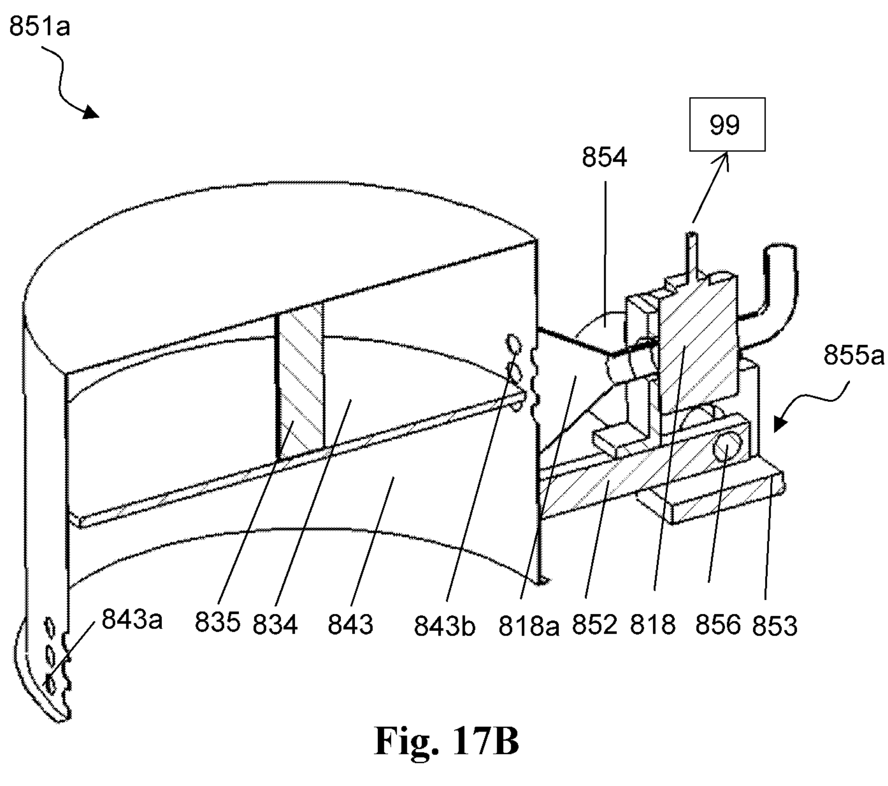


Fig. 17A



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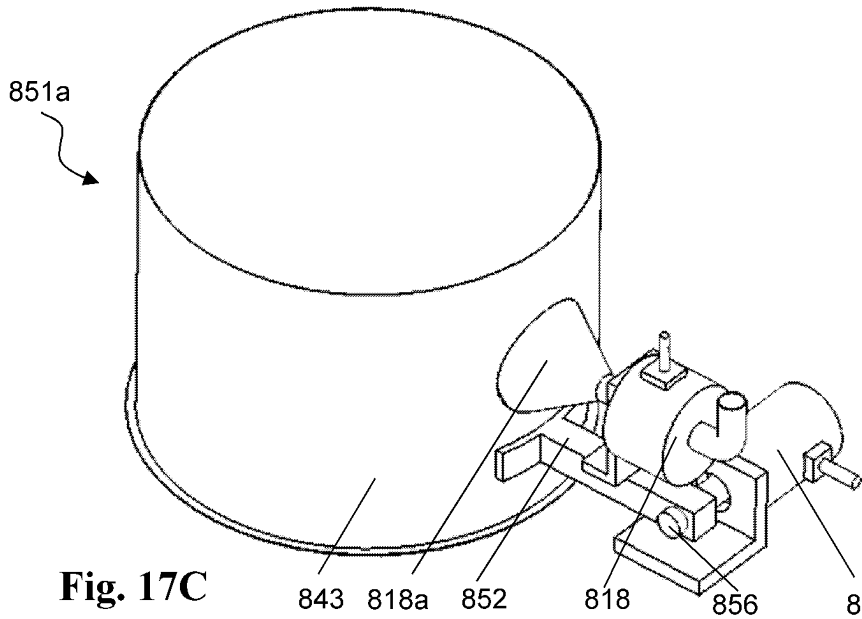

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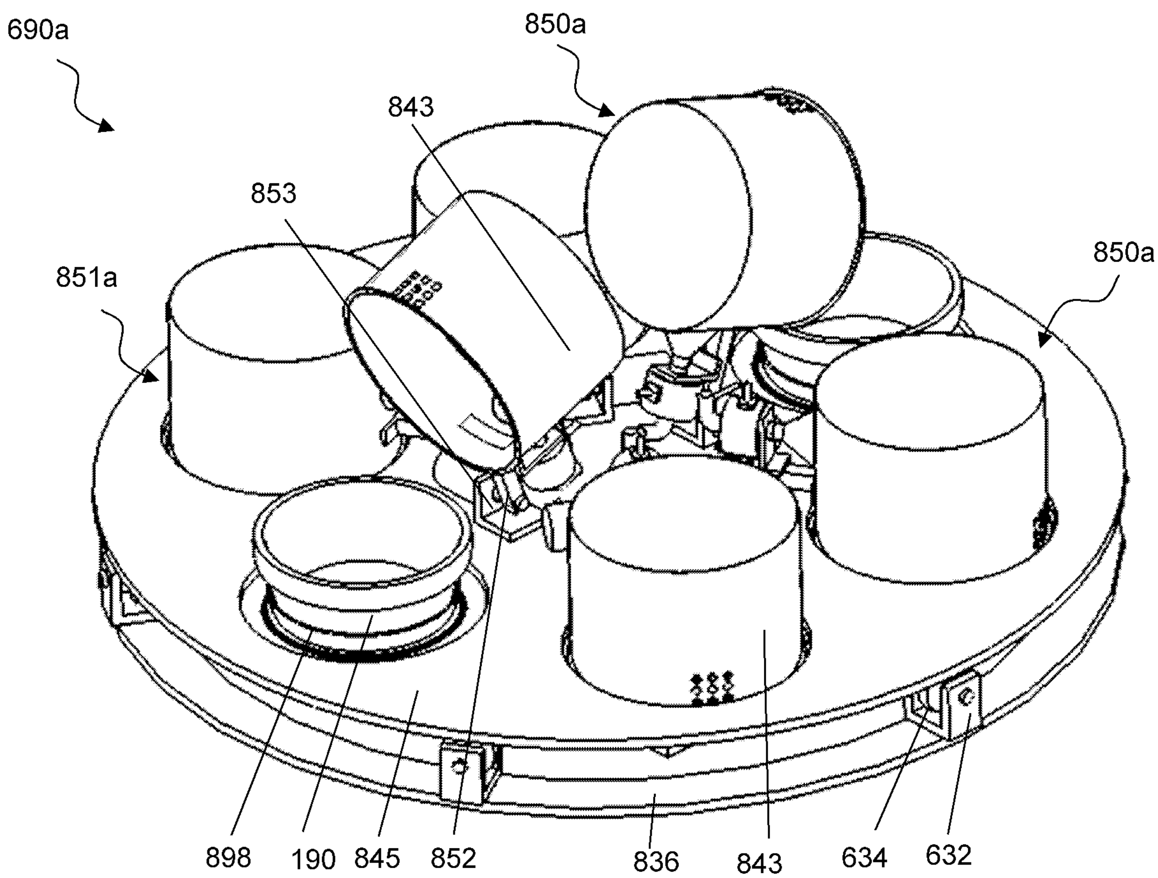

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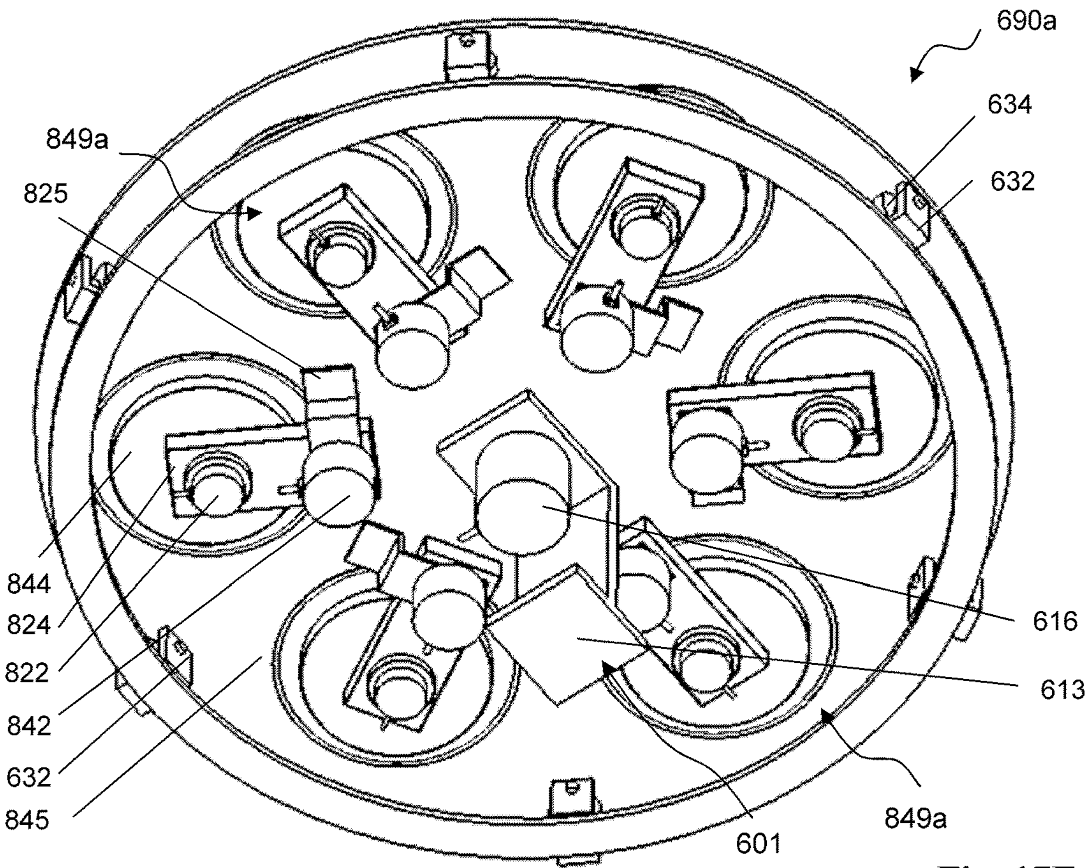

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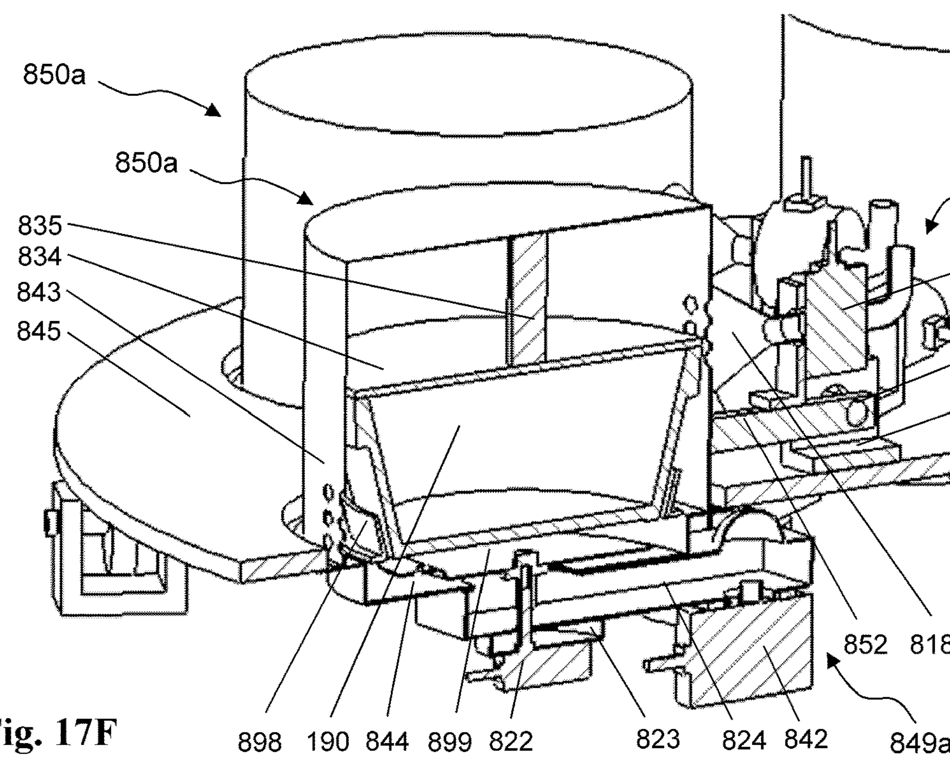

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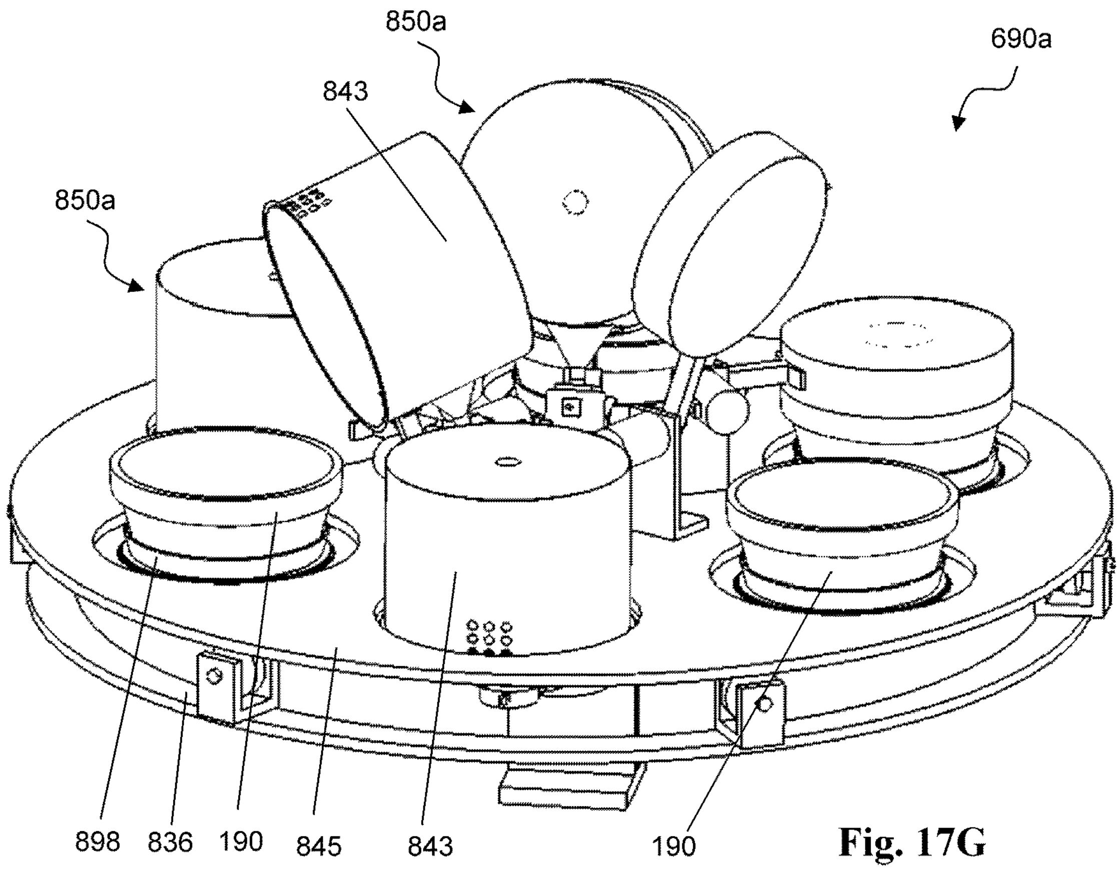

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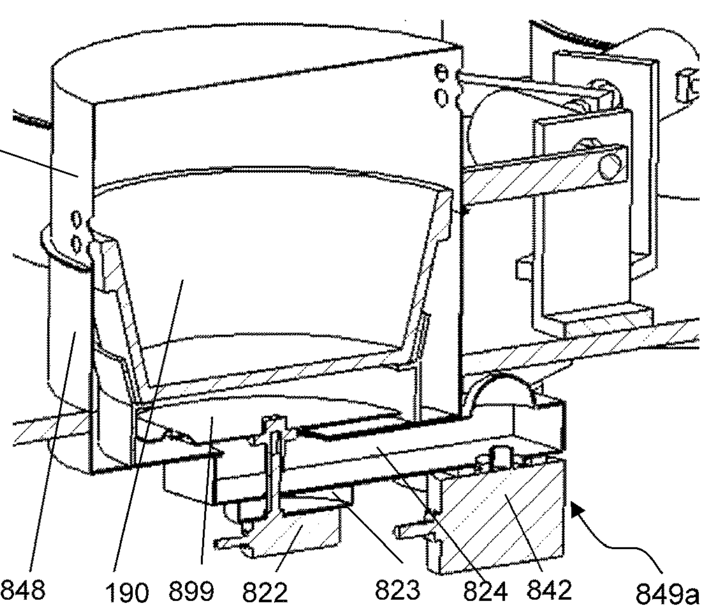

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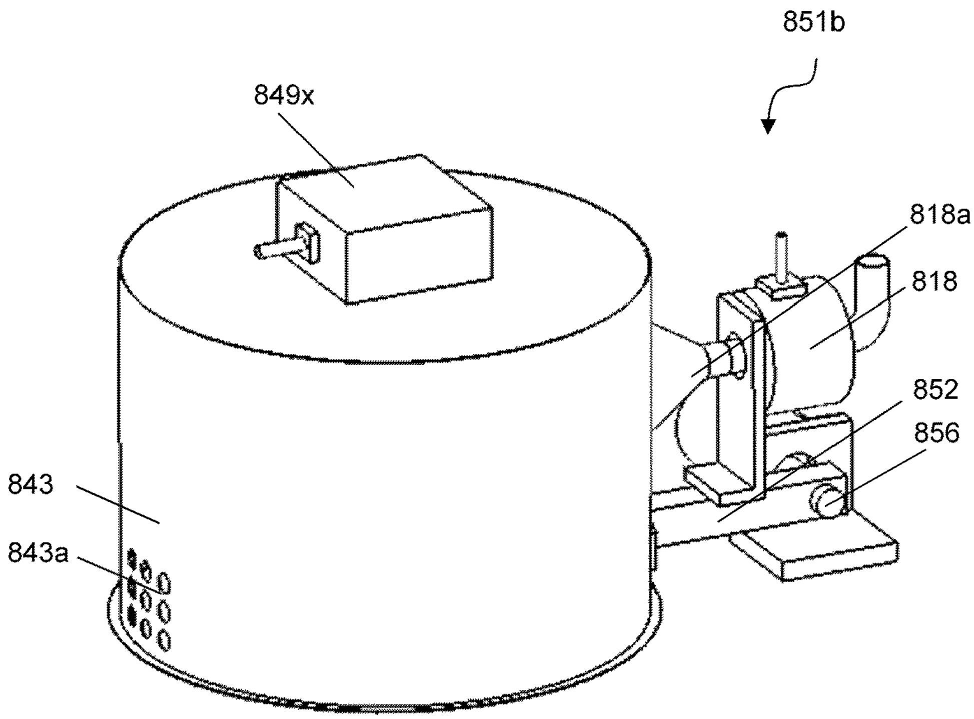

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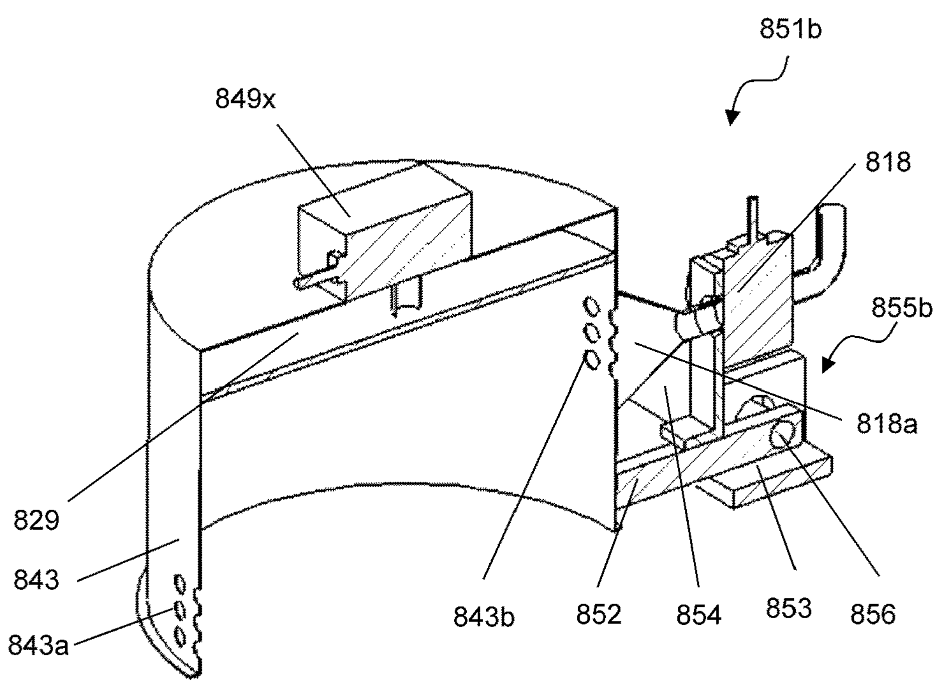

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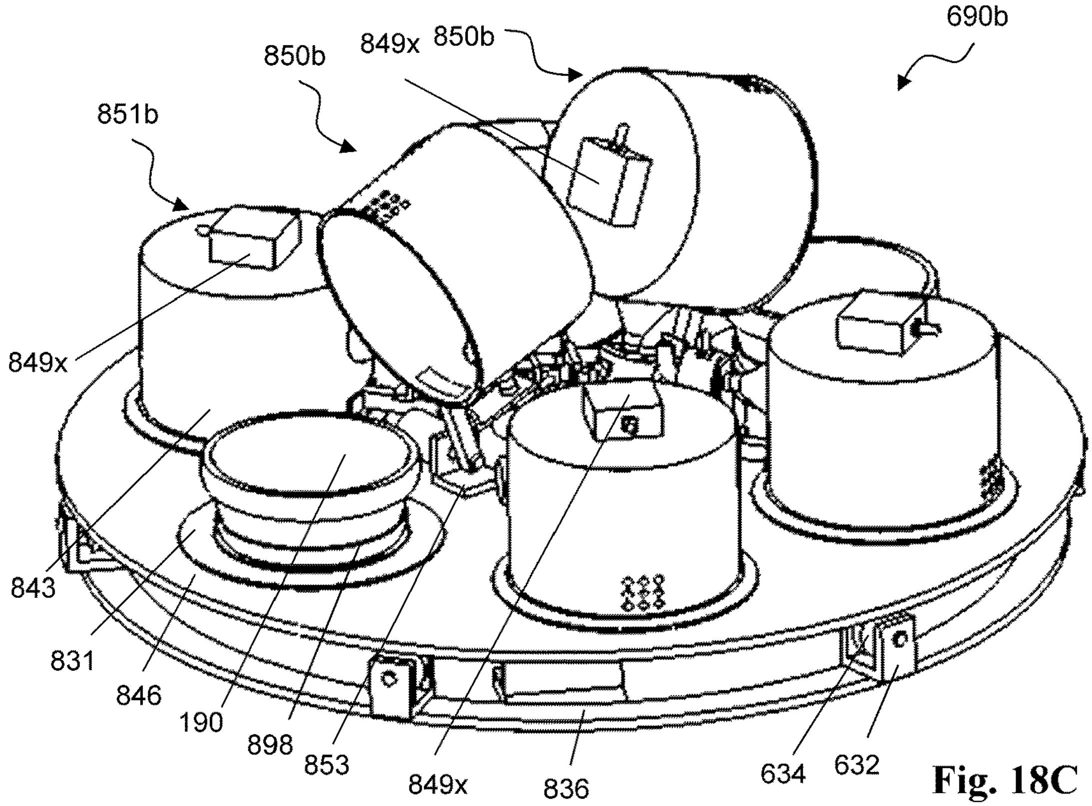

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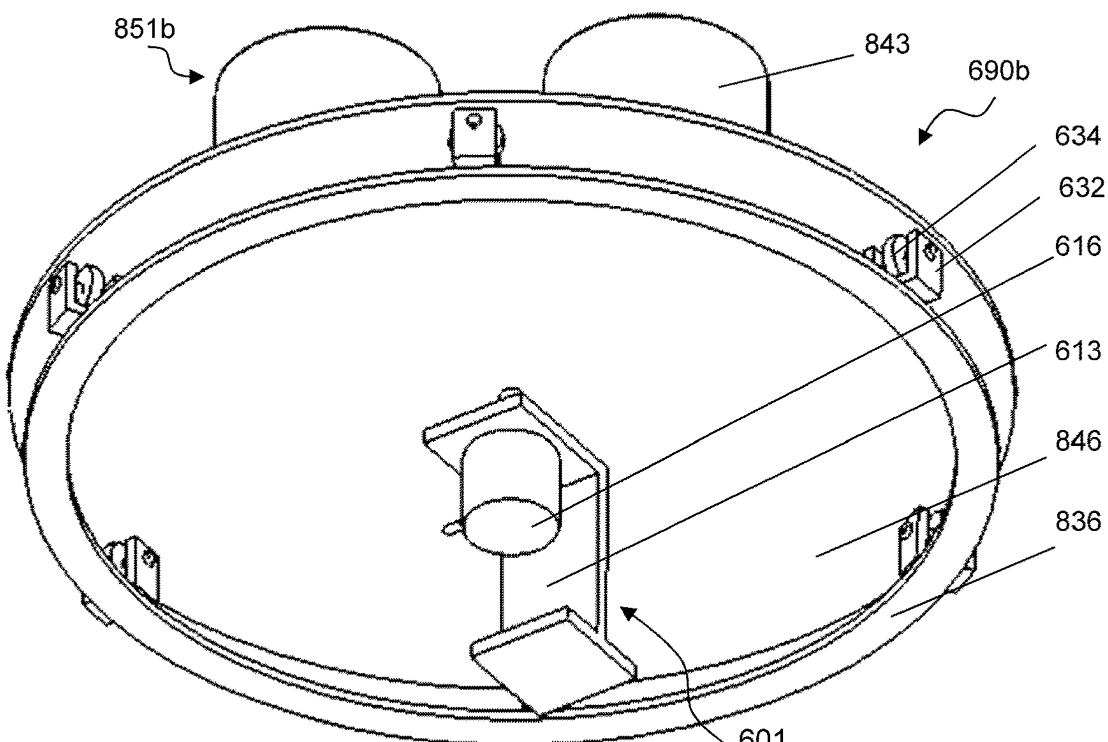

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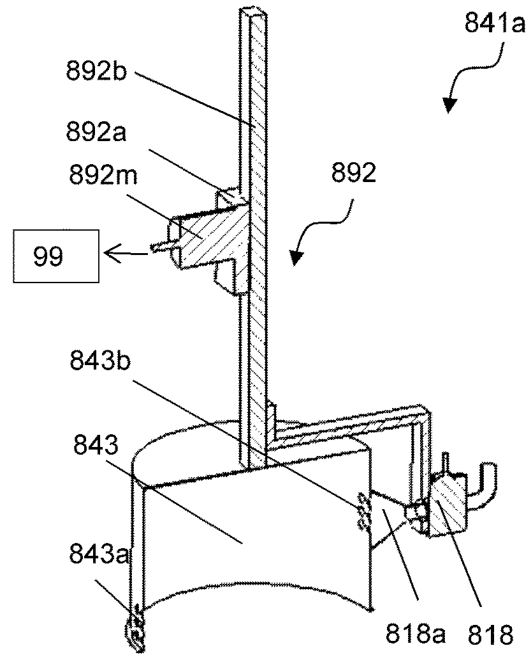

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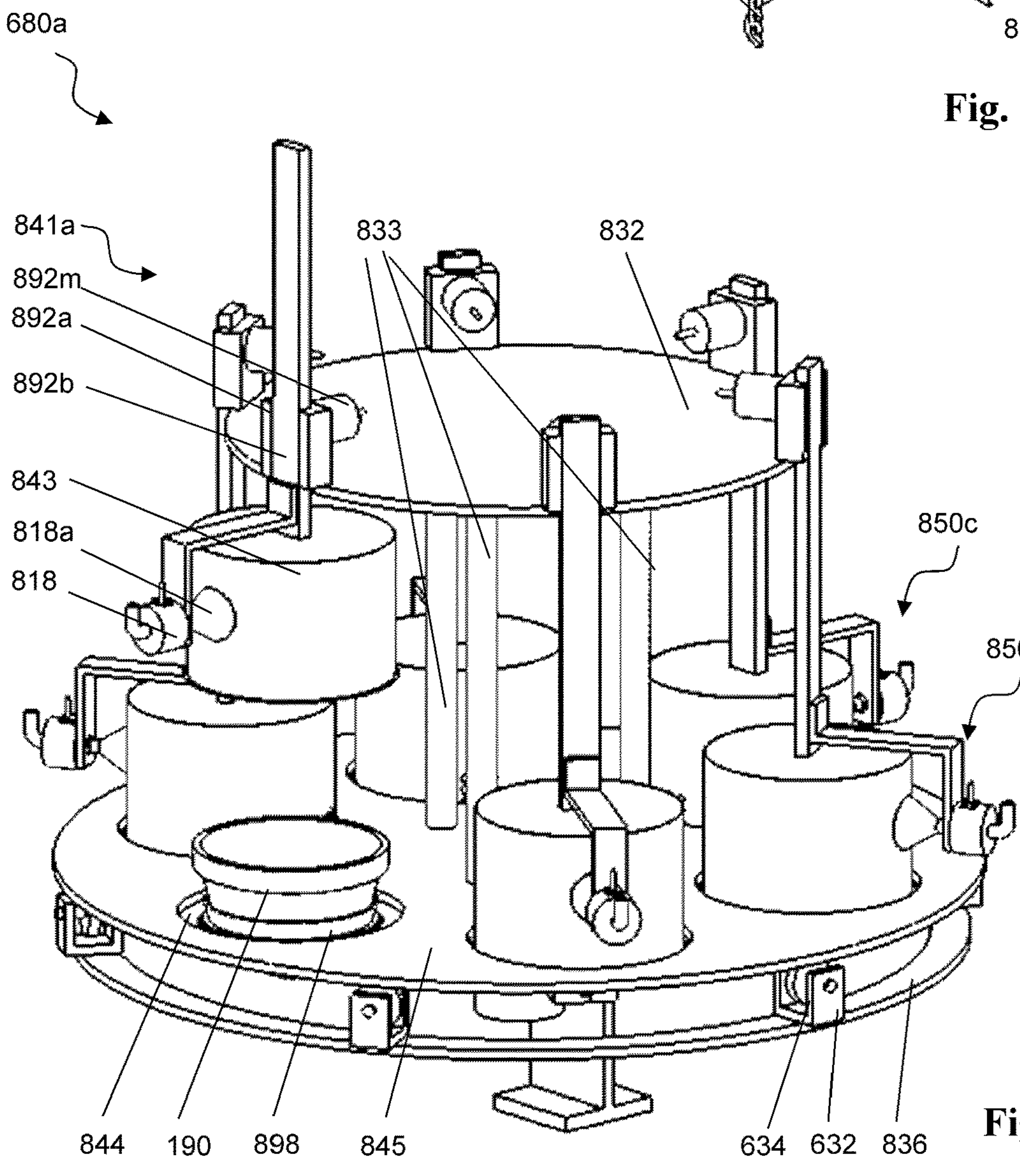


Fig. 19B

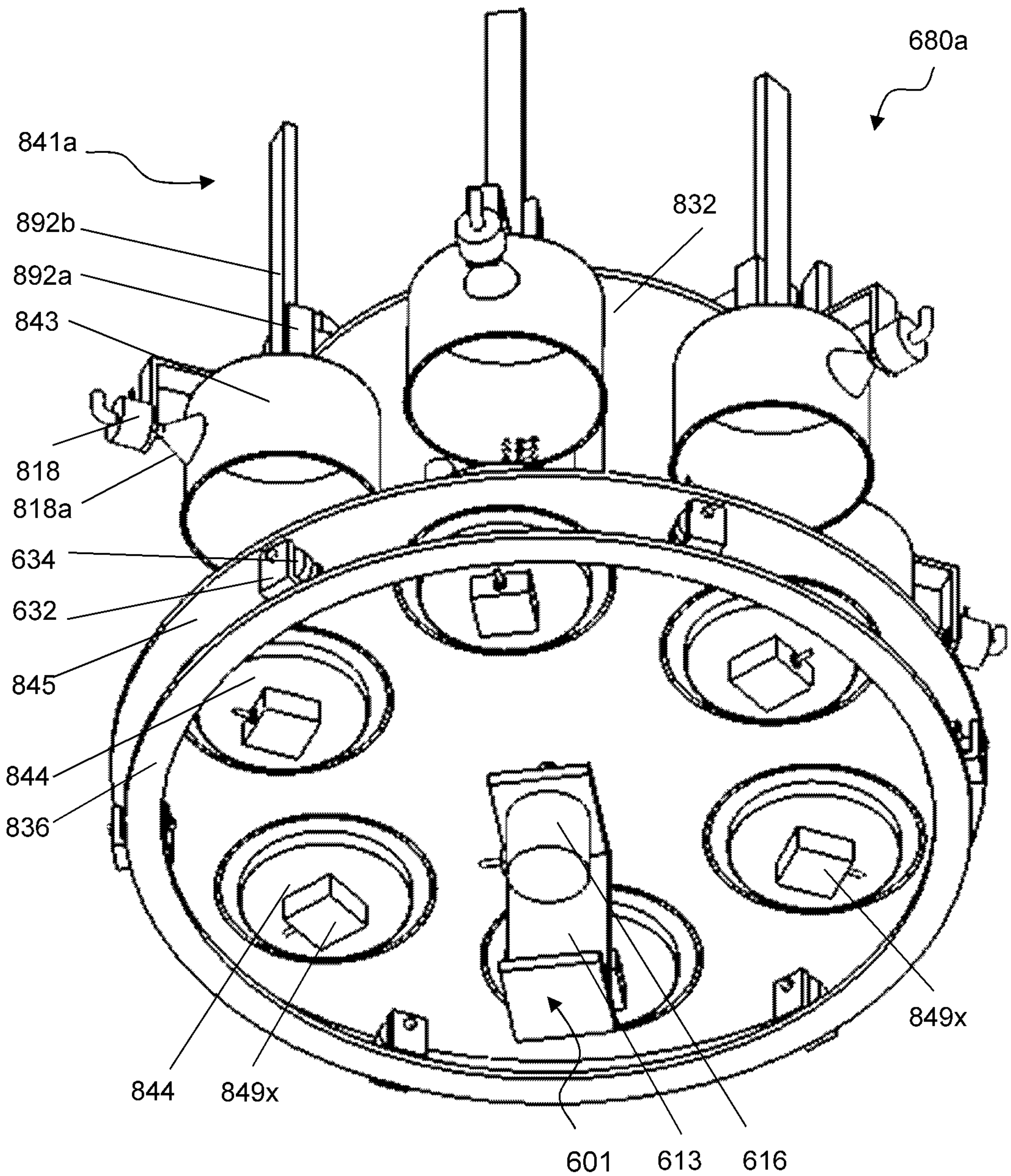

680a
841a
892b
892a
843
832
818
818a
634
632
845
844
836
844
849x
849x
601
613
616

Fig. 19C

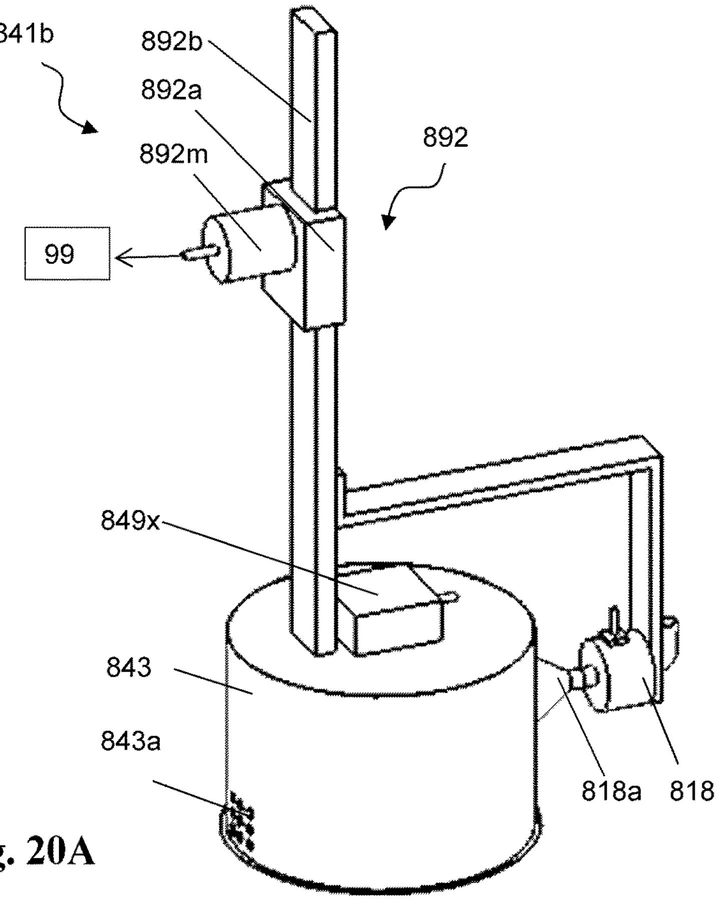


Fig. 20A

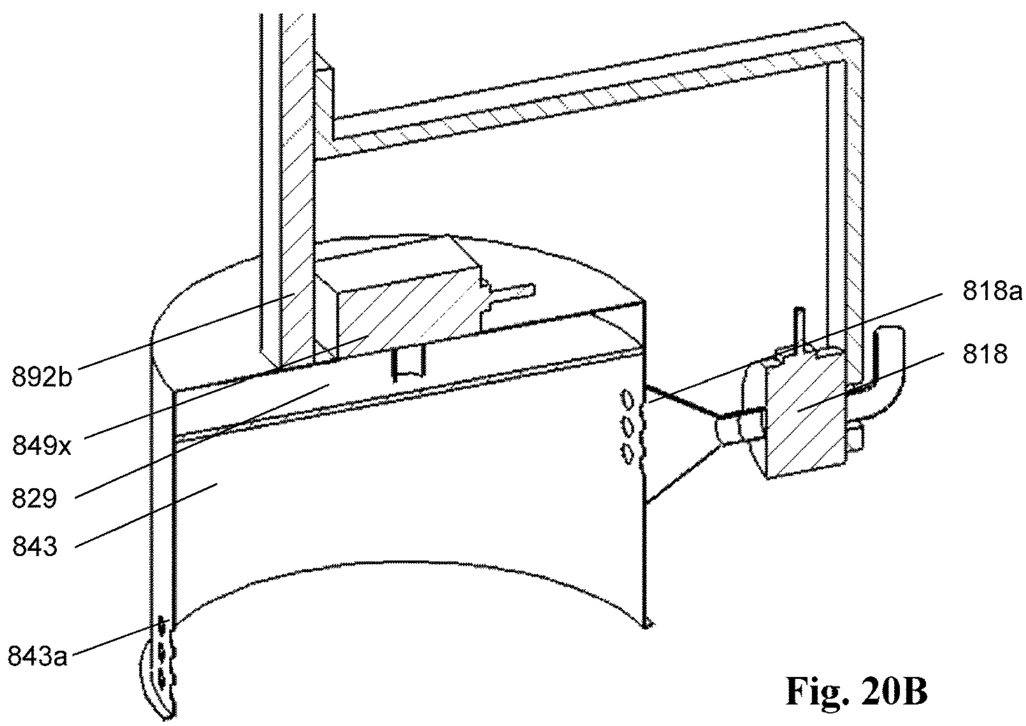


Fig. 20B

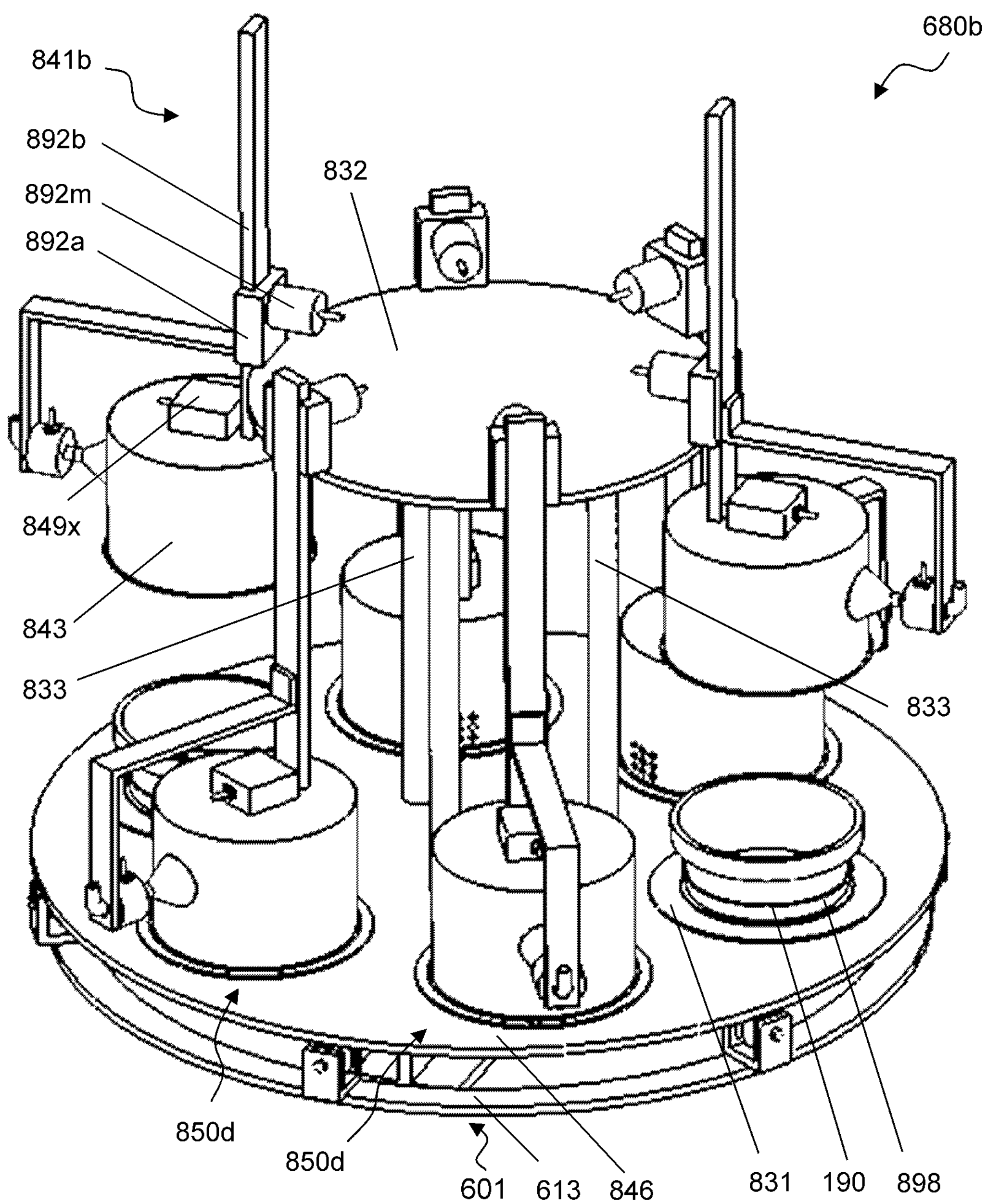

680b
841b
892b
892m
892a
832
849x
843
833
833
850d
850d
601
613
846
831
190
898

Fig. 20C

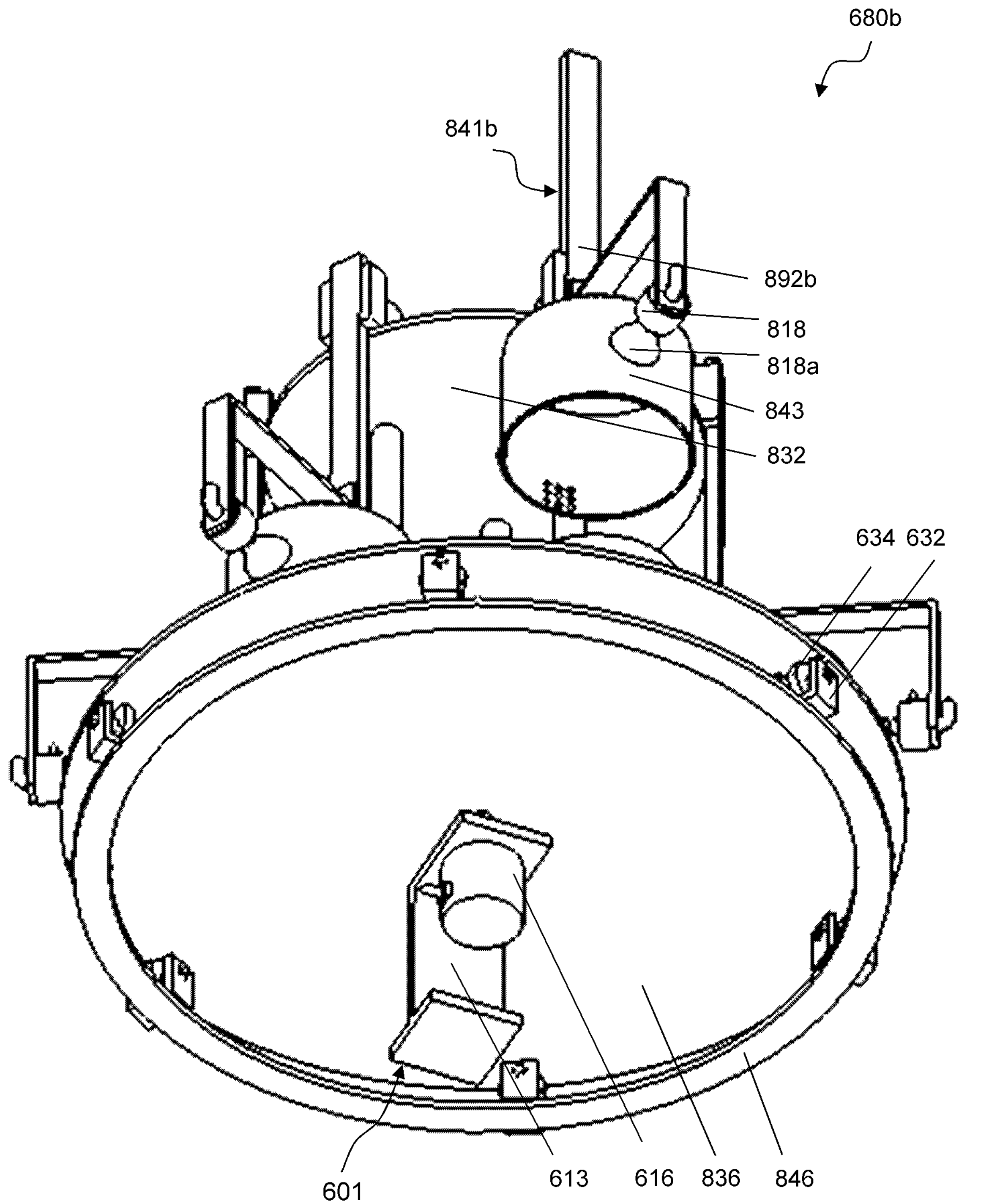


Fig. 20D

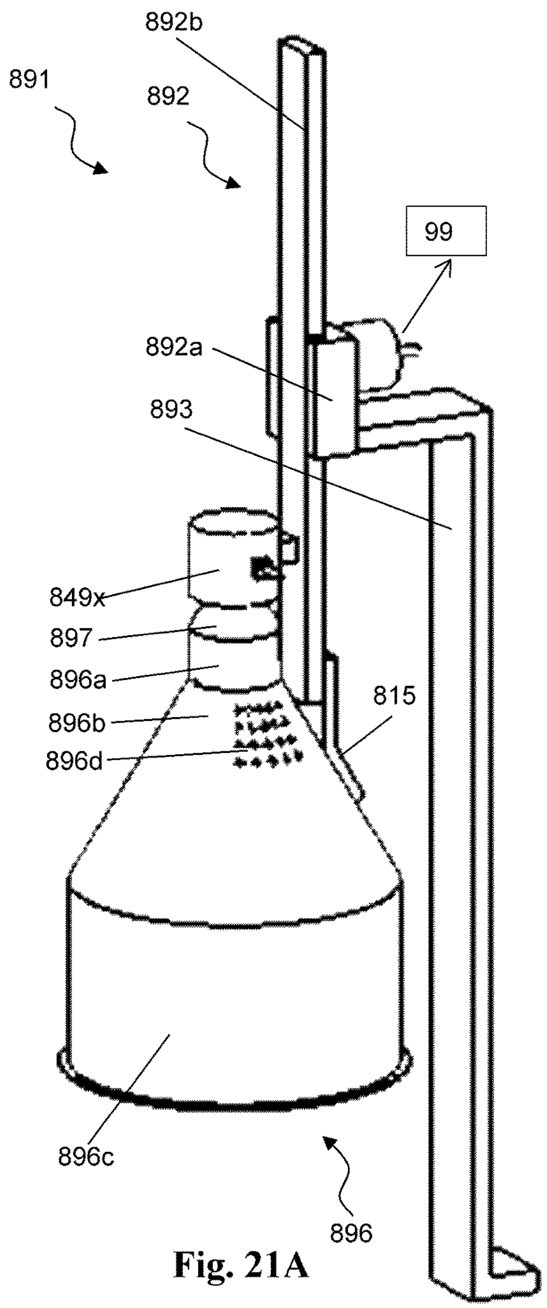


Fig. 21A

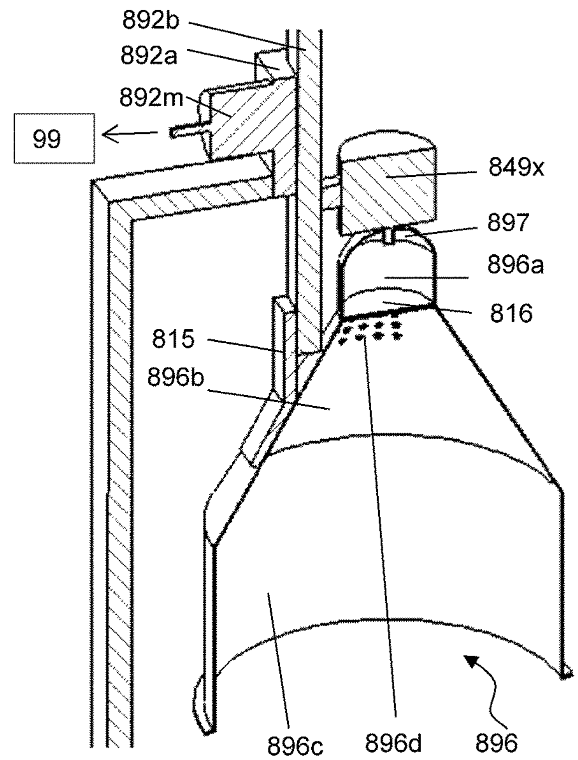


Fig. 21B

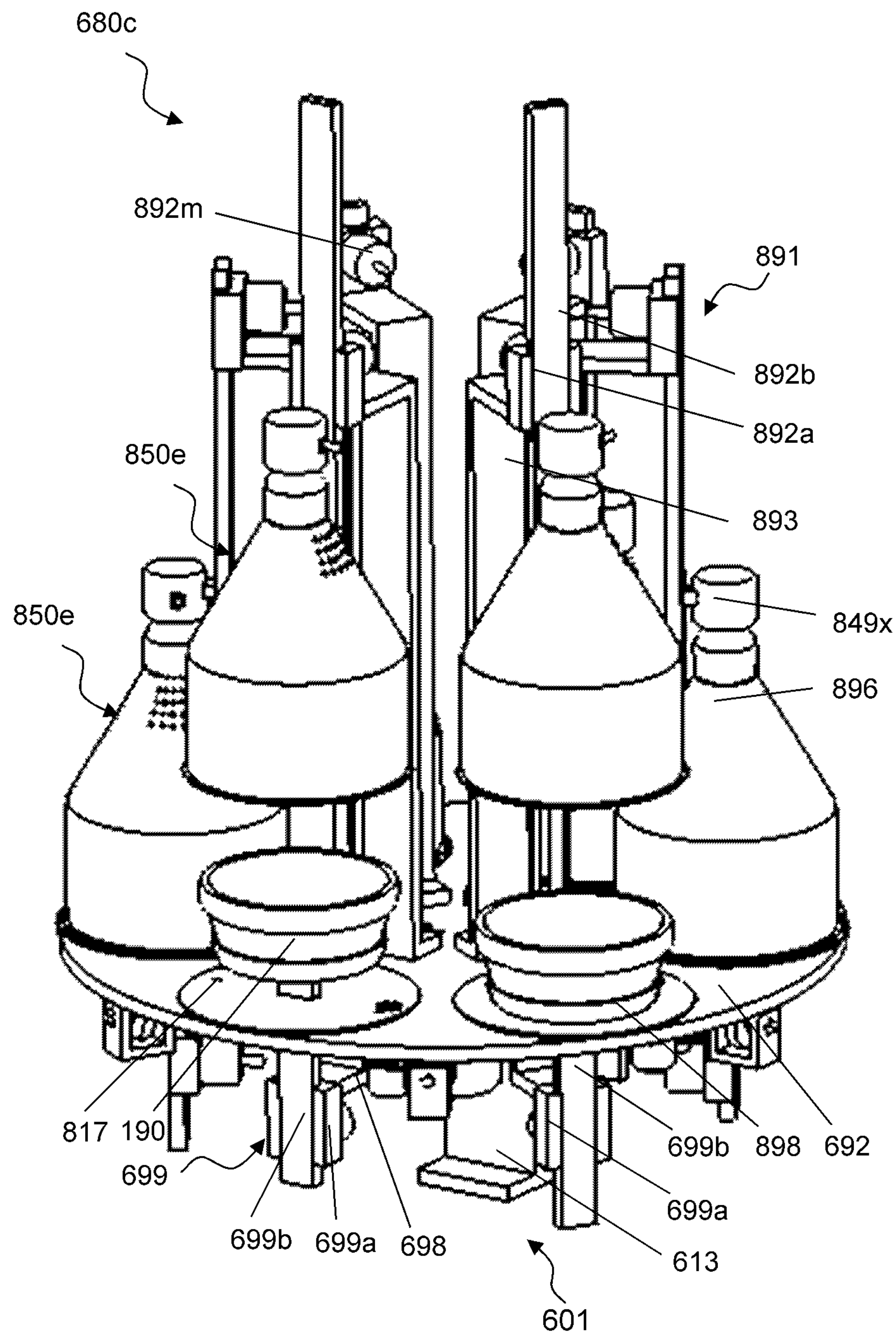

680c
892m
891
892b
892a
850e
893
850e
849x
896
817
190
699
699b
898
692
699a
699b
699a
698
613
601

Fig. 21C

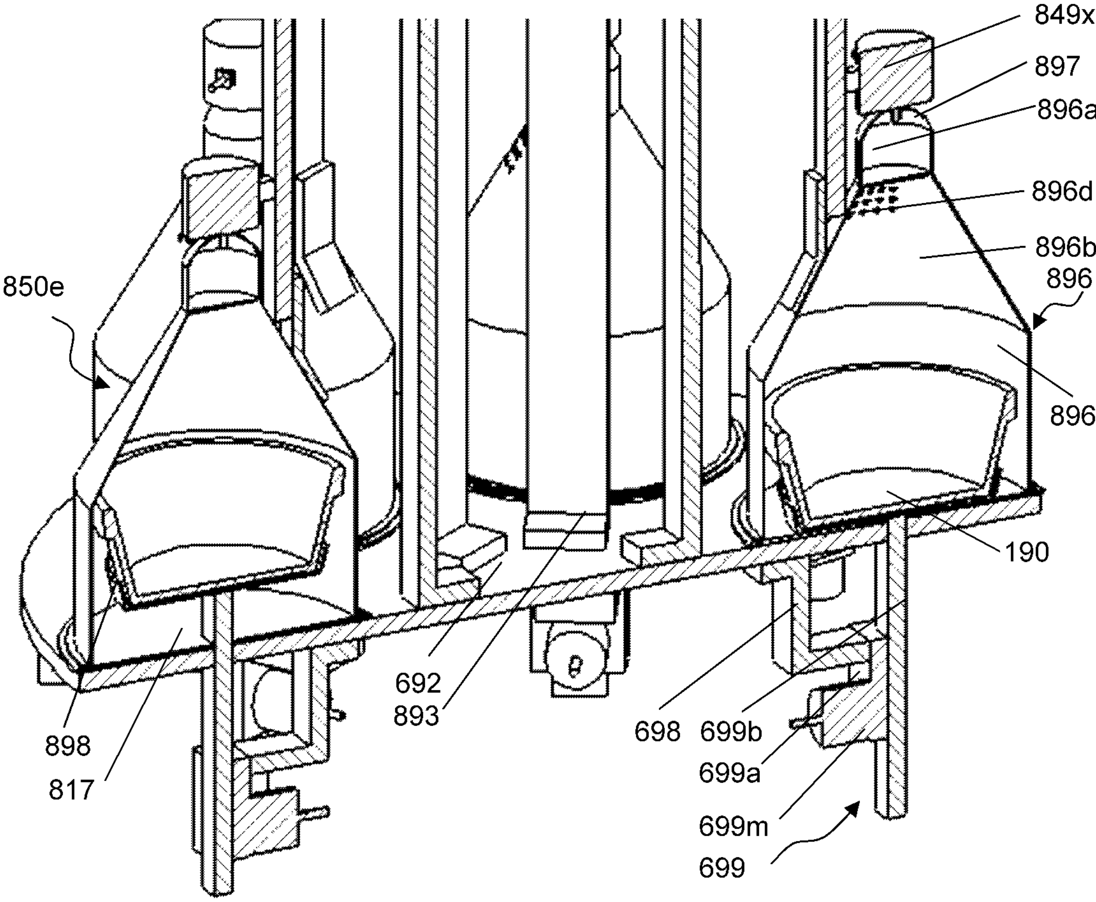
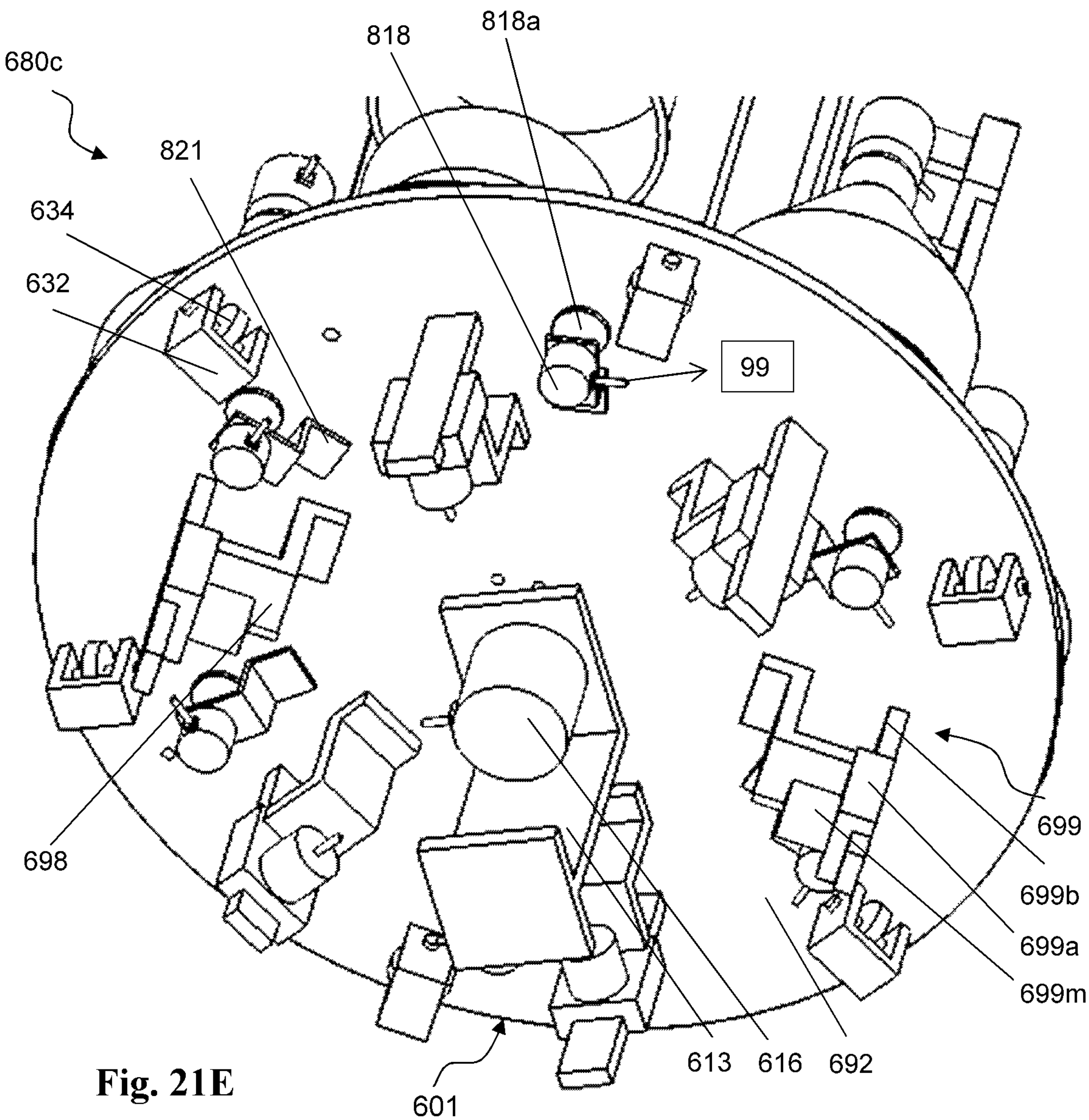

849x
897
896a
896d
896b
896
850e
896c
190
692
893
698
699b
699a
699m
699
898
817

Fig. 21D



680c
821
818
818a
634
632
99
698
699
699b
699a
699m
613
616
692
601

Fig. 21E

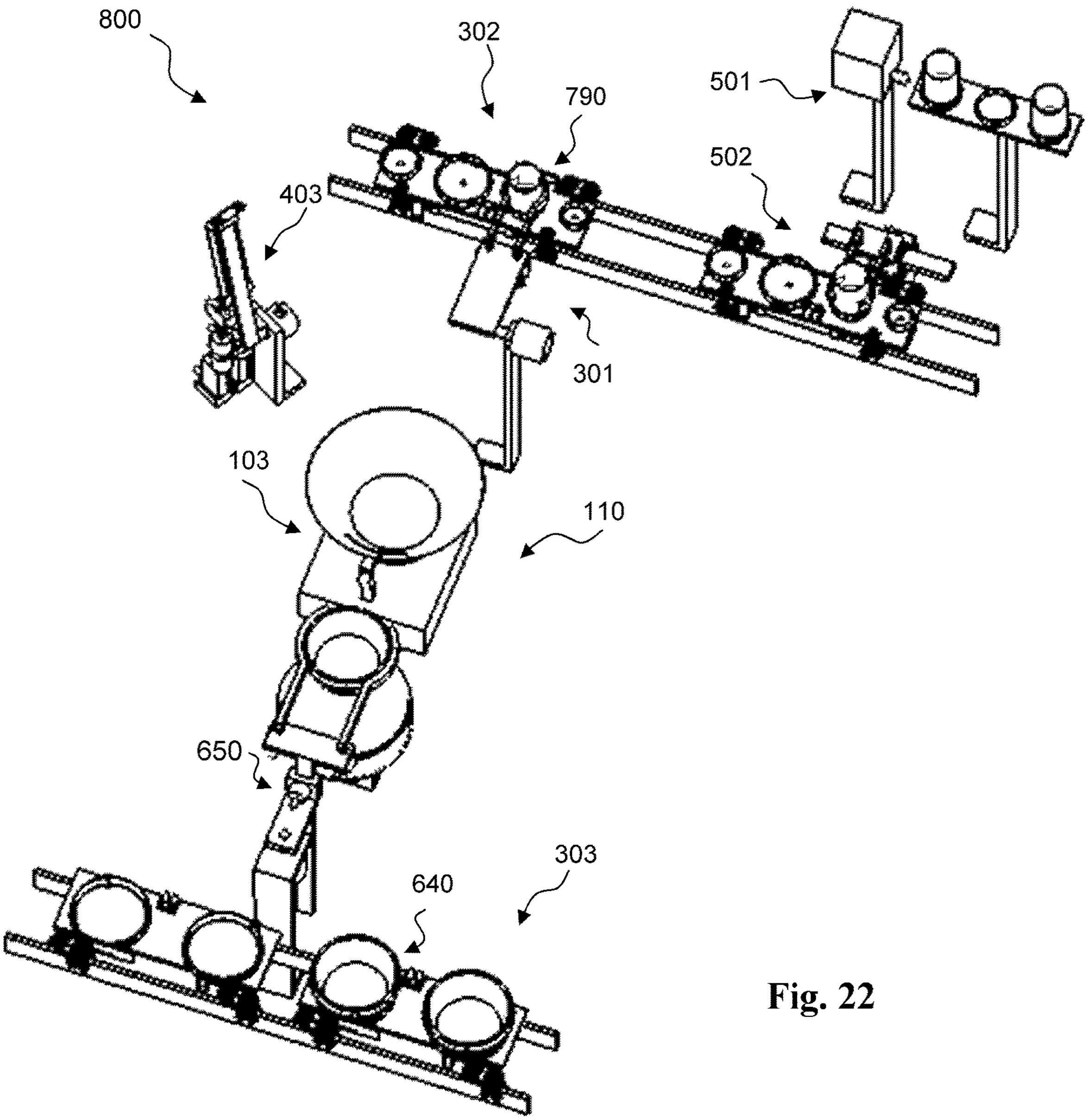
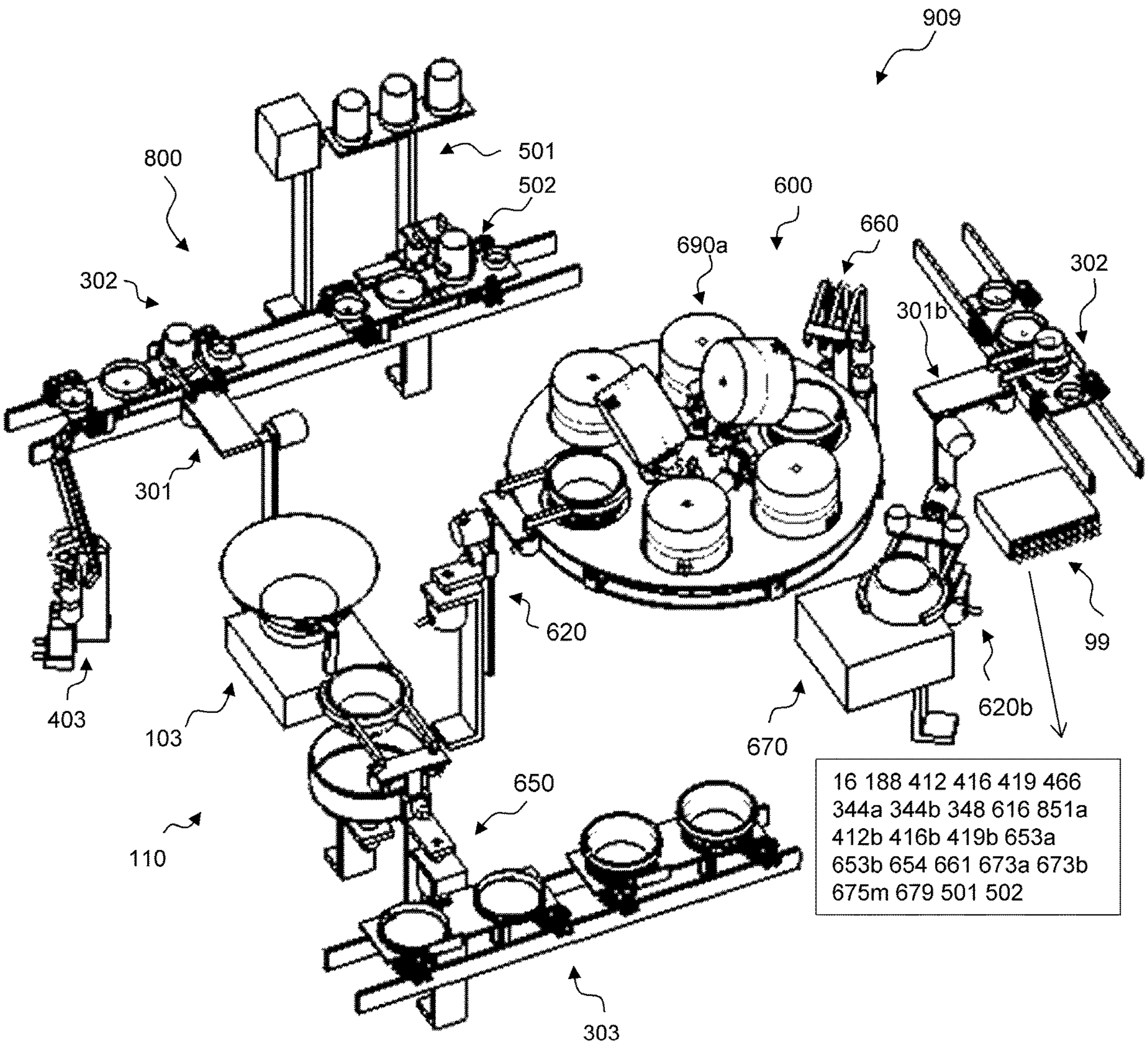


Fig. 22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800
501
502
600
660
690a
302
301b
302
301
403
103
620
99
620b
670
650
110
303
16 188 412 416 419 466
344a 344b 348 616 851a
412b 416b 419b 653a
653b 654 661 673a 673b
675m 679 501 502

Fig. 23

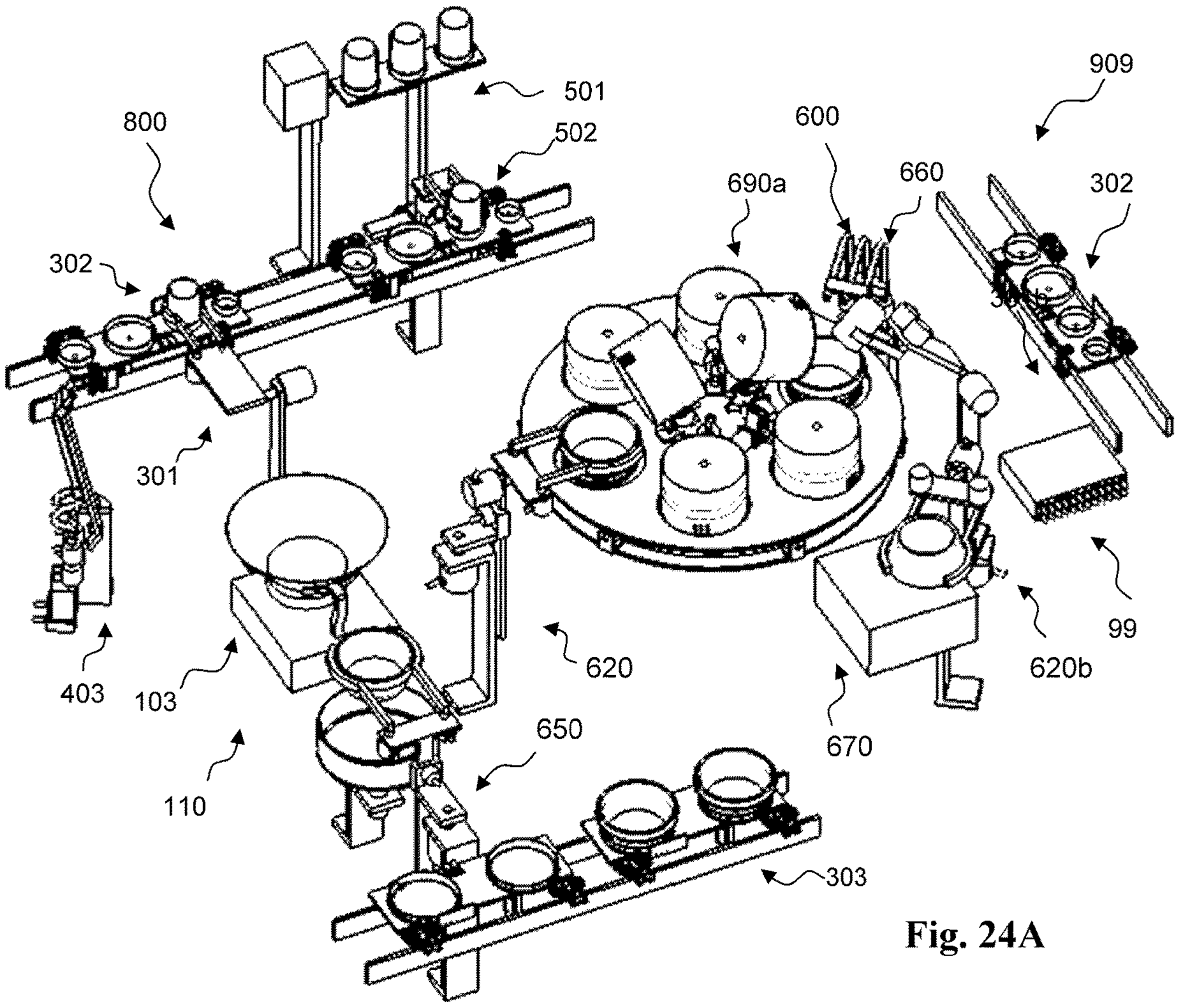

Fig. 24A

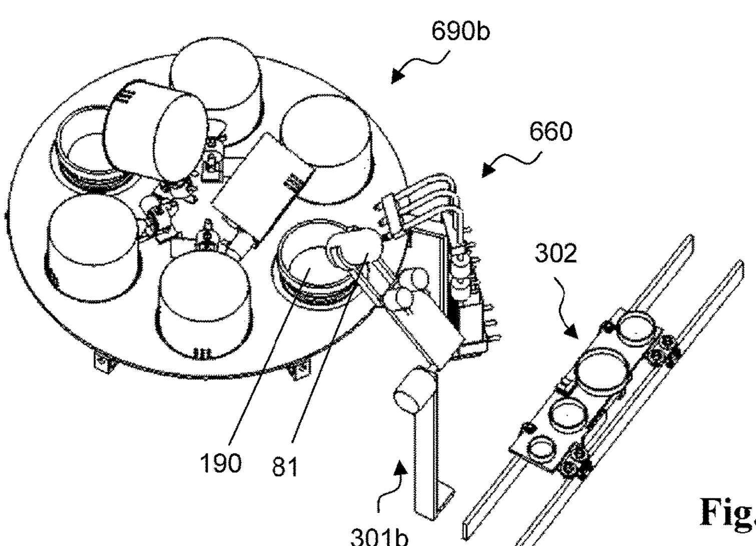

Fig. 24B

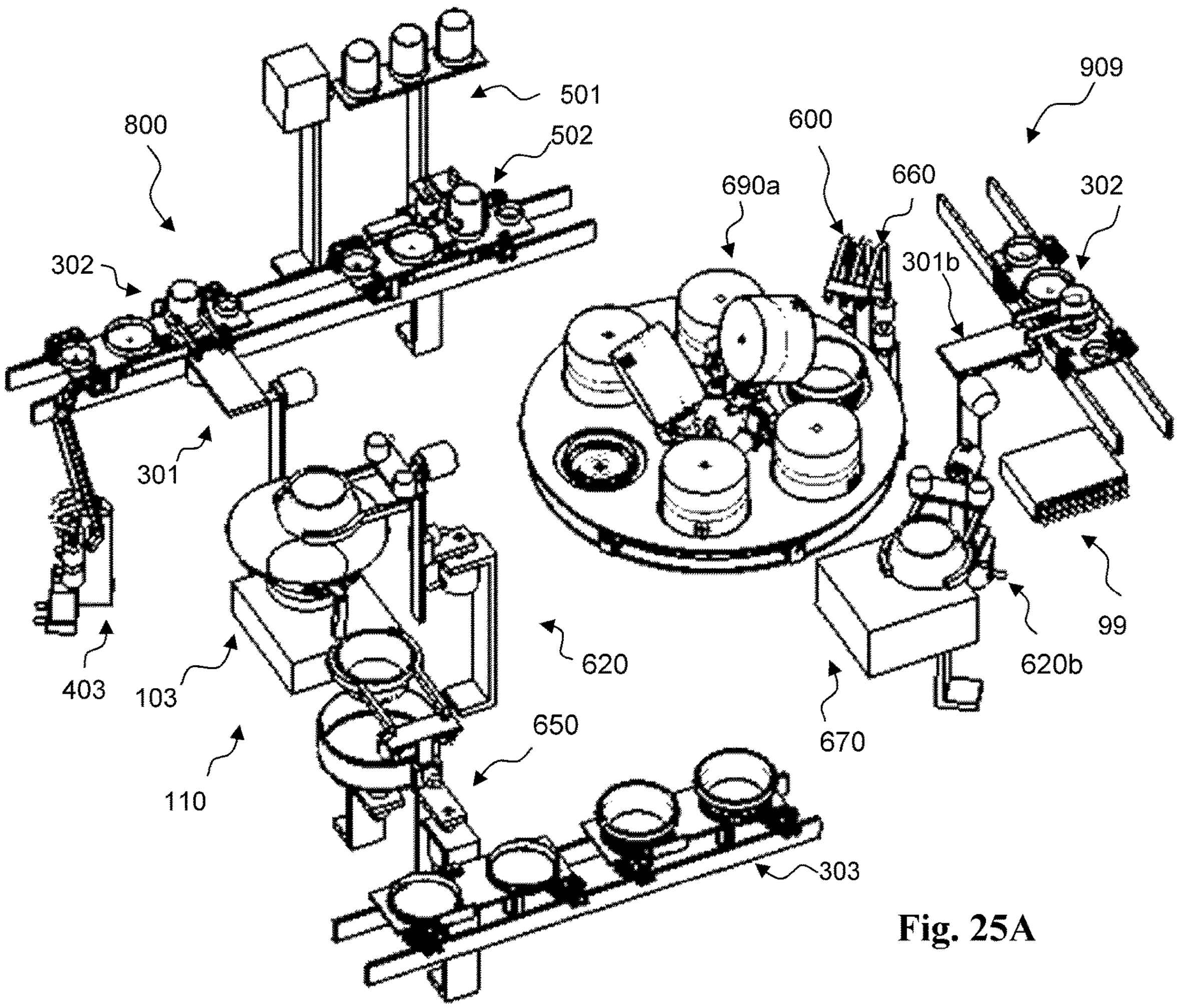

Fig. 25A

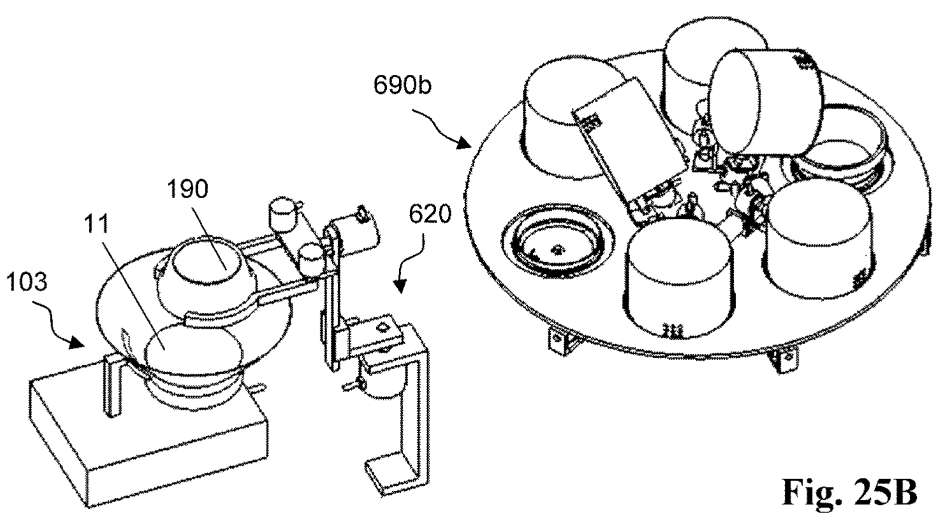

Fig. 25B

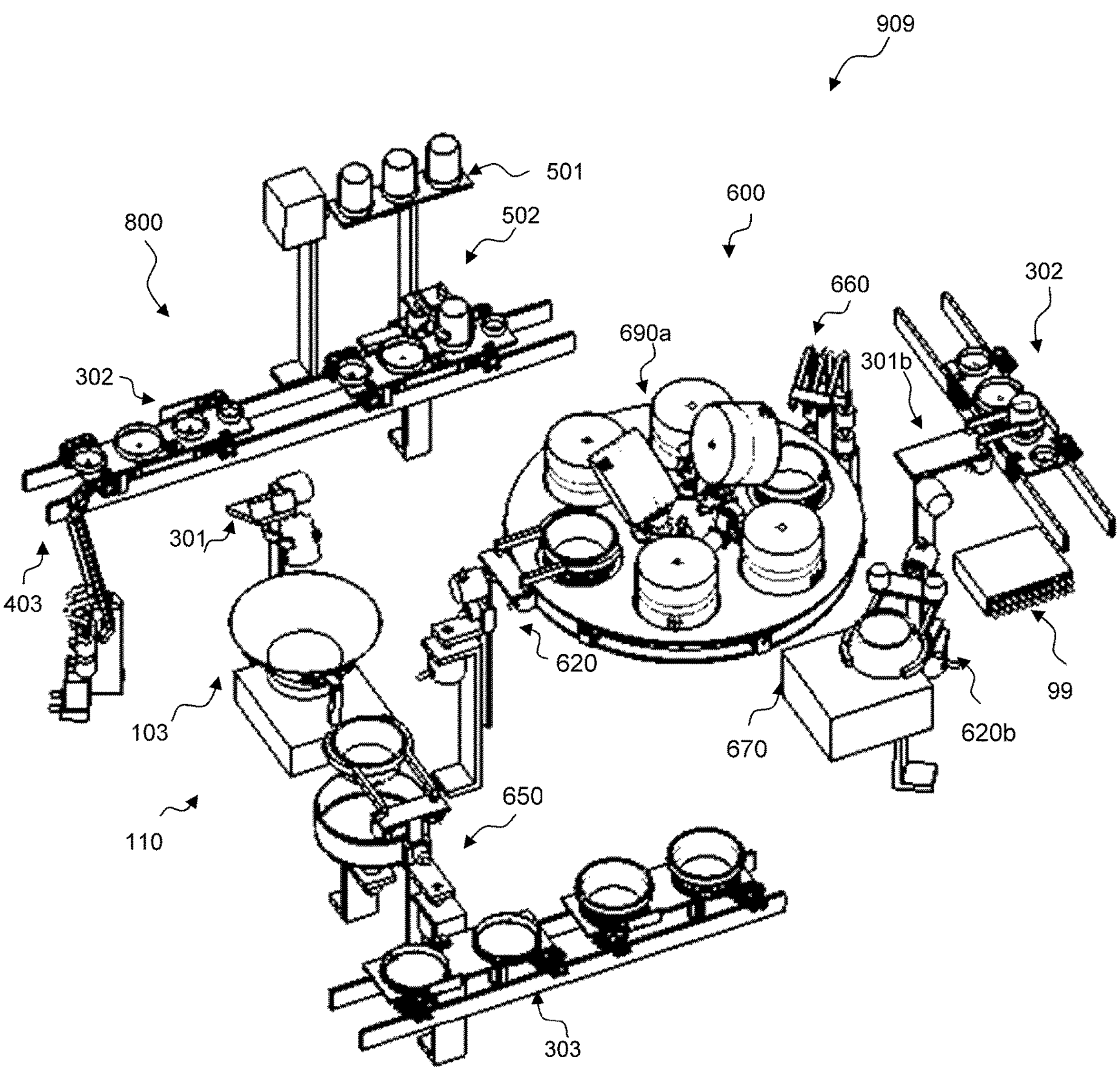
909
501
600
502
800
660
302
690a
301b
302
301
403
620
99
103
670
620b
650
110
303

Fig. 26

The computer system 99 is configured to store a program configured to send or receive signals to and from the motorized mechanisms, inductive stoves, motors, and pumps of the cooking system 909. 961

↓

The computer system 99 comprises a database. 962

↓

The computer system 99 stores the IDs of the cooking apparatuses, receiving apparatus, cookware cleaning mechanism, liquid dispensing mechanism, ingredient dispensing mechanism, transfer mechanisms, transport system. 963

↓

The computer system 99 stores the information of the structure of each mini-vehicle of the transfer system 302, including the ingredient container types that can be placed on the holders of the vehicle 790. 964

↓

The computer system 99 stores a program for controlling the transport systems 302 and 303, said program configured to control a mini-vehicle 790 of the transport system 302 to move to a position near a cooking station of the cooking system 909, so that an ingredient container on a given holder of the mini-vehicle 790 is at a dispensing position relative to the cooking apparatus at a pre-scheduled time. Said program configured to control a mini-vehicle 640 of the transport system 303 to move to a position near a receiving apparatus, so that the mini-vehicle 640 is at a food container transferring position relative to the support component of the receiving apparatus. 965

↓

The computer system 99 stores a list of food items which can be cooked by the cooking system 909. 966

↓

For each food item of the list of Step 966, the computer system 99 stores a cooking program configured to control the motorized mechanisms, inductive stoves, pumps and devices in the cooking system 909. 967

↓

For each food item of the list of Step 966, the computer system 99 stores types and quantities of food ingredients, relative timing of their dispensing and the respective ID of the destination cooking apparatus for each ingredient. 968

Fig. 27

The computer system 99 takes an order of a food item. 971

↓

The computer system 99 finds the information on the types and quantities of the ingredients needed for cooking the ordered food item, in accordance with the information stored by the computer system in Step 968. 972

↓

The computer system 99 locates the ingredient containers that contain the ingredients. 973

↓

The computer system 99 schedules cooking of the ordered food item at the cooking system 909, including the timing for running the program of Step 967 corresponding to the ordered food item, timing of dispensing of the food ingredients from each ingredient container into a respective cooking container or cookware of the cooking system 909. 974

↓

The computer system 99 controls the transport system 302 so that each ingredient container of Step 972 may be moved and stopped per the schedule of Step 974. 975

↓

The computer system 99 runs the program of Step 967 corresponding to the ordered food item, according to the schedule of Step 974, as to send or receive signals to or from the motorized mechanisms, inductive stoves, temperature sensors and pumps of the cooking system 909. The cooking of the food item is complete after all these steps. 976

Fig. 28

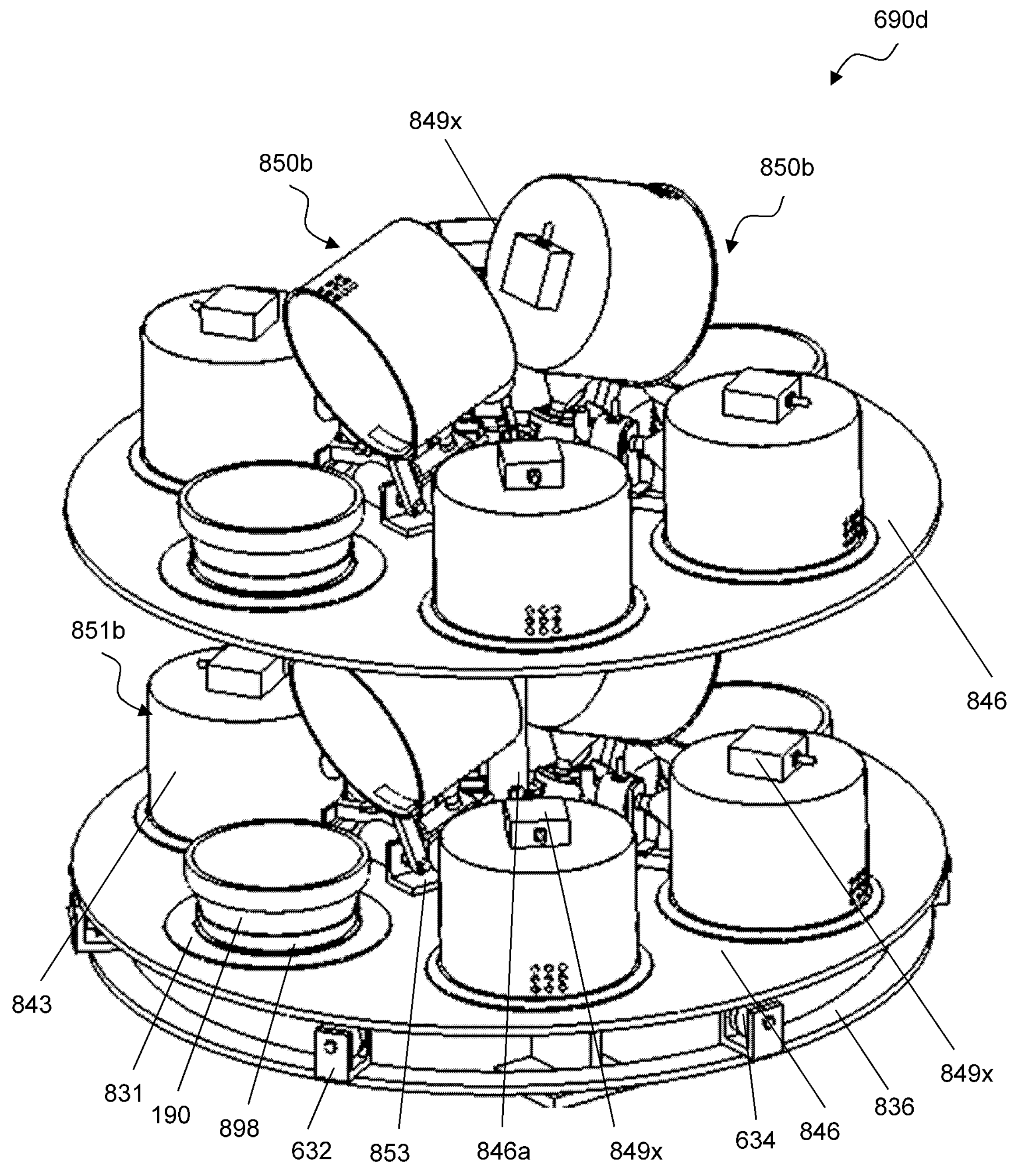
690d
849x
850b
850b
851b
846
843
831
190
898
632
853
846a
849x
634
846
836
849x

Fig. 29A

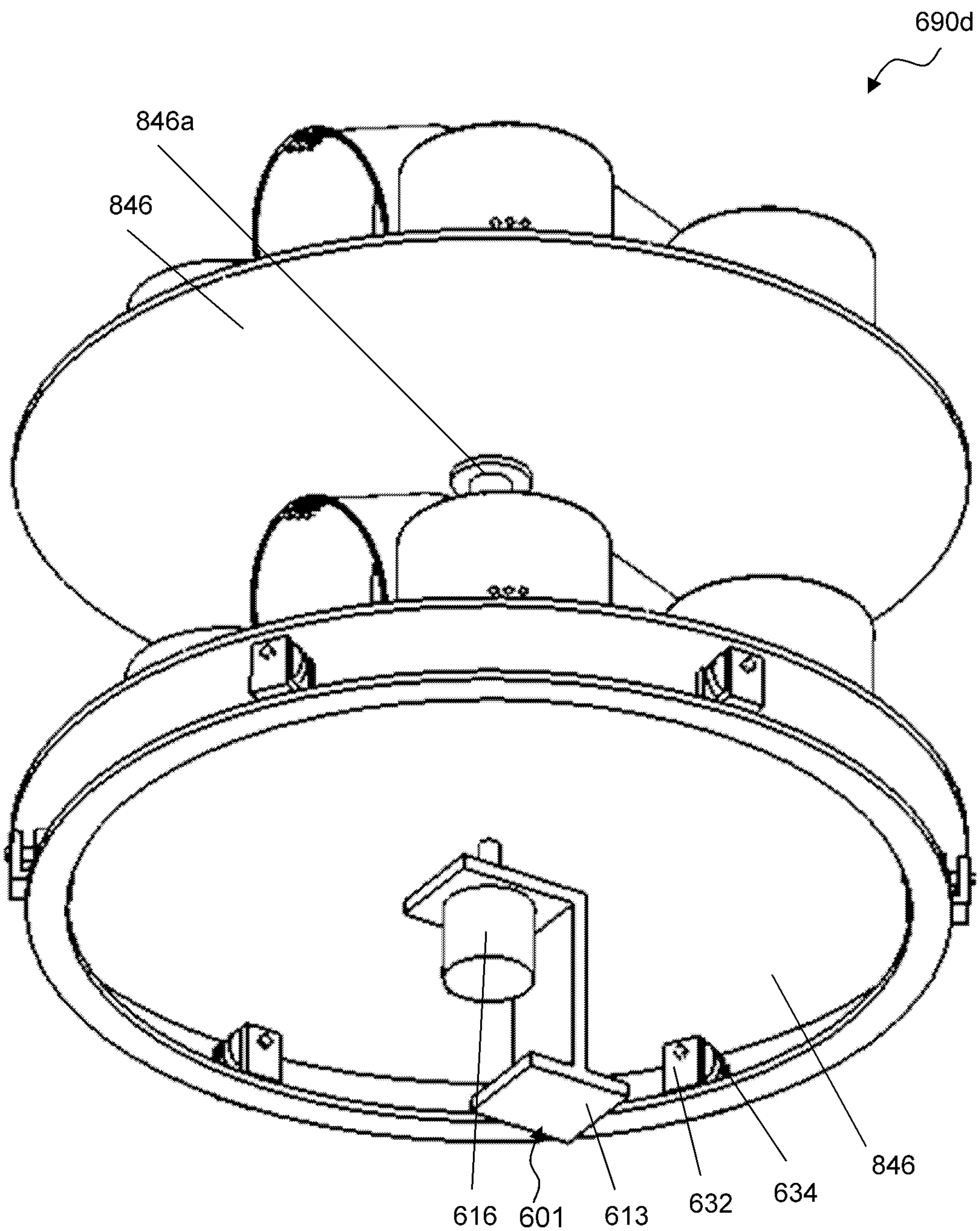

Fig. 29B

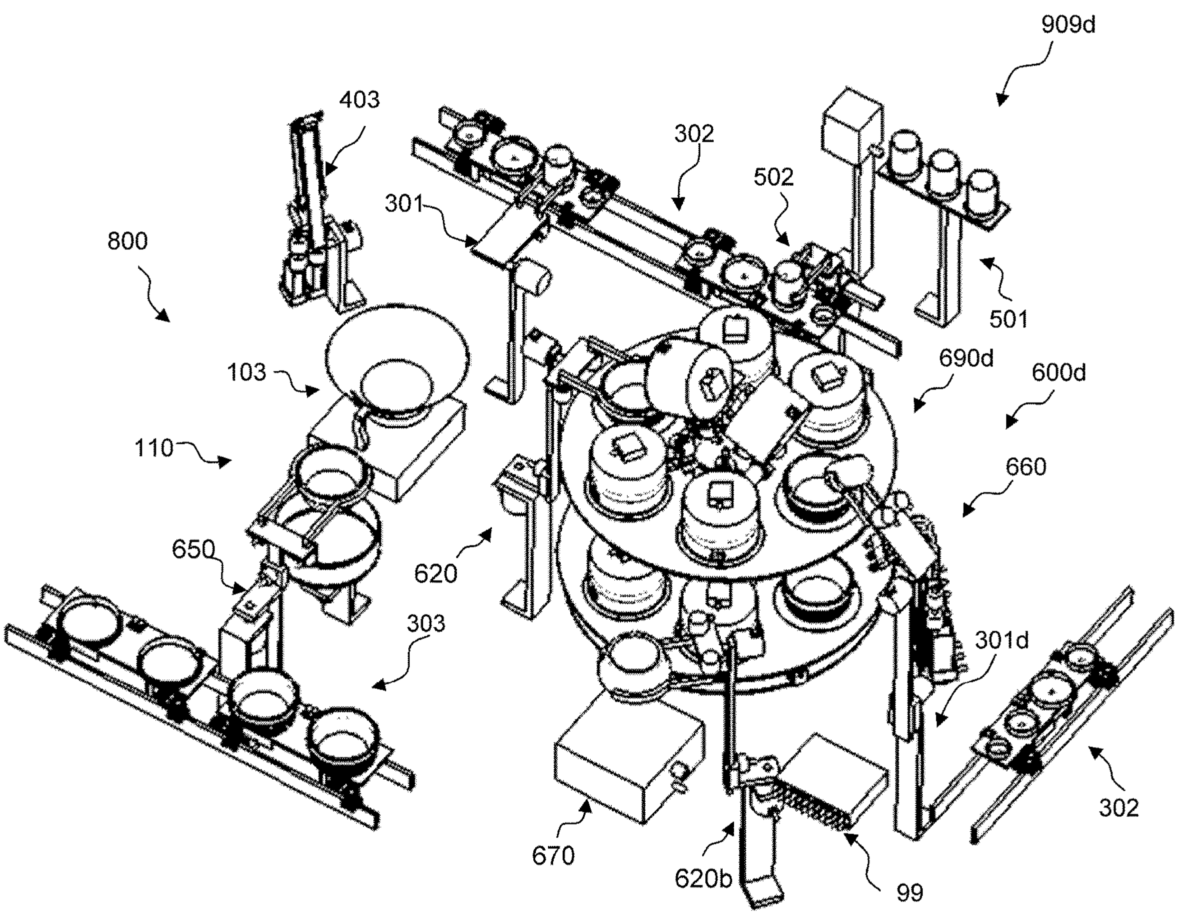
909d
403
302
502
301
800
501
690d
103
600d
110
660
650
620
303
301d
302
670
620b
99

Fig. 30

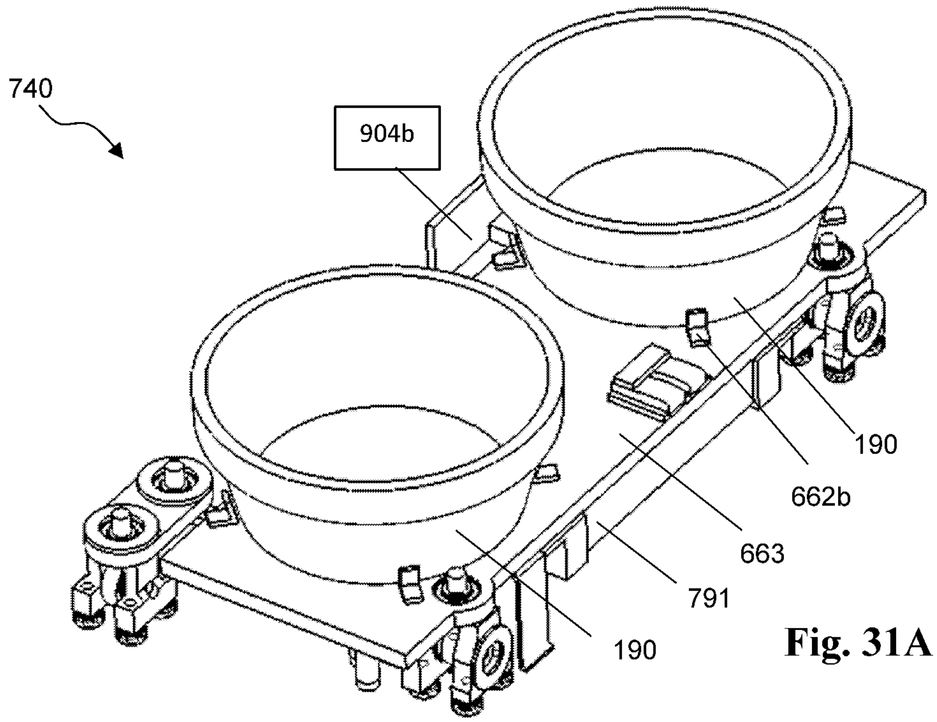

Fig. 31A

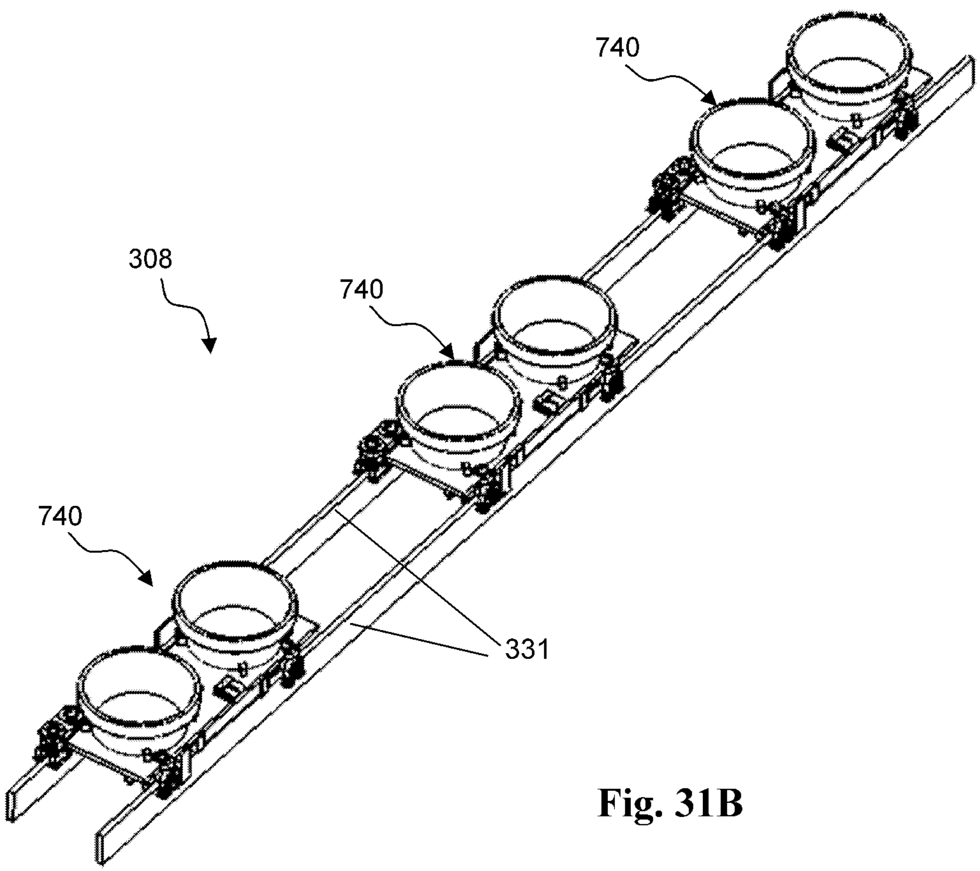

Fig. 31B

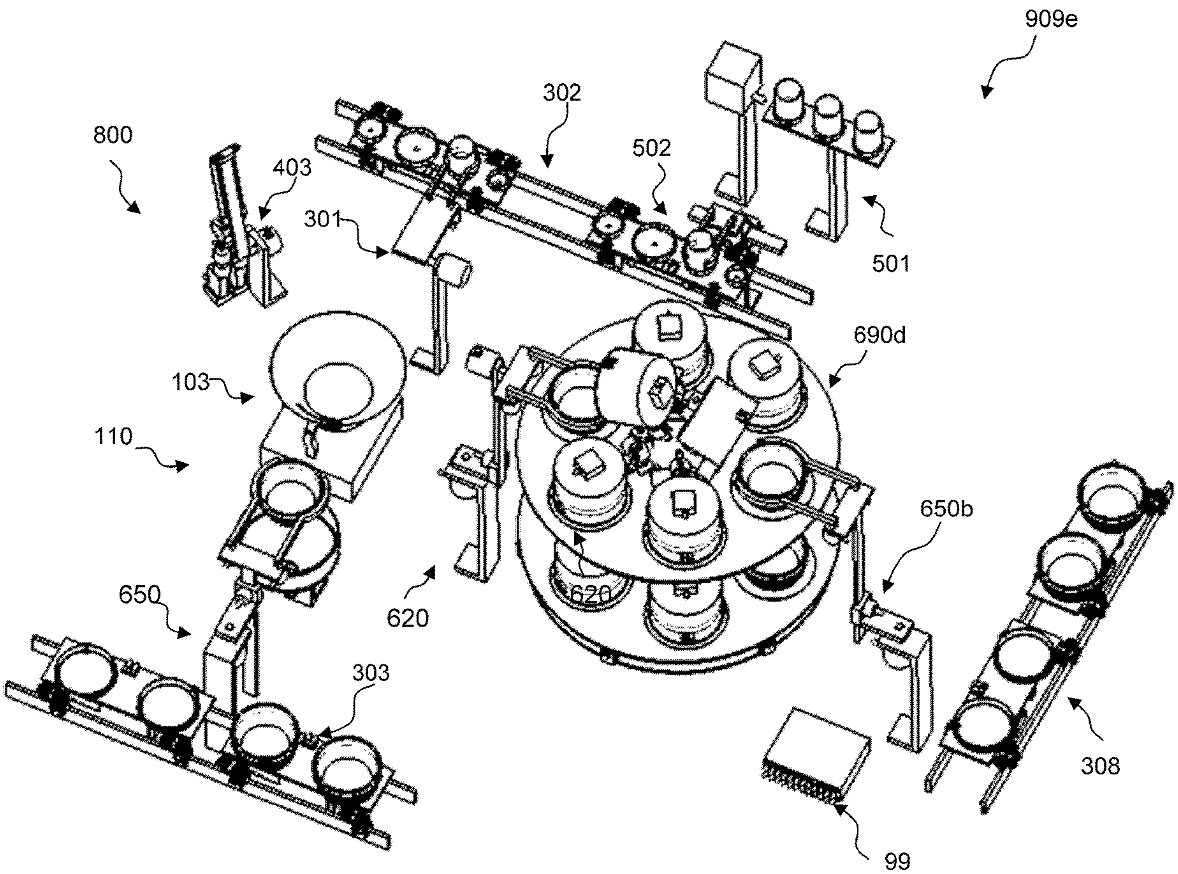
909e
800
302
403
502
301
501
690d
103
110
650b
650
620
620
303
308
99

Fig. 32

MICROWAVE COOKING SYSTEMS

This application claims the benefit of U.S. Provisional Application Ser. No. 63/057,519 filed Jul. 28, 2020, the disclosures of which are hereby incorporated herein by reference in their entireties.

CROSS-REFERENCE TO RELATED APPLICATIONS

US Provisional Patent Application

Ser. No. 63/057,519, filed Jul. 28, 2020, Inventor: Zhengxu He

US Patent Application

Ser. No. 15/706,136, filed Sep. 15, 2017, Inventor: Zhengxu He
Ser. No. 15/801,923, filed Nov. 2, 2017, Inventor: Zhengxu He
Ser. No. 15/798,357, filed Oct. 30, 2017, Inventor: Zhengxu He
Ser. No. 16/155,895, filed Oct. 10, 2018, Inventor: Zhengxu He
Ser. No. 16/510,982, filed Jul. 15, 2019, Inventor: Zhengxu He
Ser. No. 16/517,705, filed Jul. 22, 2019, Inventor: Zhengxu He
Ser. No. 16/997,196, filed Aug. 19, 2020, Inventor: Zhengxu He
Ser. No. 16/997,933, filed Aug. 20, 2020, Inventor: Zhengxu He
Ser. No. 17/069,707, filed Oct. 13, 2020, Inventor: Zhengxu He

US Patent

U.S. Pat. No. 10,455,987, issued Oct. 29, 2019, Inventor: Zhengxu He.

BACKGROUND OF THE INVENTION

The present application relates to a cooking system for producing a cooked food from food or food ingredient(s). In cooking of a food item, one or more of the following steps are required: (1) a first food or food ingredient is dispensed in a cookware such as wok; (2) a second food or food ingredient is heated in a cooking container and then dispensed in the cookware; (3) all foods or food ingredients are then stirred and/or mixed and/or heated in the cookware, to produce a cooked food; and (4) the cooked food is then dispensed into a food container, such as a dish or a bowl.

A cost-effective cooking apparatus or cooking system that does some or all of the above steps are important, as it can save labor and cost.

Furthermore, cost-effective transportation of the first and second food or food ingredient(s) to the cookware or cooking containers, respectively, is also important for the same reason.

The automation of such cooking system depends on new computer algorithms.

BRIEF SUMMARY OF THE INVENTION

A cooking system in our application comprises some cooking apparatuses and other mechanisms and/or apparatuses. The present patent application discloses a cooking system comprising a second cooking apparatus and a first cooking apparatus; wherein the first cooking apparatus comprises a plurality of microwave ovens and an intermittent motion mechanism which moves the microwave ovens.

The second cooking apparatus comprises one or more of the following parts: (1) a cookware configured to contain or otherwise hold a food or a food ingredient for the purpose of cooking a food; (2) a motion apparatus comprising a stirring motion mechanism configured to move the cookware to stir or mix the food or food ingredient contained in the cookware and a motion mechanism configured to directly or indirectly move the cookware, to dispense a cooked food into a food container; (3) a transfer apparatus configured to grip and hold a food container to allow the motion apparatus to dispense a cooked food from the cookware into a food container; and (4) a plurality of cooking containers configured to contain or otherwise hold a food or a food ingredient, a food dispensing apparatus configured to move a cooking container, said food dispensing apparatus configured to dispense a semi-cooked food held in the cooking container to the cookware. The first cooking apparatus comprises one or more of the following parts: (5) a plurality of cooking containers each configured to contain or otherwise hold a food or a food ingredient for the purpose of heating the latter; (6) a connected group of container holders each configured to position or otherwise hold a said cooking container; (7) a motion mechanism configured to produce a cyclic motion in the group of container holders; (8) above and corresponding to each of the container holders, an enclosure mechanism comprising an enclosure device configured to help enclose a cooking chamber; (9) heaters; and (10) a dispensing apparatus configured to grip and hold a said cooking container and to move it to dispense the food or food ingredient from the cooking container to the cookware of the second cooking apparatus.

The cookware may be a wok, a pan, or any container configured to contain or otherwise hold a food or a food ingredient during cooking. A cooking container may be a wok, a pan, a bowl, a basket, etc.

Implementations of our cooking system may include one or more of the following. The stirring motion mechanism may comprise a support component and a mechanism configured to produce a motion of the cookware relative to the support component, to stir or mix the food or food ingredient in the cookware. The stirring motion mechanism may comprise: a first shaft; a second shaft; a third shaft; a fourth shaft; and a fifth shaft, wherein the axes of the shafts may be configured to be parallel to each other.

Implementations of our cooking system may include one or more of the following. A motion mechanism is configured to produce an axial rotation of the support component of the stirring motion mechanism to dispense a cooked food from the cookware; wherein the axis of the axial rotation is configured to be horizontal.

Our cooking system may further comprise a dispensing apparatus which dispenses a food or a food ingredient into the cookware and/or a cooking container. The cooking system may also comprise a transfer apparatus which may move a food container containing a cooked food to an area accessible by humans.

Implementations of our cooking system may include one or more of the following. A dispensing apparatus may comprise: (1) a gripping mechanism comprising a first support component, grippers, and a motion mechanism configured to produce a controlled rotation of the gripper relative to the first support component to grab or release a container; (2) a second support component; and (3) a motion mechanism configured to produce a rotation of the first support component relative to the second support component.

A computer is used to control the above-described mechanisms and apparatuses.

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

FIG. **1** shows an aerial view of a computer system.

FIG. **2**A shows an aerial view of a motion mechanism. FIG. **2**B shows an aerial view of a linear motion mechanism FIG. **2**C shows an aerial view of a rotational motion mechanism. FIG. **2**D shows an aerial view of a rotational motion mechanism. FIG. **2**E shows an aerial view of a combination motion mechanism. FIG. **2**F shows an aerial view of a combination motion mechanism. FIG. **2**G shows an aerial view of a combination motion mechanism. FIG. **2**H shows an aerial view of a combination motion mechanism. FIG. **2**I shows an aerial view of a robot arm.

FIG. **3**A shows an aerial view of parts of a cooking apparatus.

FIG. **3**B shows an aerial view of a robot arm. FIG. **3**C shows a cooking apparatus comprising and the robot arm of FIG. **3**B.

FIG. **4**A shows an aerial view of a gripping mechanism. FIG. **4**B shows an aerial view of a gripping mechanism. FIG. **4**C shows an aerial view of a gripping mechanism. FIG. **4**D shows an aerial view of a gripping mechanism. FIG. **4**E shows an aerial view of a gripping mechanism FIG. **4**F shows an aerial view of a robotic apparatus.

FIG. **5** shows an aerial view of a transfer apparatus.

FIG. **6**A shows an aerial view of an ingredient dispensing apparatus. FIGS. **6**B-**6**C show aerial views of a cooking apparatus comprising the cooking apparatus of FIG. **3**A, the transfer apparatus of FIG. **5** and the ingredient dispensing apparatus of FIG. **6**A. FIG. **6**D shows an aerial view of another ingredient dispensing apparatus.

FIG. **7**A shows an aerial view of parts of a liquid dispensing mechanism. FIG. **7**B shows an aerial view of the liquid dispensing apparatus.

FIGS. **8**A-**8**B show aerial views of the relative positions of the cooking apparatus of FIG. **3**A and the liquid dispensing apparatus.

FIG. **9**A shows an aerial view of a transport system which includes a vehicle on tracks. FIG. **9**B shows an aerial view of some tracks, a vehicle on the tracks and the ingredient dispensing apparatus of FIG. **6**A.

FIG. **10** shows an aerial view of the relative positions of the cooking apparatus of FIG. **3**A, the ingredient dispensing apparatus of FIG. **6**A and the transport system of FIG. **9**A.

FIG. **11** shows an aerial view of a storage which can store ingredient containers.

FIG. **12** shows an aerial view of a transfer apparatus.

FIGS. **13**A-**13**D show aerial views of the relative positions of the storage apparatus of FIG. **11**, the transfer apparatus of FIG. **12**, and the transport system of FIG. **9**A.

FIG. **14** shows an aerial view of a food dispensing apparatus.

FIG. **15** shows an aerial view of a liquid dispensing apparatus.

FIG. **16**A shows an aerial view of a vehicle. FIG. **16**B shows an aerial view of a transport system comprising the vehicle of FIG. **16**A.

FIG. **17**A shows an aerial view of a rotational motion mechanism FIGS. **17**B-**17**C show views of an enclosure mechanism. FIGS. **17**D-**17**E show aerial views of a cooking apparatus comprising a plurality of enclosure mechanisms of FIGS. **17**B-**17**C. FIG. **17**F shows a cutaway view of parts of the cooking apparatus. FIG. **17**G shows an aerial view of a cooking apparatus. FIG. **17**H a cutaway view of parts of the cooking apparatus.

FIG. **18**A shows an aerial view of an enclosure mechanism FIG. **18**B shows a cutaway aerial view of the enclosure mechanism FIGS. **18**C-**18**D show aerial views of a cooking apparatus comprising a plurality of enclosure mechanisms of FIG. **18**A.

FIG. **19**A shows a cutaway view of an enclosure mechanism. FIGS. **19**B-**19**C show aerial views of a cooking apparatus comprising a plurality of enclosure mechanisms of FIG. **19**A.

FIG. **20**A shows an aerial view of an enclosure mechanism FIG. **20**B shows a cutaway aerial view of the enclosure mechanism. FIGS. **20**C-**20**D show aerial views of a cooking apparatus comprising a plurality of enclosure mechanisms of FIG. **20**A.

FIG. **21**A shows an aerial view of an enclosure mechanism FIG. **21**B shows a cutaway view of parts of the enclosure mechanism FIG. **21**C shows an aerial view of a cooking apparatus comprising a plurality of enclosure mechanisms of FIG. **21**A. FIG. **21**D shows a cutaway view of parts of the cooking apparatus. FIG. **21**E shows an aerial view of parts of the cooking apparatus.

FIG. **22** shows an aerial view of a cooking sub-system.

FIG. **23** shows an aerial view of a cooking system comprising the cooking sub-system of FIG. **22**. FIG. **24**A shows an aerial view of the cooking system showing the dispensing of a food or a food ingredient from an ingredient container into a food container. FIG. **24**B shows a close-up view of the dispensing. FIG. **25**A shows an aerial view of the cooking system showing the dispensing of a semi-cooked food from a food container into a cookware. FIG. **25**B shows a close-up view of the dispensing.

FIG. **26** shows an aerial view of the cooking system showing the dispensing of a food or a food ingredient from an ingredient container into the cookware.

FIG. **27** is a flow chart showing the procedures performed by the computer system of the cooking system of FIG. **20**B prior to cooking of a food.

FIG. **28** is a flow chart showing the procedures performed by the computer system of the cooking system of FIG. **20**B during the cooking of a food.

FIGS. **29**A-**29**B show aerial views of a cooking apparatus.

FIG. **30** shows an aerial view of a cooking system comprising the cooking apparatus of FIGS. **29**A-**29**B.

FIG. **31**A shows an aerial view of a vehicle. FIG. **31**B shows an aerial view of a transport system comprising the vehicle of the FIG. **31**A.

FIG. **32** shows an aerial view of a cooking system comprising the transport system of FIG. **31**B.

DETAILED DESCRIPTION OF THE INVENTION

For the present patent application, a food ingredient refers to any of the foods or substances that are combined to make a particular food. A food ingredient can be raw or precooked. A food ingredient can be solid, powder, liquid, or a mixture, etc. A food ingredient can be raw meat, sausage, fresh vegetable, dry vegetable, cooking oil, vinegar, soy sauce, water, or salt, etc.

For the present patent application, a computer system is meant to be any system or apparatus that includes one or more computers. A computer system may or may not include a database. A computer system may or may not include a network. A computer system may or may not include a memory shared by several computers. A computer system may include software. A single computer with software can be considered to be a computer system.

For the present patent application, a shaft always comprises an axis. A shaft can have different shapes at different sections. A sectional shape of a shaft can be round or rectangular, or of other shapes. For the present patent application, a rotational movement refers to a rotational movement around an axis. A rotational mechanism refers to any mechanism comprising two mating parts which are constrained to rotate relative to each other. An example of a rotational mechanism comprises a shaft and a bearing housing as mating parts, wherein the shaft and bearing housing are connected by bearings and accessories.

In some applications or embodiments, a motor comprises a base component (e.g., a frame) which is a stationary member of the motor, and a shaft which is a moving member of the motor, wherein a (usually rotational) motion of the shaft relative to the base component can be produced. A motor may be connected to a computer via wires, and/or through a driver, and/or a controller and/or a relay and/or a wireless communication device. The base component of a motor may be referred to as the support component of the motor.

Similarly, an encoder may comprise a base component, and a shaft which is rotatable relative to the base component, where the encoder can detect the degree of rotation of the shaft relative to the base component, and then inform a computer of the degree by sending signals to the computer.

Various parts of our cooking apparatuses and cooking systems are described below.

Referring to FIG. **1**, a computer system **99** comprises a computer **992** with I/O ports **991**. Via the I/O ports **991**, the computer **992** may be connected to other electric or electronic devices including but not limited to: motors (including motors with controllers); actuators; inductive stoves; sensors; etc., so that the computer may communicate with the devices by known techniques. The communications can optionally be one way or two-way (to and from). For example, the signals of the electrical or electronic devices may be sent to the computer **992**; the computer **992** may control the operations of the electrical or electronic devices by sending signals to the electrical or electronic devices. The connection of the computer **992** to the electric or electronic devices may comprise wires, wireless communication devices, controllers, drivers, and/or circuit boards. The computer system **99** comprises a memory. The computer system **99** may store data in the computer system's memory. The computer system **99** may control motors; actuators; stoves or heaters; and other devices by known techniques.

It should be noted that the computer system **99** may further comprise additional computers, a computer network, a database, computer programs, wireless communication ports, and/or other electric and electronic components.

A connection of the computer system **99** to an electric or electronic device may comprise a (wired or wireless) connection of a computer of the computer system to the device. Thus, a device is connected to the computer system **99** if the device is connected to a computer of the computer system.

Referring to FIG. **2**A, a motion mechanism **201** comprises a stationary member **201***a* and a moving member **201***b*, which is connected (but not rigidly connected) to the stationary member. In many applications the movement of the moving member **201***b* is constrained relative to the stationary member **201***a*. The motion mechanism **201** comprises a driving mechanism (not shown in figure) configured to produce a motion of the moving member **201***b* relative to the stationary member **201***a*. The motion mechanism **201** may be connected to the computer system **99** of FIG. **1** via wires or by wireless means and the computer system **99** may be configured to control the timing and speed of the motion mechanism **201**.

The motion mechanism **201** is a generic motion mechanism. Implicitly, the motion mechanism **201** includes a connection configured to connect the moving member to the stationary member, wherein the connection may often comprise bearings, sliders, kinematic pairs, and/or transmission mechanisms. The driving mechanism may be connected to the computer system **99** (via wires or by wireless means). The driving mechanism may be powered by electricity or other energy sources. A typical example of a driving mechanism is a motor.

Referring to FIG. **2**B, a linear motion mechanism **202** comprises a stationary member **202***a* and a moving member **202***b*, wherein the moving member **202***b* is constrained to move linearly relative to the stationary member **202***a*. The linear motion mechanism **202** comprises a driving mechanism (not shown in figure) configured to produce a linear motion of the moving member **202***b* relative to the stationary member **202***a*. The linear motion mechanism **202** may be connected to the computer system **99** of FIG. **1** via wires or by wireless means, and the computer system **99** may be configured to control the timing and speed of the linear motion mechanism **202**.

The linear motion mechanism **202** is a generic one. Examples of linear motion mechanisms include but are not limited to: a linear actuator; and a mechanism comprising linear rail, a slider configured to slide linearly on the linear rail, and a driving mechanism configured to drive the linear motion of the slider; etc.

It should be noted that the linear motion mechanism **202** may comprise an electric (or pneumatic, hydraulic) putter, or other types of putter. The linear motion mechanism **202** may include a motor which produces a rotational motion and a transmission mechanism configured to convert a rotation into a linear motion; wherein the transmission mechanism may optionally comprise a gear and rack, a screw rod and nut, or a sprocket and chain, etc.

A linear motion mechanism (such as the mechanism **202**) is called a vertical motion mechanism if the direction of the linear motion is vertical. A linear motion mechanism (such as the mechanism **202**) is called a horizontal motion mechanism if the direction of the linear motion is horizontal.

It should be noted that the rotation produced by a rotational motion mechanism may be a continuous rotation, an intermittent motion, or a rotation between two end-positions.

Referring to FIG. **2**C, a rotational motion mechanism **203** comprises a stationary member **203***a* and a moving member **203***b* which is constrained to rotate relative to the stationary member **203***a*. The rotational motion mechanism **203** comprises a driving mechanism (not shown in figure) configured to produce a rotation of the moving member **203***b* relative to the stationary member **203***a* around an axis, wherein the axis of the rotation is referred to as the axis of the rotational motion mechanism. The rotational motion mechanism **203** may be connected to the computer system **99** of FIG. **1** via wires or by wireless means, and the computer system **99** may be configured to control the timing and speed of the rotational motion mechanism **203**.

Referring to FIG. **2**D, a rotational motion mechanism **223** comprises: a bearing housing **223***a* as a stationary member;

a shaft **223***b* as a moving member; and a motor **227** as a driving member, i.e., a driving mechanism. The bearing housing **223***a* and the shaft **223***b* are connected by bearings **224** and accessories so that the shaft **223***b* is constrained to rotate relative to the bearing housing **223***a*. The motor **227** comprises a base component **227***a* and a shaft **227***b* so that the motor may produce a rotation of the shaft **227***b* relative to the base component **227***a*. The base component **227***a* of the motor is rigidly or fixedly connected to the bearing housing **223***a* via a connector **226**, and the shaft **227***b* of the motor is connected to the shaft **223***b* by a coupling **225**. It should be clear that the motor **227** may produce a rotation of the shaft **223***b* relative to the bearing housing **223***a*. The motor **227** is a driving mechanism of the rotational motion mechanism **223**.

It should be noted that the rotation produced by a rotational motion mechanism may be a continuous rotation, an intermittent motion, or a rotation between two end-positions.

Referring to FIG. **2**E, a motion mechanism **205** is a combination of two motion mechanisms **201** and **204**, which may also be referred to as motion sub-mechanisms; wherein the motion mechanism **201** is as described as in FIG. **2**A; wherein the motion mechanism **204** is a motion mechanism comprising a stationary member **204***a*, a moving member **204***b* which is connected to the stationary member **204***a*, and a driving mechanism (not shown in figure) configured to produce a motion of the moving member **204***b* relative to the stationary member **204***a*. The moving member **201***b* of the motion mechanism **201** is fixedly or rigidly connected to the stationary member **204***a* of the motion mechanism **204**, so that the motion mechanism **201** can produce a motion of the stationary member **204***a* relative to the stationary member **201***a* of the motion mechanism **201**. The combination motion mechanism **205** may be connected to the computer system **99** of FIG. **1** in the sense that the motion sub-mechanisms **201** and **204** are connected to the computer system **99** via wires or by wireless means, and the computer system **99** may be configured to control the motions produced by the motion sub-mechanisms of the combination motion mechanism **205**.

The motion mechanism **205** is referred to as a combination motion mechanism. It should be noted that the motion sub-mechanisms **201** and **204** may produce motions simultaneously. This applies to any combination motion mechanism in the following. Combination motion mechanisms are special cases of motion mechanisms.

Referring to FIG. **2**F, a combination motion mechanism **207** comprises rotational motion mechanisms **203** and **206**, referred to as motion sub-mechanisms; wherein the motion mechanism **203** is described in FIG. **2**C; wherein the motion mechanism **206** is a rotational motion mechanism comprising a stationary member **206***a*, a moving member **206***b* which is constrained to rotate relative to the stationary member **206***a*, and a driving mechanism (not shown in figure) configured to produce a rotational motion of the moving member **206***b* relative to the stationary member **206***a*. The moving member **203***b* of the motion mechanism **203** is fixedly or rigidly connected to the stationary member **206***a* of the rotational motion mechanism **206**, so that the rotational motion mechanism **203** can produce a rotation of the stationary member **206***a* relative to the stationary member **203***a* around the axis of the rotational motion mechanism **203**. The combination motion mechanisms **207** may be connected to the computer system **99** of FIG. **1** in the sense that the motion sub-mechanisms **203** and **206** are connected to the computer system **99** via wires or by wireless means, and the computer system **99** may be configured to control the motions produced by the motion sub-mechanisms of the combination motion mechanism **207**.

Referring to FIG. **2**G, a combination motion mechanism **209** comprises two linear motion mechanisms **202** and **208**, which may also be referred to as motion sub-mechanisms; wherein the motion mechanism **202** is described in FIG. **2**B; wherein the motion mechanism **208** is a linear motion mechanism comprising a stationary member **208***a*, a moving member **208***b* which is constrained to move linearly relative to the stationary member **208***a*, and a driving mechanism (not shown in figure) configured to produce a linear motion of the moving member **208***b* relative to the stationary member **208***a*. The moving member **208***b* of the linear motion mechanism **208** is rigidly or fixedly connected to the stationary member **202***a* of the linear motion mechanism **202**, so that the linear motion mechanism **208** can produce a linear motion of the stationary member **202***a* relative to the stationary member **208***a*. The combination motion mechanism **209** may be connected to the computer system **99** of FIG. **1** in the sense that the motion sub-mechanisms **202** and **208** are connected to the computer system **99** via wires or by wireless means, and the computer system **99** may be configured to control the motions produced by the motion sub-mechanisms of the combination motion mechanism **209**.

Referring to FIG. **2**H, a combination motion mechanism **210** comprises a rotational motion mechanism **203** and two linear motion mechanisms **202** and **208**; wherein motion mechanisms **203**, **202** and **208** are referred to as motion sub-mechanisms. The moving member **208***b* of the linear motion mechanism **208** is rigidly or fixedly connected to the stationary member **202***a* of the linear motion mechanism **202**, so that the linear motion mechanism **208** can produce a linear motion of the stationary member **202***a* relative to the stationary member **208***a* of the linear motion mechanism **208**. The moving member **203***b* is fixedly connected to the stationary member **208***a* of the linear motion mechanism **208**, so that the rotational motion mechanism **203** can produce a rotation of the stationary member **208***a* relative to the stationary member **203***a*. The combination motion mechanisms **210** may be connected to the computer system **99** of FIG. **1** in the sense that the motion sub-mechanisms **202**, **203** and **208** are connected to the computer system **99** via wires or by wireless means, and the computer system **99** may be configured to control the motions produced by the motion sub-mechanisms of the combination motion mechanism **210**.

Referring to FIG. **2**I, a robot arm **218** comprises a plurality of rotational motion mechanisms **211**, **213**, **215** and **217**; wherein the motion mechanisms **211**, **213**, **215** and **217** are referred to as motion sub-mechanisms. The rotational motion mechanism **211**, **213**, **215** or **217** comprises: a stationary member **211***a*, **213***a*, **215***a*, or respectively **217***a*; a moving member **211***b*, **213***b*, **215***b*, or respectively **217***b* which is constrained to rotate relative to the respective stationary member; and a driving member comprising a motor (not shown in figure) configured to drive a rotation of the respective moving member relative to the respective stationary member around an axis. The moving member **211***b* of the rotational motion mechanism **211** is rigidly connected to the stationary member **213***a* of the rotational motion mechanism **213** via a connector **212**; wherein the axis of the rotational motion mechanism **211** may optionally be perpendicular to the axis of the rotational motion mechanism **213**. Thus, the motion mechanism **211** can produce a rotation of the stationary member **213***a* relative to the stationary member **211***a*. The moving member **213***b* of the rotational motion mechanism **213** is rigidly connected to the stationary member **215***a* of the rotational motion mechanism **215** via a rigid connector **214**; wherein the axis of the rotational motion mechanism **213** may optionally be parallel to the axis of the rotational motion mechanism **215**. The rotational motion mechanism **213** can produce a rotation of the stationary member **215***a* relative to the stationary member **213***a*. The moving member **215***b* of the rotational motion mechanism **215** is rigidly connected to the stationary member **217***a* of the rotational motion mechanism **217** via a connector **216**; wherein the axis of the rotational motion mechanism **215** may optionally be perpendicular to the axis of the rotational motion mechanism **217**, and the rotational motion mechanism **215** can produce a rotation of the stationary member **217***a* relative to the stationary member **215***a*. The robot arm **218** may be connected to the computer system **99** of FIG. **1** in the sense that the motion sub-mechanisms **211**, **213**, **215** and **217** are connected to the computer system **99** via wires or by wireless means, and the computer system **99** may be configured to control the motions produced by the motion sub-mechanisms of the robot arm **218**.

The robot arm **218** is a combination motion mechanism which is a combination of the motion sub-mechanisms **211**, **213**, **215** and **217**. Any robot arm of prior art may be used as a motion mechanism for our applications. Any motion mechanism of prior art may be used for our applications.

It should be possible to construct a combination motion mechanism from a rather arbitrary sequence of motion mechanisms, referred to as motion sub-mechanisms.

More examples of motion mechanisms are described in U.S. patent application Ser. Nos. 17/070,059 and 17/072,011. The entire contents of these applications are incorporated herein by reference.

Referring to FIG. **3**A, a cooking apparatus **103** comprises: a cookware **11**; a heater (such as inductive stove, gas burner, electric burner, etc.) **16**; and a motion mechanism **104** comprising a stationary member **104***a* and a moving member **104***b*. The moving member **104***b* is rigidly, fixedly, or otherwise connected to the cookware **11** at least during time of operation. The heater **16** is configured to heat the cookware **11** and hence the food or food ingredient held in the cookware. The motion mechanism **104** may produce a motion of the cookware to stir or mix the food or food ingredient in the cookware, using known techniques. The motion mechanism **104** may also be able to produce a motion (e.g., a rotation around a horizontal axis) of the cookware **11** to dispense a cooked food from the cookware **11**, using known techniques. The motion mechanism **104** is driven by motors **104***m* and **104***n*, which are connected to the computer system **99** of FIG. **1** by wires or by wireless means. The stationary member **104***a* is referred to as the support component of the cooking apparatus **103**.

As an example, the motion mechanism **104** may comprise a robot arm, wherein a moving part of the robot arm is connected to the cookware. The connection to the cookware may be temporary or permanent, depending on the particular application.

It should be noted that the heater may optionally be fixedly connected to the cookware. See, e.g., U.S. patent application Ser. No. 15/801,923, the disclosures of which are hereby incorporated herein by reference in its entirety. In other applications, the heater may optionally be fixedly connected to the floor of the building or the ground.

The motion mechanism **104** of the cooking apparatus **103** may be substituted by the stirring motion mechanism, the unloading motion mechanism (or unloading apparatus in the terms of some patent applications), the dispensing apparatus, or the combination of the above, as disclosed in U.S. patent application Ser. Nos. 16/997,196, 15/706,136, 15/801,923, 16/155,895, and 17/069,707. The entire contents of the applications are incorporated herein by reference.

Referring to FIG. **3**B, a robot arm **140** comprises a plurality of rotational motion mechanisms **131**, **133**, **135** and **137**, referred to as motion sub-mechanisms. The rotational motion mechanism **131**, **133**, **135** or **137** comprises: a stationary member **131***a*, **133***a*, **135***a*, or respectively **137***a*; a moving member **131***b*, **133***b*, **135***b*, or respectively **137***b* which is constrained to rotate relative to the respective stationary member; and a driving member comprising a motor (not shown in figure) configured to drive a rotation of the respective moving member relative to the respective stationary member around a respective axis. The moving member **131***b* of the rotational motion mechanism **131** is rigidly connected to the stationary member **133***a* of the rotational motion mechanism **133** via a connector **132**; wherein the axis of the rotational motion mechanism **131** may optionally be perpendicular to the axis of the rotational motion mechanism **133**. Thus, the motion mechanism **131** can produce a rotation of the stationary member **133***a* relative to the stationary member **131***a*. The moving member **133***b* of the rotational motion mechanism **133** is rigidly connected to the stationary member **135***a* of the rotational motion mechanism **135** via a rigid connector **134**; wherein the axis of the rotational motion mechanism **133** may optionally be parallel to the axis of the rotational motion mechanism **135**. The rotational motion mechanism **133** can produce a rotation of the stationary member **135***a* relative to the stationary member **133***a*. The moving member **135***b* of the rotational motion mechanism **135** is rigidly connected to the stationary member **137***a* of the rotational motion mechanism **137** via a connector **136**; wherein the axis of the rotational motion mechanism **135** may optionally be perpendicular to the axis of the rotational motion mechanism **137**, and the rotational motion mechanism **135** can produce a rotation of the stationary member **137***a* relative to the stationary member **135***a*. The robot arm **140** may be connected to the computer system **99** of FIG. **1** in the sense that the motion sub-mechanisms **131**, **133**, **135** and **137** are connected to the computer system **99** via wires or by wireless means, and the computer system **99** may be configured to control the motions produced by the motion sub-mechanisms of the robot arm **140**.

Referring to FIG. **3**C, a cooking apparatus **103***b* comprises: a cookware **11** (as in FIG. **3**A) configured to contain or hold a food or a food ingredient; and a robot arm **140** (as in FIG. **3**B). The cookware **11** is fixedly or rigidly connected to the moving member **137***b* of the motion mechanism **137** of the robot arm **140** via a rigid component **138**. The robot arm **140** can produce a cyclic and oscillatory motion (of the rigid component **138** and) of the cookware **11** as to stir and distribute a food or a food ingredient evenly in the cookware **11**. The robot arm **140** can also produce a motion of (the rigid component **138** and) the cookware **11** wherein the cookware **11** may be turned by an angle in the motion so as to dispense a cooked food.

It should be noted that the robot arm **140** is capable of producing two types of motion of the cookware **11**: a stirring motion of cyclic or oscillatory type to mix and distribute a cooked food or a food ingredient in the cookware **11**; and an unloading motion to turn the cookware **11** in order to dispense a cooked food from the cookware **11**.

The cooking apparatus **103***b* may substitute the cooking apparatus **103** in any cooking system, kitchen system, or automated restaurant described in the present patent application. This applies to the cooking system, kitchen system, or automated restaurant described in the following.

It should be further noted that the robot arm **140** in the cooking apparatus **103***b* may be substituted by other type of robot arm.

Referring to FIG. **4**A, a gripping mechanism **221** comprises: grippers **261***a* and **261***b* which can optionally be rigid or elastic components and rotational motion mechanisms **263** and **264**. The rotational motion mechanism **263** comprises a stationary member **263***a* and a moving member **263***b*; and the rotational motion mechanism **263** is configured to produce a rotation of the moving member **263***b* relative to the stationary member **263***a*. The rotational motion mechanism **264** comprises a stationary member **264***a* and a moving member **264***b*; and the rotational motion mechanism **264** is configured to produce a rotation of the moving member **264***b* relative to the stationary member **264***a*. The stationary members **263***a* and **264***a* are fixedly connected to a support component **262**. The gripper **261***a* is rigidly or fixedly connected to the moving member **264***b*. The rotational motion mechanism **264** can produce a rotation of the gripper **261***a* around the axis of the rotational motion mechanism **264** relative to the stationary member **264***a*. Similarly, the gripper **261***b* is rigidly or fixedly connected to the moving member **263***b*. The rotational motion mechanism **263** can produce a rotation of the gripper **261***b* around the axis of the rotational motion mechanism **263** relative to the stationary member **263***a*. As the gripper **261***a* or **261***b* is rigidly connected to the moving member **264***b* or respectively **263***b*, the rotational motion mechanism **264** or **263** can produce a rotation of the gripper **261***a* or respectively **261***b*. The axis of rotation of the rotational motion mechanism **264** is parallel to the axis of rotation of the rotational motion mechanism **263**, and the rotational motion mechanisms **264** and **263** are configured to rotate the respective grippers **261***a* and **261***b* in opposite directions simultaneously. Thus, the grippers **261***a* and **261***b* can be rotated anti-synchronously around a pair of parallel axes. Each of the grippers **261***a* and **261***b* is rotated between a first end-position and a second end-position. When at the first end-positions, the grippers **261***a* and **261***b* may together grip a container or other object. When at the second end-positions, the grippers **261***a* and **261***b* can open up and release the container or object. The motion mechanism **263** or **264** is driven by a motor **263***m* or respectively **264***m*. The gripping mechanism **221** may be connected to the computer system **99** of FIG. **1** in the sense that the motors **263***m* and **264***m* are connected to the computer system **99** via wires or by wireless means, and the computer system **99** may be configured to control the motions of the grippers **261***a* and **261***b* produced by the motion mechanisms **263** and **264**. Thus, the gripping mechanism **221** may be controlled by the computer system **99** to grip or release a container or other object.

Referring to FIG. **4**B, a gripping mechanism **221***a* comprises: grippers **261***a* and **261***b* which are optionally rigid or elastic components; a rotational motion mechanism **267** comprising a stationary member **267***a* and a moving member (a shaft) **267***b*; a rotational mechanism **265** comprising a first mating part **265***a* and a second mating part (a shaft) **265***b* which is constrained to rotate relative to the first mating part **265***a*. The rotational motion mechanism **267** is configured to produce a rotational motion of the moving member **267***b* relative to the stationary member **267***b*. The stationary members **267***a* and the first mating part **265***a* are rigidly or fixedly connected to a support component **262**. The gripper **261***a* is rigidly or fixedly connected to the moving member **267***b*. The gripper **261***b* is rigidly or fixedly connected to the second mating part (a shaft) **265***b*. The axis of rotation of the rotational motion mechanism **267** and the axis of the rotational mechanism **265** are configured to be parallel to each other. A transmission mechanism **266** is configured to connect the rotational motion mechanism **267** and the rotational mechanism **265**, so that a rotation of the shaft **267***b* relative to the stationary member **267***a* is transmitted to an anti-synchronous rotation of the shaft **265***b*. Thus, the grippers **261***a* and **261***b* can be rotated anti-synchronously around a pair of parallel axes. Each of the grippers **261***a* and **261***b* is rotated between a first end-position and a second end-position. When at the first end-positions, the grippers **261***a* and **261***b* may together grip a container or other object. When at the second end-positions, the grippers **261***a* and **261***b* can open up and release the container or object. The motion mechanism **267** is driven by a motor **267***m*. The gripping mechanism **221***a* may be connected to the computer system **99** of FIG. **1** in the sense that the motor **267***m* is connected to the computer system **99** via wires or by wireless means, and the computer system **99** may be configured to control the motions of the grippers **261***a* and **261***b* produced by the motion mechanism **267**. Thus, the gripping mechanism **221***a* may be controlled by the computer system **99** to grip or release a container or other object.

For examples of gripping mechanisms that may substitute the gripping mechanism **221***a* described above, see FIGS. **39**A-**39**B ("gripping mechanism **701**"), or FIGS. **47**A-**47**C ("gripping mechanism **905**"), of U.S. patent application Ser. No. 16/517,705. The entire content of the US patent application is hereby incorporated herein by reference.

Referring to FIG. **4**C, a gripping mechanism **221***b* comprises: a support component **262** which is a rigid component; grippers **261***a* and **261***b*; a linear motion mechanism **260**; a rigid component **274**; shafts **273***a* and **273***b*; links **272***a* and **272***b*; and shafts **271***a* and **271***b*. The linear motion mechanism **260** comprises a stationary member **260***a* and a moving member **260***b* which is constrained to move linearly (along a horizontal direction) relative to the stationary member **260***a*. A pair of shafts **268***a* and **268***b* are constrained to rotate relative to the support component **262** respectively around a pair of vertical axes. The shaft **273***a* (or respectively **273***b*) connects the link **272***a* (or respectively **272***b*) to the rigid component **274** so that the link **272***a* (or respectively **272***b*) is constrained to rotate relative to the rigid component **274** around the axis of the shaft **273***a* (or respectively **273***b*). The shaft **271***a* (or **271***b*) connects the link **272***a* (or respectively **272***b*) to the gripper **261***a* (or respectively **261***b*) so that the gripper **261***a* (or respectively **261***b*) is rotatable relative to the link **272***a* (or respectively **272***b*). The gripper **261***a* (or **261***b*) is rigidly or fixedly connected to the shaft **268***a* (or respectively **268***b*). Thus, the gripper **261***a* (or respectively **261***b*) is constrained to rotate relative to the support component **262** around the axis of the shaft **268***a* (or respectively **268***b*). The parts **268***a*, **271***a*, **272***a*, and **273***a* are mirror images of the parts **268***b*, **271***b*, **272***b*, and **273***b* about a vertical plane which is parallel to the direction of the linear motion of the moving member **260***b* relative to the stationary member **260***a*; wherein the stationary member **260***a* is rigidly or fixedly connected to the support component **262**. The rigid component **274** is rigidly or fixedly connected to the moving member **260***b*. Thus, the linear motion mechanism **260** may produce a horizontal motion of the rigid component **274** and hence anti-synchronous rotations in the grippers **261***a* and **261***b*. Each of the grippers **261***a* and **261***b* is rotated between a first end-position and a second end-position. When at the first end-positions, the grippers **261***a* and **261***b* may together grip a container or other object. When at the second end-positions, the grippers **261***a* and **261***b* can open up and release the container or object. The motion mechanism **260** is driven by a motor **260***m*. The gripping mechanism **221***b* may be connected to the computer system **99** of FIG. **1** in the sense that the motor **260***m* is connected to the computer system **99** via wires or by wireless means, and the computer system **99** may be configured to control the motions of the grippers **261***a* and **261***b* produced by the motion mechanism **260**. Thus, the gripping mechanism **221***b* may be controlled by the computer system **99** to grip or release a container or other object.

A gripping mechanism may also be referred to as a gripper mechanism.

Referring to FIG. **4**D, a gripping mechanism **241** comprises a support component (or base component) **244** and a plurality of gripper sub-mechanisms **243** which are referred to as robotic fingers. Each gripper sub-mechanism **243** comprises: grippers **243***d* and **243***b* wherein the gripper **243***d* is rotatable relative to the gripper **243***b* and the gripper **243***b* is rotatable relative to the support component **244**; a motion mechanism comprising a motor (hidden in figure) which drives a rotation of the gripper **243***d* relative to the gripper **243***b*; and a motion mechanism comprising a motor (hidden in figure) which drives a rotation of the gripper **243***b* relative to the support component **244**. (It should be noted that an optional transmission mechanism may be used to link the rotation of the grippers **243***d* and **243***b* and then only one motor is needed to drive the rotations of both grippers.) The gripping mechanism **241** may be connected to the computer system **99** of FIG. **1** in the sense that all motors are connected to the computer system **99** via wires or by wireless means, and the computer system **99** may be configured to control the motions produced by the motors in the gripping mechanism **241**. The gripping mechanism **241** may be controlled by the computer system **99** to grip or release a container or other object.

Referring to FIG. **4**E, a gripping mechanism **251** comprises: a support component (or base component) **252** and a plurality of gripper sub-mechanisms **253** which are referred to as robotic fingers. Each gripper sub-mechanism **253** comprises: grippers **253***a*, **253***b* and **253***c*, wherein the gripper **253***c* is rotatable relative to the gripper **253***b*, the gripper **253***b* is rotatable relative to the gripper **253***a*, and the gripper **253***a* is rotatable relative to the support component **252**; a motion mechanism comprising a motor (hidden in figure) which drives a rotation of the gripper **253***c* relative to the gripper **253***b*; a motion mechanism comprising a motor (hidden in figure) which drives a rotation of the gripper **253***b* relative to the gripper **253***a*; and a motion mechanism comprising a motor (hidden in figure) which drives a rotation of the gripper **253***a* relative to the support component **252**. (It should be noted that an optional transmission mechanism may be used to link the rotation of the grippers **253***a*, **253***b* and **253***c* and then only one motor is needed to drive the rotations of the grippers). The gripping mechanism **251** may be connected to the computer system **99** of FIG. **1** in the sense that all motors are connected to the computer system **99** via wires or by wireless means, and the computer system **99** may be configured to control the motions produced by the motors in the gripping mechanism **251**. The gripping mechanism **251** may be controlled by the computer system **99** to grip or release a container or other object.

The gripping mechanisms **241** (FIG. **4**D) and **251** (FIG. **4**E) are commonly referred to as robot hands. The gripper sub-mechanisms **243** and **253** are referred to as robot fingers. In fact, any robot hand may be used as a gripping mechanism for our purposes here. Robot hands may also be referred to as robot end effectors. Similarly, any robot arm may be used as a motion mechanism for our purpose.

Referring to FIG. **4**F, a robotic apparatus **222** comprises a robot arm **218** and a gripping mechanism **241**. The gripping mechanism **241** is configured to grip or release a container or other object. The support component **244** of the gripping mechanism **241** is fixedly connected to the moving member **217***b* of the rotational motion mechanism **217** of the robot arm **218**, so that the robot arm can move the gripping mechanism **241**. When the gripping mechanism **241** grips a container or other object, the robotic apparatus **222** can transfer the container or object to another position. The robot arm **218** and the gripping mechanism **241** may be connected to the computer system **99** of FIG. **1** via wires or by wireless means, and the computer system **99** may be configured to control the motions produced the motion mechanism in the robotic apparatus **222**. The robotic apparatus **222** may be controlled by the computer system **99** to grip a container or other object, and then move the container or object, and then release the container or object at a different position. The robotic apparatus **222** may substitute a transfer apparatus to grip and move a container or other object. The robotic apparatus **222** may also substitute a dispensing apparatus to grip and move a container to dispense food or food ingredient(s) from the container.

It should be noted that the gripping mechanism **241** of the robotic apparatus **222** may be substituted by the gripping mechanism **221***a* (or **221***b*) or other gripping mechanism.

It should be noted that the gripping mechanisms **221**, **221***a*, **221***b*, **241** and **251** are some realizations of gripping mechanisms. They may be substituted by other types of gripping mechanism such as an electric gripper, a pneumatic gripper, etc.

Referring to FIG. **5**, a transfer apparatus **650** comprises a gripping mechanism **607** comprising: a rigid component **672** referred to as a support component; grippers **671***a* and **671***b* which can optionally be rigid or elastic components; shafts **674***a* and **674***b*; and motors **673***a* and **673***b*. Each of the motors **673***a* and **673***b* comprises a base component which is fixedly connected to the rigid component **672**. The gripper **671***a* is rigidly or fixedly connected to the shaft **674***a*. The motor **673***a* can produce a rotation of the shaft **674***a*, and hence of the gripper **671***a*, around the axis of the shaft **674***a* relative to the rigid component **672**. Similarly, the gripper **671***b* is rigidly or fixedly connected to the shaft **674***b*. The motor **673***b* can produce a rotation of the shaft **674***b*, and hence of the gripper **671***b*, around the axis of the shaft **674***b* relative to the rigid component **672**. The motors **673***a* and **673***b* are configured to rotate the respective grippers **671***a* and **671***b* anti-synchronously around a pair of parallel axes. The gripper **671***a* or **671***b* is rotated between two end-positions. At some first end-positions, the grippers **671***a* and **671***b* may grip a food container **182** under the condition that the food container is placed in a certain position relative to the rigid component **672**. At some second end-positions, the grippers **671***a* and **671***b* can open and release the food container **182**.

The transfer apparatus **650** further comprises a motion mechanism **650***a* comprising: a vertical motion mechanism **608** and a rotational motion mechanism **609**. The vertical motion mechanism **608** comprises a stationary member **675**; a moving member **669** which is constrained to move vertically relative to the stationary member **675**; and a motor **675***m* configured to drive a motion of the moving member relative to the stationary member. The rotational motion mechanism **609** comprises: a stationary member **678**; a shaft **677** referred to as a moving member which is constrained to rotate around the axis of the shaft **677** relative to the stationary member **678**; and a motor **679** configured to drive a rotation of the moving member relative to the stationary member. The rigid component **672** of the gripping mechanism **607** is rigidly, fixedly, or otherwise connected to the moving member **669**, so that the vertical motion mechanism **608** is configured to produce a vertical motion of the moving member **669** and the rigid component **672** between two end-positions. A connector **676** rigidly, fixedly, or otherwise connects the shaft **677** to the stationary member **675**. The motor **679** comprises a base component which is fixedly connected to the stationary member **678**. The motor **679** can produce a rotation of both the shaft **677** and the connector **676** between two end-positions. The transfer apparatus **650** may be configured to grip a food container **182** and transfer it combination of vertical motion, and rotational motion to another position. The motors **673***a*, **673***b*, and **679** are all connected to the computer system **99** of FIG. **1**, so that the computer system **99** may control timings and speeds of their produced motions. The vertical motion mechanism **608** is driven by a motor **675***m*, which is connected to the computer system **99** by wires or by wireless means. The stationary member **678** may be referred to as the support component of the transfer apparatus **650**.

Referring to FIG. **6**A, an ingredient dispensing apparatus **301** comprises a gripping mechanism **311** comprising: a rigid component **345** referred to as a support component; grippers **341***a* and **341***b* which can optionally be rigid or elastic components; shafts **343***a* and **343***b*; and motors **344***a* and **344***b*. Each of the motors **344***a* and **344***b* comprises a base component which is fixedly connected to the rigid component **345**. The gripper **341***a* is rigidly or fixedly connected to the shaft **343***a*. The motor **344***a* is configured to produce a rotation of the shaft **343***a* and hence of the gripper **341***a* around the axis of the shaft **343***a* relative to the rigid component **345**. Similarly, the gripper **341***b* is rigidly or fixedly connected to the shaft **343***b*. The motor **344***b* is configured to produce a rotation of the shaft **343***b* and hence of the gripper **341***b* around the axis of the shaft **343***b* relative to the rigid component **345**. The shafts **343***a* and **343***b* have parallel axes, and the motors **344***a* and **344***b* are configured to rotate the respective grippers **341***a* and **341***b* anti-synchronously around the parallel axes. The gripper **341***a* or **341***b* is rotated between two end-positions. At some first end-positions, the grippers **341***a* and **341***b* are configured to work together to grip an ingredient container **81** under the condition that the ingredient container is placed in a certain position relative to the rigid component **345**; wherein the ingredient container **81** is configured to hold a food or a food ingredient. The ingredient container is not part of the ingredient dispensing apparatus **301**.

The ingredient dispensing apparatus **301** further comprises a motion mechanism **312** comprising: a rigid connector **349** referred to as a stationary member; a shaft **347** referred to as a moving member which is constrained to rotate around the axis of the shaft **347** relative to the stationary member **349**; and a motor **348** configured to drive a rotation of the moving member relative to the stationary member. The shaft **347** comprises a horizontal axis and the axis is perpendicular to the axes of the shafts **343***a* and **343***b*, although these are not strict requirements. The motor **348** comprises a base component which is fixedly connected to the rigid connector **349**. The rigid connector **349** is referred to as the support component of the ingredient dispensing apparatus **301**.

The shaft **347** is rigidly, fixedly, or otherwise connected to the rigid component **345** of the gripping mechanism **311**, and the motor **348** can produce a rotation of the rigid component **345** between two end-positions. When the rigid component **345** is rotated to a first end-position such that the axes of the shafts **343***a* and **343***b* are vertical, and when the grippers **341***a* and **341***b* are rotated to their first end-positions relative to the rigid component **345**, the grippers **341***a* and **341***b* are configured to grip an ingredient container **81** under the condition that the ingredient container is at a certain position relative to the support component **349**. The position of the ingredient container is referred to as the dispensing position relative to the support component **349**. Then, the rigid component **345** is rotated to the second end-position while the ingredient container is gripped by the grippers **341***a* and **341***b*, so that the ingredient container is turned to dispense the food or food ingredient into a cookware. Virtually the entire contents of the ingredient container are dispensed by the turning of the ingredient container. The angle between the first end-position and the second end-position in the rotation of the rigid component **345** is usually between 90 degrees and 180 degrees. The motors **344***a*, **344***b* and **348** are connected to the computer system **99** of FIG. **1** via wires, so that the computer system **99** may control the timing and speed of the motors.

Referring to FIGS. **6**B-**6**C, a cooking apparatus **110** (to be referred to as a second cooking apparatus) comprises: a cooking apparatus **103** (as in FIG. **3**A); a transfer apparatus **650** (as in FIG. **5**); a sink **192**; a garbage disposal **192***d* connected to the sink wherein the garbage disposal is below the sink; and an ingredient dispensing apparatus **301** (as in FIG. **6**A). The sink **192** and the garbage disposal **192***d* are fixedly connected to the floor of the building or the ground via a connector **191**. A liquid pipe (not shown in figures) connects an exit of the garbage disposal **192***d* to a sewage or wastewater tank. The positions of the support component **678** of the transfer apparatus **650**, the connector **191**, the support component of the cooking apparatus **103** and the support component **349** of the dispensing apparatus **301** are fixed relative to each other. When the rigid component **345** of the ingredient dispensing apparatus **301** is rotated to the second end-position, the food or food ingredient in the ingredient container **81** gripped by the ingredient dispensing apparatus **301** can be dispensed into the cookware **11** of the cooking apparatus **103** (see FIG. **6**B). When a food container **182** gripped by the gripping mechanism **607** is moved by the motion mechanism **650***a* to a receiving position relative to the support component of the cooking apparatus **103**, the cookware **11** can be rotated to a certain "dispensing position" by the motion mechanism **104** of the cooking apparatus **103**, so that a cooked food held in the cookware **11** can be dispensed into the food container **182** (optionally through a funnel) (see FIG. **6**C). When the food container **182** gripped by the gripping mechanism **607** is moved away from the receiving position, if the cookware **11** contains wastewater (say, from a cleaning after a dish is cooked), the cookware **11** of the cooking apparatus **103** can be rotated to the dispensing position by the motion mechanism **104**, to dispense the wastewater held in the cookware **11** into the sink **192**.

The shaft **347** is rigidly, fixedly, or otherwise connected to the rigid component **345** of the gripping mechanism **311**, so the motor **348** can produce a rotation of the rigid component **345** between two end-positions. When the rigid component

345 is rotated to a first end-position such that the axes of the shafts **343***a* and **343***b* are vertical, and when the grippers **341***a* and **341***b* are rotated to their first end-positions relative to the rigid component **345**, the grippers **341***a* and **341***b* are configured to grip an ingredient container under the condition that the ingredient container is at a certain position relative to the support component **349**. The position of the ingredient container is referred to as the dispensing position relative to the support component **349**. Then, the rigid component **345** is rotated to the second end-position while the ingredient container is gripped by the grippers **341***a* and **341***b*, so that the ingredient container is turned to dispense the food or food ingredient into a cookware. Virtually the entire contents of the ingredient container are dispensed by the turning of the ingredient container. The angle between the first end-position and the second end-position in the rotation of the rigid component **345** is usually between 90 degrees and 180 degrees. The motors **344***a*, **344***b* and **348** are connected to the computer system **99** of FIG. **1** via wires, so that the computer system **99** may control the timing and speed of the motors.

Referring to FIGS. **6**B-**6**C, a cooking apparatus **110** (to be referred to as a second cooking apparatus) comprises: a cooking apparatus **103** (as in FIG. **3**A); a transfer apparatus **650** (as in FIG. **5**); a sink **192**; a garbage disposal **192***d* connected to the sink wherein the garbage disposal is below the sink; and an ingredient dispensing apparatus **301** (as in FIG. **6**A). The sink **192** and the garbage disposal **192***d* are fixedly connected to the floor of the building or the ground via a connector **191**. A liquid pipe (not shown in figures) connects an exit of the garbage disposal **192***d* to a sewage or wastewater tank. The positions of the support component **678** of the transfer apparatus **650**, the connector **191**, the support component of the cooking apparatus **103** and the support component **349** of the dispensing apparatus **301** are fixed relative to each other. When the rigid component **345** of the ingredient dispensing apparatus **301** is rotated to the second end-position, the food or food ingredient in the ingredient container **81** gripped by the ingredient dispensing apparatus **301** can be dispensed into the cookware **11** of the cooking apparatus **103** (see FIG. **6**B). When a food container **182** gripped by the gripping mechanism **607** is moved by the motion mechanism **650***a* to a receiving position relative to the support component of the cooking apparatus **103**, the cookware **11** can be rotated to a certain "dispensing position" by the motion mechanism **104** of the cooking apparatus **103**, so that a cooked food held in the cookware **11** can be dispensed into the food container **182** (optionally through a funnel) (see FIG. **6**C). When the food container **182** gripped by the gripping mechanism **607** is moved away from the receiving position, if the cookware **11** contains wastewater (say, from a cleaning after a dish is cooked), the cookware **11** of the cooking apparatus **103** can be rotated to the dispensing position by the motion mechanism **104** to dispense the wastewater held in the cookware **11** into the sink **192**.

Referring to FIG. **6**D, an ingredient dispensing apparatus **301***d* comprises a gripping mechanism **311** (as in FIG. **6**A) and a motion mechanism **312***d*. As explained earlier, the grippers **341***a* and **341***b* of the gripping mechanism **311** may be rotated between two end-positions (a first end-position and a second end-position) to grip or release an ingredient container **81**. The motion mechanism **312***d* comprises: a support component **332**; a shaft **347** which is constrained to rotate relative to the support component **332**; and a motor **348** configured to drive the rotation of the shaft **347** relative to the support component **332**. The shaft **347** comprises a horizontal axis, which is perpendicular to the axes of the shafts **343***a* and **343***b* of the gripping mechanism **311**, although these are not strict requirements. The shaft **347** is rigidly, fixedly, or otherwise connected to the rigid component **345** of the gripping mechanism **311** so that the motor **348** can produce a rotation of the rigid component **345** between two end-positions. When the rigid component **345** is rotated to a first end-position, the axes of the shafts **343***a* and **343***b* are vertical so that the gripping mechanism **311** may grip an ingredient container which is at upright position. When the rigid component **345** is rotated to the second end-position, the gripped ingredient container can be turned to dispense the food or food ingredient (into a cookware). Virtually the entire contents of the ingredient container are dispensed by the turning of the ingredient container. The angle between the first end-position and the second end-position in the rotation of the rigid component **345** is usually between 90 degrees and 180 degrees. The ingredient container is not part of the ingredient dispensing apparatus **301***d*.

The ingredient dispensing apparatus **301***d* further comprises a vertical motion mechanism **351** comprising: a stationary member **349***e*; a moving member **349***d* which is constrained to move vertically relative to the stationary member **349***e* wherein the direction of motion is vertical; and a motor **351** configured to drive the motion of the moving member **349***d* relative to the stationary member **349***e*. The support component **332** of the motion mechanism **312***d* is fixedly or rigidly connected to the moving member **349***d*, so that the vertical motion mechanism **351** can drive a vertical linear motion of the support component **332** between two end-positions. The motor **352** is referred to as a driving member. The ingredient dispensing apparatus **301***d* further comprises a support component **353** which is rigidly or fixedly connected to the stationary member **349***e*. The support component **353** may be referred to as the support component of the dispensing apparatus **301***d*. The dispensing apparatus **301***d* is configured to grip an ingredient container **81** and transfer it via a combination of vertical motion and rotational motion to dispense the food or food ingredient in the ingredient container **81** into a cookware.

The motors **344***a*, **344***b*, **348** and **352** of the ingredient dispensing apparatus **301***d* are all connected to the computer system **99** of FIG. **1**, so that the computer system **99** may control timings and speeds of their produced motions.

Referring to FIGS. **7**A-**7**B, a liquid dispensing mechanism **403** comprises a rotational motion mechanism **401** comprising: a support component **463** referred to as a stationary member; a shaft **464** referred to as a moving member which is constrained to rotate around the axis of the shaft **464** relative to the stationary member **463**; and a motor **466** configured to drive a rotation of the moving member relative to the stationary member. The axis of the shaft **464** is configured to be horizontal. A rotatable component **462** is rigidly or fixedly connected the shaft **464**. The motor **466** comprises a base component which is fixedly connected to the support component **463**. The motor **466** can produce a rotation of the shaft **464** and the rotatable component **462** between two end-positions relative to the support component **463**. The motor **466** is connected to the computer system **99** of FIG. **1** via wires, so that the computer system **99** may control the timing and speed of the motor **466**.

The liquid dispensing mechanism **403** further comprises: a plurality of sprayers **411**; a plurality of liquid pipes **420**; a plurality of flexible pipes **410**; and a plurality of liquid containers **414**. Each liquid container **414** is configured to contain a liquid ingredient, e.g., cooking oil, vinegar, soy sauce, or water. Each flexible pipe **410** connects a sprayer **411** to a corresponding liquid pipe **420** and each liquid pipe **420** is inserted into a corresponding liquid container **414**, and a pump **412** can pump liquid contained in the liquid container **414** to the outlet of the sprayer **411** wherein the flow may be measured by a flowmeter **419**. Each liquid container **414** is positioned on an electronic scale **416** and the electronic scale **416** can weigh the corresponding liquid container **414**. A plurality of connectors **417** are configured to fixedly connect the flexible pipes **410** to improve stability of the pipes. The connectors **417** are fixedly or rigidly connected to the rotatable component **462**, so that the rotational motion mechanism can produce a rotation of the shaft, the connectors **417**, the flexible pipes **410** and the sprayers **411** between two end-positions relative to the support component **463** around the axis of the shaft **464**. A connector **418** fixedly connects the flexible pipes **410** to the support component **463**. The pumps **412**, flowmeters **419**, and electronic scales **416** are connected by wired or wireless means to the computer system **99** of FIG. **1**, so that the computer system may control the timing and amount of liquid to be drawn from the corresponding liquid container.

It should be noted that the electronic scale **416** may be substituted by other types of scale, such as electronic balances.

It should be noted the flowmeters **419**, the pumps **412**, and the electronic scale **416** may be fixedly connected to the floor of the building or the ground.

FIGS. **8**A-**8**B show two end-positions of the rotatable component **462** of the liquid dispensing mechanism **403** relative to the cookware **11** of a cooking apparatus **103**. When the rotatable component **462** is rotated to a first end-position, the open end of the sprayers **411** is to be positioned above the cookware **11** when the cookware **11** is at the upright position (see FIG. **8**A), so that the liquid may flow to and be dispensed into the cookware **11**. The liquid dispensing mechanism **403** is used to dispense a plurality of liquid ingredients into the cookware **11** of a cooking apparatus **103** when the rotatable component **462** is at the first end-position and the cookware **11** is at the upright position.

When the rotatable component **462** is rotated to the second end-position, the rotatable component **462** and the sprayers **411** are all away from the cookware **11**. The angle between the first end-position and the second end-position may be about 90 degrees, although this is not a strict requirement.

Referring to FIG. **9**A, a vehicle **790** comprises: a support component **786**; a computer **904**; a plurality of wheels; motors configured to drive rotations of some of the wheels; a rechargeable battery **791**; and a plurality of container holders **785***a*, **785***b* and **785***c* wherein each container holder **785***a*, **785***b* or **785***c* is rotationally symmetric with a vertical axis. Each container holder **785***a*, **785***b* or **785***c* is configured to hold an ingredient container of a specific diameter. Each container holder **785***a* on a vehicle **790** may hold an ingredient container **81** so that the movement of the ingredient container may be restricted or limited when the vehicle **790** is moving.

It should be noted that the vehicle **790** may move on a pair of curved rail tracks whose widths are smaller than the widths of straight rail tracks. The vehicle **790** can carry and transport a plurality of ingredient containers. When the vehicle **790** moves, then the vehicle **790** can transport the ingredient containers held by the container holders of the vehicle.

It should be noted that the any of container holders in the vehicle **790** may be substituted by a solid that is shaped to position or hold an ingredient container.

Referring to FIG. **9**A, a transport system **302** comprises tracks each comprising pairs of mini-rails **331** and a plurality of vehicles **790**. Each mini-rail **331** of the transport system **302** is fixedly connected to the floor of the building or the ground. The vehicle **790** and the container holders **785***a* on the vehicle **790** may move along the mini-rails **331**. The transport system **302** is configured to transfer ingredient containers. The computer **904** is connected to the computer system **99** of FIG. **1** via wireless means, and the computer system **99** is configured to control the timing and speed of the vehicles **790**.

See U.S. patent application Ser. Nos. 16/517,705 and 16/997,933 for more details of the vehicle **790**. The entire contents of these applications are hereby incorporated herein.

Referring to FIG. **9**B, a vehicle **790** in the transport system **302** may move an ingredient container **81** to a dispensing position relative to the support component **349** of the ingredient dispensing apparatus **301**. Then the support component **345** of the gripping mechanism **311** may be rotated to the first end-position relative to the support component **349** while the grippers **341***a* and **341***b* are kept at their second end-positions, and then the grippers **341***a* and **341***b* are rotated to their first end-position to grip the ingredient container **81**.

FIG. **10** shows the relative positions of the cooking apparatus **103**, the ingredient dispensing apparatus **301** and the transport system **302**. A vehicle **790** of the transport system **302** moves an ingredient container **81**, which contains a food or a food ingredient, to a dispensing position relative to the support component **349** of the ingredient dispensing apparatus **301**. The rigid component **345** of the ingredient dispensing apparatus **301** may be rotated to the first end-position, and then the grippers **341***a* and **341***b* can move to their first end-positions to grip the ingredient container **81**. Then the rigid component **345** is rotated to the second end-position so that the ingredient container **81** is moved to dispense the food or food ingredient from the ingredient container **81** into the cookware **11** of the cooking apparatus **103**. The ingredient dispensing apparatus **301** is configured to dispense virtually the entire contents held in the ingredient container **81** into the cookware **11**; wherein exception (to the "virtually entire contents") may be a very small quantity of ingredients which are undesirably stubbornly sticking to a surface of the ingredient container **81** and this small quantity of ingredients will be waste. Afterwards, the rigid component **345** is rotated back to the first end-position, and after that, the grippers **341***a* and **341***b* can move to their second end-positions to release the emptied container **81** on the container holders **785***a* of the vehicle **790**. It should be noted that the vehicle **790** is braked during the time of the above procedures.

Referring to FIG. **11**, a storage **501** comprises: a plurality of container holders **511**; and a support component **512**; wherein each container holder **511** is configured to position or otherwise hold one or more ingredient containers **81**. Each container holder **511** is fixedly connected to the support component **512**. The support component **512** is fixedly connected to the floor of the building or the ground by a rigid connector **513**. The storage **501** also comprises a refrigeration mechanism **514** configured to refrigerate the ingredient containers **81** to keep the food ingredient in the ingredient containers fresh. The refrigeration mechanism

514 is fixedly connected to the floor of the building or the ground by a rigid connector **515**.

Note that the storage **501** may be substituted by the storage system **560** of FIGS. 20A-24 of U.S. patent application Ser. No. 16/517,705 and similar storage system disclosed in U.S. patent application Ser. No. 16/997,933. The entire contents of the application are incorporated herein by reference.

Referring to FIG. **12**, a transfer apparatus **502** comprises a gripping mechanism **517** comprising: a rigid component **539** referred to as a support component; grippers **521***a* and **521***b* which can optionally be rigid or elastic components; shafts **531***a* and **531***b*; and motors **530***a* and **530***b*. Each of the motors **530***a* and **530***b* comprises a base component which is fixedly connected to the rigid component **539**. The gripper **521***a* is rigidly or fixedly connected to the shaft **531***a*. The motor **530***a* can produce a rotation of the shaft **531***a* and hence of the gripper **521***a* around the axis of the shaft **531***a* relative to the rigid component **539**. Similarly, the gripper **521***b* is rigidly or fixedly connected to the shaft **531***b*. The motor **530***b* can produce a rotation of the shaft **531***b* and hence of the gripper **521***b* around the axis of the shaft **531***b* relative to the rigid component **539**. The motors **530***a* and **530***b* are configured to rotate the respective grippers **521***a* and **521***b* anti-synchronously around a pair of parallel axes. The gripper **521***a* or **521***b* is rotated between two end-positions. At some first end-positions, the grippers **521***a* and **521***b* may grip an ingredient container **81** under the condition that the food container is placed in a certain position relative to the rigid component **539**. At some second end-positions, the grippers **521***a* and **521***b* can open and release the ingredient container **81**.

The transfer apparatus **502** further comprises a motion mechanism **502***a* comprising: a vertical motion mechanism **518**; a linear motion mechanism **519**; and a rotational motion mechanism **520**. The vertical motion mechanism **518** comprises a stationary member **523**; a moving member **522** which is constrained to move vertically relative to the stationary member **523**; and a motor **523***m* configured to drive a motion of the moving member relative to the stationary member. The linear motion mechanism **519** comprises a stationary member **535**; a moving member **534** which is constrained to move linearly relative to the stationary member **535**; and a motor **535***m* configured to drive a motion of the moving member relative to the stationary member. The rotational motion mechanism **520** comprises: a rigid component **529***y* referred to as a stationary member; a shaft **525** referred to as a moving member which is constrained to rotate relative to the stationary member **529***y*; and a motor **528** configured to drive a rotation of the moving member relative to the stationary member. The rigid component **539** of the gripping mechanism **517** is fixedly or rigidly connected to the moving member **522**, so that the vertical motion mechanism **518** is configured to produce a vertical motion of the moving member **522** and the rigid component **539** between two end-positions. A connector **533** fixedly or rigidly connects the stationary member **523** and the moving member **534**, so that the linear motion mechanism **519** is configured to produce a linear motion of the moving member **534** and the stationary member **523** between two end-positions. A connector **524** fixedly or rigidly connects the shaft **525** and the stationary member **535** so that the rotational motion mechanism **520** can produce a rotation of the shaft **525** and the stationary member **535** between two end-positions. The transfer apparatus **502** can grip one or more ingredient containers **81**, move them to another position, and then release them to place them at the "another position." The motors **530***a*, **530***b*, **523***m*, **535***m*, and **528** are all connected to the computer system **99** of FIG. **1**, so that the computer system **99** may control timings and speeds of their produced motions.

FIGS. **13**A-**13**D show the relative positions of the storage **501**, the transfer apparatus **502** and the transport system **302**. If a vehicle **790** of the transport system **302** moves to a certain position relative to the support component **529***y* of the transfer apparatus **502**, if the shaft **525** (or the horizontal motion mechanism **519**) is rotated to a first end-position by the motor **528**, if the moving member **522** is at the lower end-position in the vertically linear sliding produced by the vertical motion mechanism **518**, if the moving member **534** is moved to a certain position by the horizontal motion mechanism **519**, and if an ingredient container **81** (which contains a food or a food ingredient and which is at an upright position) is at a certain position relative to the support component **529***y*, the grippers **521***a* and **521***b* can be moved to their first end-positions to grip the ingredient container (see FIG. **13**A). Then, the moving member **522** is moved to the upper end-position while the ingredient container is being gripped by the grippers **521***a* and **521***b* (see FIG. **13**B). Then, the shaft **525** and the stationary member **535** of the horizontal motion mechanism **519** are rotated by the motor **528** to the second end-position while the ingredient container is being gripped by the grippers **521***a* and **521***b* (see FIG. **13**C). Then, the moving member **522** is moved to the lower end-position and then, the grippers **521***a* and **521***b*, when rotated to their second end-positions, may release the ingredient container to a container holder **785***a* of a vehicle **790** (see FIG. **13**D). The computer system **99** may control the timing and speed of the motor **528**, the horizontal motion mechanism **519** and the vertical motion mechanism **518**. It should be noted that the vehicle **790** is braked during the time of the above procedures.

Referring to FIG. **14**, a food dispensing apparatus **620** comprises a gripping mechanism **603** comprising: a rigid component **652** referred to as a support component; grippers **651***a* and **651***b* which can optionally be rigid or elastic components; shafts **665***a* and **665***b*; motors **653***a* and **653***b*. Each of the motors **653***a* and **653***b* comprises a base component which is fixedly connected to the rigid component **652**. The gripper **651***a* is rigidly or fixedly connected to the shaft **665***a*. The motor **653***a* can produce a rotation of the shaft **665***a* and hence of the gripper **651***a* around the axis of the shaft **665***a* relative to the rigid component **652**. Similarly, the gripper **651***b* is rigidly or fixedly connected to the shaft **665***b*. The motor **653***b* can produce a rotation of the shaft **665***b* and hence of the gripper **651***b* around the axis of the shaft **665***b* relative to the rigid component **652**. The motors **653***a* and **653***b* are configured to rotate the respective grippers **651***a* and **651***b* anti-synchronously around a pair of parallel axes. The gripper **651***a* or **651***b* is rotated between two end-positions. At some first end-positions, the grippers **651***a* and **651***b* may grip a cooking container **190** under the condition that the cooking container is placed in a certain position relative to the rigid component **652**, wherein the cooking container **190** is configured to contain or hold a food or a food ingredient. At some second end-positions, the grippers **651***a* and **651***b* can open and release the cooking container.

The food dispensing apparatus **620** further comprises a motion mechanism **602** comprising: a rotational motion mechanism **604** referred to as a first motion sub-mechanism; a vertical motion mechanism **605** referred to as a second motion sub-mechanism; and a rotational motion mechanism **606** referred to as a third motion sub-mechanism. The rotational motion mechanism **604** comprises: a shaft **645** referred to as a moving member, wherein the shaft **645** comprises a horizontal axis; and a motor **654** configured to drive a rotation of the shaft **645**, wherein the motor **654** comprises a base component. The vertical motion mechanism **605** comprises a stationary member **655**; a moving member **656** which is constrained to move vertically relative to the stationary member **655**; and a motor **655***m* configured to drive a motion of the moving member relative to the stationary member. The rotational motion mechanism **606** comprises: a rigid component **659** referred to as a stationary member; a shaft **658** referred to as a moving member which is constrained to rotate relative to the stationary member around the axis of the shaft **658**, wherein the shaft **658** comprises a vertical axis; and a motor **661** configured to drive a rotation of the moving member relative to the stationary member. The rigid component **652** of the gripping mechanism **603** is rigidly, fixedly, or otherwise connected to the shaft **645** so that the rotational motion mechanism **604** can produce a rotation of the shaft **645** and the rigid component **652** between two end-positions relative to the base component of the motor **654**. When the rigid component **652** is at the first end-position, the cooking container **190** gripped by the gripping mechanism **603** is in the upright position. When the rigid component **652** is at the second end-position, the cooking container **190** gripped by the gripping mechanism **603** is turned by about 180 degrees to dispense the food or food ingredient into the cookware **11**. The base component of the motor **654** is rigidly, fixedly, or otherwise connected to the moving member **656** so that the vertical motion mechanism **605** can produce a vertical motion of the moving member **656** and components attached on it between two end-positions. A connector **657** is rigidly, fixedly, or otherwise connected to the shaft **658** of the rotational motion mechanism **606** and the stationary member **655** so that the rotational motion mechanism **606** can produce a rotation of the connector **657** and components attached on it between two end-positions. The rigid component **659** is fixedly connected to the base component of the motor **661** and the floor of the building or the ground. The food dispensing apparatus **620** is configured to grip a cooking container **190** and transfer it via a combination of vertical motion, linear motion and rotational motion to another position. The motors **653***a*, **653***b*, **654**, **655***m* and **661** are all connected to the computer system **99** of FIG. **1**, so that the computer system **99** may control timings and speeds of their produced motions. The rigid component **659** may be referred to as the support component of the food dispensing apparatus **620**.

Referring to FIG. **15**, a liquid dispensing apparatus **660** comprises: a plurality of sprayers **428**; a plurality of liquid pipes **410***b*, **420***b*; a plurality of liquid containers **414***b*; and a rigid component **429** referred to as a support component. Each liquid container **414***b* is configured to hold a liquid ingredient, e.g., cooking oil, vinegar, or water. Each liquid pipe **410***b* connects a sprayer **428** to a corresponding liquid pipe **420***b* and each liquid pipe **420***b* is inserted into a corresponding liquid container **414***b*, and a pump **412***b* can pump liquid contained in the liquid container **414***b* to the outlet of the sprayer **428** wherein the flow may be measured by a flowmeter **419***b*. Each liquid container **414***b* is positioned on an electronic scale **416***b* and the electronic scale **416***b* can weigh the corresponding liquid container **414***b*. Connectors **417***b* and **418***b* are configured to fixedly connect the liquid pipes **410***b* to improve stability of the liquid pipes. The connectors **417***b*, **418***b* and the base component of each electronic scale **416***b* are all fixedly connected to the rigid component **429**. The pumps **412***b*, flowmeters **419***b*, and electronic scales **416***b* are connected by wired or wireless means to the computer system **99** of FIG. **1**, so that the computer system may control the timing and amount of liquid to be drawn from the corresponding liquid container. The liquid dispensing apparatus **660** can be used to dispense a liquid ingredient contained in a liquid container **414***b* into a cookware or cooking container.

Referring to FIG. **16**A, a vehicle **640** is similarly configured to the vehicle **790** except that: the support component **786** is substituted by a support component **663**; the container holders **785***a*, **785***b* and **785***c* are substituted by a plurality of container holders **662**; and the computer **904** is substituted by a computer **904***b*. The other part numbers in the vehicle **640** are the same as the corresponding part numbers in the vehicle **790**. The container holders **662** are configured to hold a food container **182** so that the movement of the food container **182** may be restricted or limited when the vehicle is moving. The computer **904***b* may control the operations of the electrical or electronic devices of the vehicle **640** by sending signals to the electrical or electronic device. The computer **904***b* may communicate with the computer system **99** of FIG. **1** via a wireless communication device.

Referring to FIG. **16**B, a transport system **303** comprises tracks each comprising pairs of mini-rails **331** and a plurality of vehicles **640**. Each mini-rail **331** is fixedly connected to the floor of the building or the ground. The vehicles **640** and the food containers **182** held by the container holders **662** on the vehicles **640** may move along the mini-rails **331**. The transport system **303** can transfer food containers **182**. The computer **904***b* is connected to the computer system **99** of FIG. **1** via wireless means, so that the computer system **99** may control the timing and speed of the vehicles **640**.

It should be noted that the vehicle **640** may comprise additional components for the purpose of staying on track.

It should be noted that the transport system **303** may comprise track switch mechanisms. The vehicles **640** or **790** may move on different mini rails by means of a track switch mechanism.

Referring to FIG. **17**A, a rotational motion mechanism **601** comprises: a support component **613**; a shaft **611** (as a moving member) which is constrained to rotate relative to the support component **613** around the axis of the shaft **611**; and a motor **616** which drives the rotation of the shaft **611** relative to the support component **613**. The shaft **611** may be moved from one angular position to another among a plurality of angular positions. The motor **616** is connected to the computer system **99** of FIG. **1** via wires, and the computer system **99** is configured to control the timing and speed of the motion of the motor **616**. The axis of the shaft **611** is configured to be vertical or nearly vertical. The support component **613** is optionally fixedly connected to a building floor or to the ground. The axis of the rotation (i.e., the axis of the shaft **611**) is configured to be vertical.

The rotation of the shaft **611** is an intermittent rotation, as it is rotated from one angular position to another among a plurality of angular positions. Thus, the rotational motion mechanism **601** may be referred to as an intermittent motion mechanism.

Referring to FIGS. **17**B-**17**C, an enclosure mechanism **851***a* comprises: an enclosure device **843**, referred to as a second enclosure component; and a motion mechanism **855***a* as follows. The motion mechanism **855***a* comprises: a support component **853**; a shaft **856** which is constrained to rotate relative to the support component **853** between two end-positions; and a motor **854** configured to drive a rotation of the shaft **856** relative to the support component **853**. The enclosure device **843** optionally has metal surfaces (e.g., stainless steel) and the enclosure device **843** optionally has the shape of a cap. These surfaces are smooth so they can efficiently reflect electromagnetic waves. The enclosure device **843** comprises a plurality of holes **843***a* and **843***b*, wherein the holes **843***b* are above the holes **843***a*. The enclosure device **843** is rigidly or fixedly connected to the shaft **856** via a connector **852**. Thus, the motion mechanism **855***a* can produce a rotation of the enclosure device **843** between two end-positions: a first end-position; and a second end-position. The motor **854** is connected to the computer system **99** of FIG. **1** via wired or wireless means, and the computer system **99** may control timings and speeds of its produced motion.

The enclosure mechanism **851***a* further comprises: an air pump **818**; and a pipe connector **818***a*, wherein the air pump **818** is fixedly connected to the connector **852**, wherein the pipe connector **818***a* is fixedly connected to the inlet of the air pump **818**. The other end of the pipe connector **818***a* is connected to the exterior side of the enclosure device **843** and the interior of the pipe connector **818***a* may draw air through the holes **843***b*. Thus, the pump **818** can draw air through the holes **843***b* from the cooking chamber enclosed by the enclosure device **843** (and another enclosure component, see below). At the same time, fresh air can flow from outside into the cooking chamber through the holes **843***a*. The enclosure mechanism **851***a* further comprises a cover **834** which may optionally be a round plate of non-metallic material. The cover **834** is configured to be rigidly, elastically, or fixedly connected to the enclosure device **843** via a connector **835**.

It should be noted that the enclosure device **843** may have other shape. The enclosure device **843** may consist of a lid.

Referring to FIGS. **17**D-**17**F, a cooking apparatus **690***a* comprises: a rotatable component **845** comprising a rigid component; and a rotational motion mechanism **601** (as in FIG. **17**A). The shaft **611** of the rotational motion mechanism **601** is fixedly connected to the rotatable component **845** so that the rotational motion mechanism **601** may produce a rotational motion of the rotatable component **845** around the axis of the shaft **611** wherein the axis of the shaft **611** is vertical. The rotatable component **845** may comprise a turntable but this is not a requirement.

A plurality of wheels **634** are mounted on a plurality of support components **632**, which are rigidly or fixedly connected to a support component **836**; and the support component **836** is optionally rigidly or fixedly connected to the floor of the building or the ground (the connection is not shown in figures but is easy to construct). The wheels **634** are used to touch and provide support to the rotatable component **845**.

The cooking apparatus **690***a* further comprises a plurality of microwave ovens **850***a* which are mounted on the rotatable component **845**. Each microwave oven **850***a* comprises a microwave generator **849***a* comprising: a magnetron **842**; a waveguide **824**; a stirrer **899** comprising fan blades which are rotatable relative to a base component **823**; and a motor **822** configured to drive the rotation of the stirrer **899**. The microwave oven **850***a* further comprises a first enclosure component **844** which is fixedly connected to the rotatable component **845**. The magnetron **842**, the waveguide **824**, the base component **823**, and a stationary member of the motor **822** are all fixedly connected to first enclosure component **844**. A container holder **898** is fixedly connected to the first enclosure component **844** wherein the container holder **898** is configured to hold a cooking container **190** so that the movement of the cooking container **190** relative to the rotatable component **845** may be restricted or limited when the rotatable component **845** is moved. The container holder **898** is located inside a cooking chamber (see below).

The microwave oven **850***a* further comprises an enclosure mechanism **851***a* (as in FIGS. **17**B-**17**C), wherein the support component **853** of the enclosure mechanism **851***a* is fixedly connected to the rotatable component **845**. A motor **854** can produce a rotation of the enclosure device **843** between two end-positions: a first end-position; and a second end-position. When the enclosure device **843** is rotated to the first end-position, the enclosure device **843** is configured to touch on the first enclosure component **844**. At this time, the enclosure device **843** (as the second enclosure component), the first enclosure component **844** can enclose a cooking chamber; the cover **834** can cover the cooking container **190** (if any) which is positioned or held by the container holder **898**; and the cover **834**, the container holder **898**, and the cooking container **190** are all inside the cooking chamber, and the microwave oven may cook the food or food ingredient contained within or otherwise held by the cooking container **190**. It should be noted that the cover **834** is used to prevent the food or food ingredient in the cooking container **190** from splashing out during the cooking process (as in FIG. **17**F). When the enclosure device **843** is rotated to the second end-position, the cooking chamber is opened. The magnetron **842** comprises an emitter which is inserted into a corresponding waveguide **824**, so that electromagnetic waves produced by the magnetron **842** can pass through the interior of the waveguide **824**.

The waveguide **824** is rigidly connected to the first enclosure component **844**, and electromagnetic waves can transfer from the waveguide **824** to the cooking chamber enclosed by the first enclosure component **844** and the enclosure device **843**. The motor **822** is configured to rotate the blades of the stirrer **899**. The magnetron **842** can produce an electromagnetic wave of fixed or variable frequencies; and via the waveguide and the stirrer, the electromagnetic waves are directed into the cooking chamber to heat the food or food ingredient contained in a cooking container **190** positioned inside the cooking chamber.

The microwave ovens **850***a* are optionally configured to be cyclically and symmetrically positioned around the rotational axis of the rotatable component **845**, wherein the rotational axis is vertical. This is not a strict requirement. It is possible that some microwave ovens are bigger than others. It is also possible that some microwave ovens are substituted by other ovens (see FIG. **17**G).

It should be noted that the support component **613** of the rotational motion mechanism **601** is referred to the base support component of the cooking apparatus **690***a*.

It should be noted that the holes **843***b* may be drilled through the top of the enclosure device **843** and the air pump may be mounted on the top of the cap.

It should be noted that the first enclosure component **844** and the enclosure device **843** (as the second enclosure component) may be substituted by other shapes of solids, as long as they are structured to enclose a cooking chamber. For example, the first enclosure component **844** may comprise a flat bottom and a cylindrical wall, and the second enclosure component is a solid flat lower surface. As another example, the first enclosure component **844** may be substituted by a solid **848** comprising a flat bottom and a cylindrical wall, and the second enclosure component **843** may be substituted by a solid **847** comprising a flat bottom and a cylindrical wall, wherein both cylindrical walls comprise the same sectional shapes, see FIG. **17**H. The interior side of the enclosure device **843** (or the first enclosure component) comprises a metal (e.g., stainless steel) surface.

It should be noted that each microwave generator **849***a* may be substituted by another type of generator of electromagnetic waves. Generators of electromagnetic waves may be constructed by known techniques.

Referring to FIGS. **18**A-**18**B, an enclosure mechanism **851***b* comprises: an enclosure device (referred to as a second enclosure component) **843** (as described above); and a motion mechanism **855***b* as follows. The motion mechanism **855***b* comprises: a support component **853**; a shaft **856** which is constrained to rotate relative to the support component **853** between two end-positions; and a motor **854** configured to drive a rotation of the shaft **856** relative to the support component **853**. The enclosure device **843** is rigidly or fixedly connected to the shaft **856** via a connector **852**. Thus, the motion mechanism **855***b* can produce a rotation of the enclosure device **843** between two end-positions: a first end-position; and a second end-position. The motor **854** is connected to the computer system **99** of FIG. **1** via wires or wireless means, and the computer system **99** may control timings and speeds of its produced motion.

The enclosure mechanism **851***b* further comprises an air pump **818** and a pipe connector **818***a*, wherein the air pump **818** is fixedly connected to the connector **852**, wherein the pipe connector **818***a* is fixedly connected to the inlet of the air pump **818**. Another end of the pipe connector **818***a* is connected to the exterior side of the enclosure device **843** and the interior of the pipe connector **818***a* may draw air through the holes **843***b*. Thus, the pump **818** can draw air from the cooking chamber enclosed by the enclosure device **843** and other components through the holes **843***b*. At the same time, fresh air can flow from outside into the cooking chamber through the holes **843***a*.

The enclosure mechanism **851***b* further comprises a solid component **829** of non-metallic material. The component **829** optionally has the shape of a round board. The component **829** is fixedly mounted inside the enclosure device **843**, above the holes **843***a* and **843***b*. The component **829** can prevent or limit steam and air to pass through it.

Referring to FIGS. **18**C-**18**D, a cooking apparatus **690***b* (which may be referred to as a first cooking apparatus) comprises: a rotatable component **846**; and a rotational motion mechanism **601** (as in FIG. **17**A). The shaft **611** of the rotational motion mechanism **601** is fixedly connected to the rotatable component **846** so that the rotational motion mechanism **601** may produce a rotational motion of the rotatable component **846** around the axis of the shaft **611** wherein the axis of the shaft **611** is vertical. (The rotatable component **846** may comprise a turntable but this is not a requirement.)

A plurality of wheels **634** are mounted on a plurality of support components **632**, which are rigidly or fixedly connected to the support component **836**; and the support component **836** is optionally rigidly or fixedly connected to the floor of the building or the ground (the connection is not shown in figures but is easy to construct). The wheels **634** are used to touch and provide support to the rotatable component **846**.

The cooking apparatus **690***b* further comprises a plurality of microwave ovens **850***b* which are mounted on the rotatable component **846**. Each microwave oven **850***b* comprises a microwave generator **849***x*. The microwave generator **849***x* is configured the same way as the microwave generator **849***a*. Thus, the microwave generator **849***x* comprises: a magnetron **842**; a waveguide **824**; a stirrer **899** comprising fan blades which are rotatable relative to a base component **823**; and a motor **822** configured to drive the rotation of the stirrer **899**.

The microwave oven **850***b* further comprises an enclosure mechanism **851***b* (as in FIGS. **18**A-**18**B) and a first enclosure component **831**, wherein the support component **853** and the first enclosure component **831** are all fixedly connected to the rotatable component **846**. The microwave generator **849***x* is fixedly or rigidly connected to the enclosure device **843** of the enclosure mechanism **851***b*. The motor **854** of the enclosure mechanism **851***b* can produce a rotation of the enclosure device **843** between two end-positions: a first end-position; and a second end-position. When the enclosure device **843** is rotated to the first end-position, the enclosure device **843** is configured to touch on the first enclosure component **831**. At this time, the enclosure device **843** and the first enclosure component **831** are configured to enclose a cooking chamber. When the enclosure device **843** is rotated to the second end-position, the cooking chamber is opened. A container holder **898** is fixedly connected to the first enclosure component **831** and is located inside the cooking chamber. The container holder **898** is configured to hold a cooking container **190** in the cooking chamber so that the movement of the cooking container **190** relative to the rotatable component **846** may be restricted or limited when the rotatable component **846** is moved.

In the microwave oven **850***b*, the electromagnetic waves produced by the magnetron **842** of the microwave generator **849***x* can pass through (with help of the stirrer **899**) the interior of the waveguide **824** and then transfer into the cooking chamber enclosed by the first enclosure component **831** and the enclosure device **843**. The electromagnetic waves in the cooking chamber can heat the food or food ingredient contained in a cooking container **190** positioned inside the cooking chamber.

The microwave ovens **850***b* are optionally configured to be cyclically and symmetrically positioned around the rotational axis of the rotatable component **846**, wherein the rotational axis is vertical. This is not a strict requirement. It is possible that some microwave ovens are bigger than others. It is also possible that some microwave ovens are substituted by other ovens.

It should be noted that the support component **613** of the rotational motion mechanism **601** is referred to the base support component of the cooking apparatus **690***b*.

Referring to FIG. **19**A, an enclosure mechanism **841***a* comprises: an enclosure device (referred to as a second enclosure component) **843** (as described above); and a linear actuator **892**. The linear actuator **892** comprises: a base component **892***a* (as a stationary member); a rod **892***b* (as a moving member) which is constrained to move vertical relative to the base component **892***a* between two end-positions; and a motor **892***m* configured to drive a motion of the rod **892***b* relative to the base component **892***a*. The end of the rod **892***b* is fixedly connected to the enclosure device **843**, so that the linear actuator **892** can move the enclosure device **843** vertically relative to the base component **892***a* between two end-positions. The motor **892***m* is connected to the computer system **99** of FIG. **1** by wires or by wireless means and the computer system **99** may control timings and speeds of its produced motion.

The enclosure mechanism **841***a* further comprises: an air pump **818**; and a pipe connector **818***a*, wherein the air pump **818** is fixedly connected to the moving member **892***b*, wherein the pipe connector **818***a* is fixedly connected to the inlet of the air pump **818**. The other end of the pipe connector **818***a* is connected to the exterior side of the enclosure device **843** and the interior of the pipe connector **818***a* may draw air through the holes **843***b*. Thus, the pump **818** can draw air through the holes **843***b* from the cooking chamber enclosed by the enclosure device **843** and other components. At the same time, fresh air can flow from outside into the cooking chamber through the holes **843***a*.

Referring to FIGS. **19**B-**19**C, a cooking apparatus **680***a* (which may be referred to as a first cooking apparatus) comprises: a rotatable component **845**; a support component **832**; and a rotational motion mechanism **601** (as in FIG. **17**A). The support component **832** is fixedly connected to the rotatable component **845** via a plurality of connectors **823**. The shaft **611** of the rotational motion mechanism **601** is fixedly connected to the rotatable component **845** so that the rotational motion mechanism **601** may produce a rotational motion of the rotatable component **845** and the support component **832** around the axis of the shaft **611** wherein the axis of the shaft **611** is vertical. The rotatable component **845** may comprise a turntable but this is not a requirement.

A plurality of wheels **634** are mounted on a plurality of support components **632**, which are rigidly or fixedly connected to the support component **836**; and the support component **836** is optionally rigidly or fixedly connected to the floor of the building or the ground (the connection is not shown in figures but is easy to construct). The wheels **634** are used to touch and provide support to the rotatable component **845**.

The cooking apparatus **680***a* further comprises a plurality of microwave ovens **850***c* which are mounted on the rotatable component **845**. Each microwave oven **850***c* comprises: a first enclosure component **844** which is fixedly connected to the rotatable component **845**; a microwave generator **849***x* fixedly connected to first enclosure component **844**; and an enclosure mechanism **841***a* (as in FIG. **19**A), wherein the base component **892***a* of the enclosure mechanism **841***a* is fixedly connected to the support component **832**. The linear actuator **892** of the enclosure mechanisms **841***a* can produce a vertical motion of the enclosure device **843** between two end-positions: a first end-position which is lower; and a second end-position which is higher. When the enclosure device **843** is moved to the lower position (i.e., the first end-position), the enclosure device **843** is configured to touch on the first enclosure component **844**. At this time, the enclosure device **843** and the first enclosure component **844** are configured to enclose a cooking chamber. When the enclosure device **843** is moved to the higher position (the second end-position), the cooking chamber is opened.

A container holder **898** is fixedly connected to the first enclosure component **844** and is located inside the cooking chamber. The container holder **898** is configured to hold a cooking container **190** in the cooking chamber so that the movement of the cooking container **190** relative to the rotatable component **845** may be restricted or limited when the rotatable component is moved.

In the microwave oven **850***c*, the electromagnetic waves produced by the magnetron **842** can pass through (with help of the stirrer **899**) the interior of the waveguide **824** and then transfer into the cooking chamber enclosed by the first enclosure component **844** and the enclosure device **843**. The electromagnetic waves in the cooking chamber can heat the food or food ingredient contained in a cooking container **190** positioned inside the cooking chamber.

The microwave ovens **850***c* are optionally configured to be cyclically and symmetrically positioned around the rotational axis of the rotatable component **845**, wherein the rotational axis is vertical. This is not a strict requirement. It is possible that some microwave ovens are bigger than others. It is also possible that some microwave ovens are substituted by other ovens (e.g., light wave ovens).

It should be noted that the support component **613** of the rotational motion mechanism **601** is referred to the base support component of the cooking apparatus **680***a*.

Referring to FIGS. **20**A-**20**B, an enclosure mechanism **841***b* comprises: a linear actuator **892**; and an enclosure device (referred to as a second enclosure component) **843** (as described above). The enclosure device **843** is fixedly connected to the moving member **892***b*, so that the linear actuator **892** can move the enclosure device **843** vertically relative to the base component **892***a* between two end-positions.

The enclosure mechanism **841***b* further comprises: an air pump **818**; and a pipe connector **818***a*, wherein the air pump **818** is fixedly connected to the moving member **892***b*, wherein the pipe connector **818***a* is fixedly connected to the inlet of the air pump **818**. The other end of the pipe connector **818***a* is connected to the exterior side of the enclosure device **843** and the interior of the pipe connector **818***a* may draw air through the holes **843***b*. Thus, the pump **818** can draw air from the cooking chamber enclosed by the enclosure device **843** and other components through the holes **843***b*. At the same time, fresh air can flow from outside into the cooking chamber through the holes **843***a*.

The enclosure mechanism **841***b* further comprises a solid component **829** of non-metallic material (configured the same way as the one described in FIG. **18**B). The component **829** is fixedly mounted inside the enclosure device **843**, above the holes **843***a* and **843***b*. The component **829** is used to prevent or limit steam and air to pass through it.

Referring to FIGS. **20**C-**20**D, a cooking apparatus **680***b* (which may be referred to as a first cooking apparatus) comprises: a rotatable component **846**; a support component **832**; and a rotational motion mechanism **601** (as in FIG. **17**A). The shaft **611** of the rotational motion mechanism **601** is fixedly connected to the rotatable component **846** so that the rotational motion mechanism **601** may produce a rotational motion of the rotatable component **846** around the axis of the shaft **611** wherein the axis of the shaft **611** is vertical. The rotatable component **846** may comprise a turntable but this is not a requirement. The support component **832** is fixedly connected to the rotatable component **846** via a plurality of connectors **833**.

A plurality of wheels **634** are mounted on a plurality of support components **632**, which are rigidly or fixedly connected to the support component **836**; and the support component **836** is optionally rigidly or fixedly connected to the floor of the building or the ground (the connection is not shown in figures but is easy to construct). The wheels **634** are used to touch and provide support to the rotatable component **846**.

The cooking apparatus **680***b* comprises a plurality of microwave ovens **850***d* which are mounted on the rotatable component **846**. Each microwave oven **850***d* comprises: a first enclosure component **831** which is fixedly connected to the rotatable component **846**; and an enclosure mechanism **841***b* (as in FIGS. **20**A-**20**B), wherein the base component **892***a* of the enclosure mechanism **841***b* is fixedly connected to the support component **832**; and a microwave generator **849***x* fixedly connected to the enclosure device **843** of the enclosure mechanism **841***b*. The linear actuator **892** of the enclosure mechanisms **841***b* can produce a vertical motion of the enclosure device **843** between two end-positions: a first end-position which is lower; and a second end-position which is higher. When the enclosure device **843** is moved to the lower position (i.e., the first end-position), the enclosure device **843** is configured to touch on the first enclosure component **831**. At this time, the enclosure device **843** and the first enclosure component **831** are configured to enclose a cooking chamber. When the enclosure device **843** is moved to the higher position (the second end-position), the cooking chamber is opened. A container holder **898** is fixedly connected to the first enclosure component **831** and is located inside the cooking chamber. The container holder **898** is configured to hold a cooking container **190** in the cooking chamber so that the movement of the cooking container **190** relative to the rotatable component **846** may be restricted or limited when the rotatable component **846** is moved.

In the microwave oven **850***d*, the electromagnetic waves produced by the magnetron **842** of the microwave generator **849***x* can pass through (with help of the stirrer **899** of the microwave generator **849***x*) the interior of the waveguide **824** and then transfer into the cooking chamber enclosed by the first enclosure component **831** and the enclosure device **843** to heat the food or food ingredient contained in the cooking container **190** positioned inside the cooking chamber. The electromagnetic waves in the cooking chamber can heat the food or food ingredient contained in a cooking container **190** positioned inside the cooking chamber.

The microwave ovens **850***d* are optionally configured to be cyclically and symmetrically positioned around the rotational axis of the rotatable component **846**, wherein the rotational axis is vertical. This is not a strict requirement. It is possible that some microwave ovens are bigger than others. It is also possible that some microwave ovens can be substituted by other ovens.

It should be noted that the support component **613** of the rotational motion mechanism **601** is referred to the base support component of the cooking apparatus **680***b*.

Referring to FIGS. **21**A-**21**B, an enclosure mechanism **891** comprises: a linear actuator **892**; a support component **893**; and an enclosure device (referred to as a second enclosure component) **896** in the shape of a cap. The enclosure device **896** comprises cylindrical surfaces **896***a* and **896***c* which are joined to a cone surface **896***b*; wherein the radius of the cylindrical surface **896***c* is larger than that of the cylindrical surface **896***a*. The enclosure device **896** comprises a paraboloidal surface **897** at the top. The enclosure device **896** may be configured to be a rigid component. All surfaces **896***a*, **896***b*, **896***c* and **897** of the enclosure device **896** are configured to be metal surfaces preferably made of stainless-steel sheets. These surfaces need to be smooth so they can efficiently reflect electromagnetic waves. The enclosure device **896** comprises a plurality of holes **896***d* (through the cone surface **896***b*) so air can flow through the holes. A round component **816** of non-metallic material is fixedly mounted inside the cap, above the holes **896***d*. The component **816** can prevent or limit steam and air to pass through it. The enclosure device **896** is fixedly or rigidly connected to the moving member **892***b* of the linear actuator **892** by a connector **815** and the base component **892***a* of the linear actuator **892** is fixedly or rigidly connected to the support component **893**. Thus, the linear actuator **892** can produce a vertical linear motion of the enclosure device **896** relative to the support component **893** between two end-positions.

It should be noted that the paraboloidal surface **897** may be substituted by another type of curved surface or a flat surface.

Referring to FIGS. **21**C-**21**E, a cooking apparatus **680***c* comprises: a rotatable component **692**; and a rotational motion mechanism **601** (as in FIG. **17**A). The shaft **611** of the rotational motion mechanism **601** is fixedly connected to the rotatable component **692** so that the rotational motion mechanism **601** may produce a rotational motion of the rotatable component **692** around the axis of the shaft **611** wherein the axis of the shaft **611** is vertical. The rotatable component **692** may comprise a turntable but this is not a requirement.

A plurality of wheels **634** are mounted on a plurality of support components **632**, which are rigidly or fixedly connected to the support component **836**; and the support component **836** is optionally rigidly or fixedly connected to the floor of the building or the ground (the connection is not shown in figures but is easy to construct). The wheels **634** are used to touch and provide support to the rotatable component **692**.

The cooking apparatus **680***c* comprises a plurality of microwave ovens **850***e* which are mounted on the rotatable component **692**. Each microwave oven **850***e* comprises: a first enclosure component **817**; and an enclosure mechanism **891** (as in FIG. **21**A) wherein the support component **893** of the enclosure mechanism **891** is fixedly connected to the rotatable component **692**. The first enclosure component **817** is rigidly or fixedly connected to the rotatable component **692**. In the microwave oven **850***e*, the linear actuator **892** of the enclosure mechanism **891** can produce a vertical motion of the enclosure device **896** between two end-positions: a first end-position which is lower; and a second end-position which is higher. When the enclosure device **896** is moved to the lower position (i.e., the first end-position), the enclosure device **896** is configured to touch on the first enclosure component **817**. The enclosure device **896** and the first enclosure component **817** are configured to enclose a cooking chamber. When the enclosure device **896** is moved to the higher position (the second end-position), the cooking chamber is opened.

The microwave oven **850***e* further comprises a linear actuator **699**. The linear actuator **699** comprises: a base component **699***a* referred to as a stationary member; a moving member **699***b* which is constrained to move vertically relative to the base component **699***a* between two end-positions; and a motor **699***m* configured to drive a motion of the moving member relative to the stationary member. The base component **699***a* is fixedly connected to the rotatable component **692** via a connector **698**. A container holder **898** is fixedly connected to (the top end of) the moving member **699***b* (see FIG. **21**D) and is located inside the cooking chamber. The container holder **898** is configured to position or otherwise hold a cooking container **190** so that the movement of the cooking container **190** relative to the moving member **699***b* may be restricted or limited. The linear actuator **699** is configured to produce a vertical movement in the corresponding container holder **898** and the cooking container **190**. The motor **699***m* is connected to the computer system **99** of FIG. **1** by wires or by wireless means and the computer system **99** may control timings and speeds of its produced motion.

The microwave oven **850***e* further comprises a microwave generator **849***x* fixedly connected to the moving member **892***b*. Microwaves produced by the corresponding magnetron **842** of the microwave generator **849***x* can heat the food or food ingredient in the corresponding cooking container in the cooking chamber and during this period of time, the linear actuator **699** can produce a motion in the container holder **898** so that the food or food ingredient(s) in a cooking container **190** can be evenly exposed to the electromagnetic waves. When the enclosure device **896** is moved to the higher position (the second end-position), the enclosure device **896** is away from the cooking container **190** so that a cooking container **190** can be moved to or from the container holder **898**.

An air pump **818** is fixedly connected to the rotatable component **692** via a connector **821**, and a pipe connector **818***a* is fixedly connected to the outlet of the air pump **818**. The other end of the pipe connector **818***a* is joined with the exterior side of the first enclosure component **817** and the pump **818** is configured to pump air into a corresponding cooking chamber, wherein the first enclosure component **817** comprises holes to let air flow through. Air in the cooking chamber can escape to the outside through the holes **896***d* of the enclosure device **896**.

It should be noted that the support component **613** of the rotational motion mechanism **601** is referred to the base support component of the cooking apparatus **680***c*.

In some embodiments, referring to FIG. **22**, a cooking sub-system **800** comprises: a transport system **302** (as in FIG. **9**A); a cooking apparatus **110** referred to as a second cooking apparatus (as in FIGS. **6**B-**6**C); and a transport system **303** (as in FIG. **16**B). The ingredient dispensing apparatus **301** of the cooking apparatus **110** is positioned next the cooking apparatus **103** of the cooking apparatus **110**. The ingredient dispensing apparatus **301** can grip an ingredient container from a vehicle **790** of the transport system **302** and then dispense the food or food ingredient from the ingredient container into the cookware **11** of the cooking apparatus **103**. The transfer apparatus **650** of the cooking apparatus **110** is positioned next to the cooking apparatus **103**. The transfer apparatus **650** can grip a food container **182** from a vehicle **640** of the transport system **303** and move it to a receiving position relative to the support component of the cooking apparatus **103** to receive a cooked food from the cookware **11**.

The cooking sub-system **800** further comprises: a liquid dispensing apparatus **403** (as in FIG. **7**B); a transfer apparatus **502** (as in FIG. **12**); and a storage **501** (as in FIG. **11**) which may store a plurality of ingredient containers **81**, wherein the ingredient containers **81** may contain or otherwise hold a food or a food ingredient. The liquid dispensing apparatus **403** is positioned next to the cooking apparatus **103** and can dispense liquid ingredients into the cookware **11**. The transfer apparatus **502** may load an ingredient container **81** (which contains or otherwise holds a food or a food ingredient) from the storage **501** to a vehicle **790** of the transport system **302**. The vehicle **790** may move the ingredient container to a location next to the cooking apparatus **103** to be gripped and held by the ingredient dispensing apparatus **301**.

The cooking sub-system **800** may cook and transfer food or food ingredient(s) as follows. The transfer apparatus **502** loads one or more ingredient containers **81** (which contain or otherwise hold a food or a food ingredient) from the storage **501** to a vehicle **790** of the transport system **302**. Then, the vehicle **790** can transport the (one or more) ingredient containers to a location next to the cooking apparatus **103** so that the ingredient containers may be successively gripped and moved by the ingredient dispensing apparatus **301** to dispense the food or food ingredient from the ingredient containers to the cookware **11** of the cooking apparatus **103**, when the cookware **11** is already rotated to the upright position (or first end-position). The liquid dispensing apparatus **403** may dispense liquid ingredients into the cookware **11**. As we will see below, another cooking apparatus may be used to produce a semi-cooked food and the semi-cooked food may be dispensed into the cookware **11** as well. Then, a cooked food is produced in the cookware **11** from the dispensed food or food ingredient(s), the dispensed liquid ingredients and the semi-cooked food. The cooked food in the cookware can be dispensed into a food container **182**, which is gripped by the transfer apparatus **650**. The transfer apparatus **650** can then load the food container **182** to a vehicle **640** (of the transport system **303**) which is moved to a certain position relative to the support component **678** of the transfer apparatus **650**. The vehicle **640** can transport the food container **182** to a location that is closer to customers.

It should be noted that the cooking sub-system **800** may further comprise a cleaning apparatus. The cookware **11** may be cleaned by the cleaning apparatus after the cooked food is dispensed. See U.S. patent application Ser. No. 17/069,707 for examples of the cleaning apparatus. The entire contents of the application are hereby incorporated herein. There are many known cleaning apparatuses that can be used for the function of the cleaning apparatus.

In some embodiments, referring to FIGS. **23**-**26**, a cooking system **909** comprises: a cooking sub-system **800** (as in FIG. **22**); and a cooking apparatus **600**. The cooking apparatus **600** comprises: a cooking apparatus **690***a* (as in FIG. **17**G) referred to as a first cooking apparatus; a food dispensing apparatus **620** (as in FIG. **14**); a liquid dispensing apparatus **660** (as in FIG. **15**); and an ingredient dispensing apparatus **301***b*, wherein the ingredient dispensing apparatus **301***b* is a copy of the ingredient dispensing apparatus **301** but is positioned next to the cooking apparatus **690***a*. The part numbers in the mechanism **301***b* are the same as the corresponding parts of the mechanism **301**. In the cooking apparatus **600**, the ingredient dispensing apparatus **301***b* can grip and move an ingredient container from a vehicle **790** of the transport system **302** of the cooking sub-system **800** to dispense the food or food ingredient from the ingredient container into one of the cooking containers **190** of the cooking apparatus **690***a*, and the ingredient container is returned to the vehicle after the food or food ingredient are dispensed. The food dispensing apparatus **620** is positioned between the cooking apparatus **103** of the cooking sub-system **800** and the cooking apparatus **690***a* to grip and move a cooking container **190** which is positioned or otherwise held by a container holder **898** (of the cooking apparatus **690***a*) to dispense semi-cooked food from the cooking container **190** into the cookware **11** of the cooking apparatus **103**. The liquid dispensing apparatus **660** is positioned next to the cooking apparatus **690***a* to dispense liquid ingredients into a cooking container **190** of the cooking apparatus **690***a*.

The cooking system **909** further comprises: a container moving apparatus **620***b*; a food container cleaning mechanism **670** (described below); and a computer system **99** (as in FIG. **1**). The container moving apparatus **620***b* is a copy of the food dispensing apparatus **620** but is positioned next to the food container cleaning mechanism **670**. The part numbers in the container moving apparatus **620***b* are the same as the corresponding parts of the food dispensing apparatus **620**. The container moving apparatus **620***b* can grip and move a cooking container **190** from the cooking apparatus **690***a* and turn the container **190** (optionally by 180 degrees) to a certain cleaning position so that the cooking container **190** can get cleaned by the food container cleaning mechanism **670**. Then the cleaned cooking container **190** can be moved back to the rotatable component **845** of the cooking apparatus **690***a*.

In the cooking system **909**, when the rotatable component **845** of the cooking apparatus **690***a* is rotated by the rotational motion mechanism **601** to a certain position, one or more of the following processes may be completed: (1) a motion mechanism **855***a* rotates the corresponding enclosure device **843** to the second end-position so that the ingredient dispensing apparatus **301***b* may move an ingredient container to dispense the food or food ingredient from the ingredient container into a cooking container **190** which is held by the container holder **898** corresponding to the motion mechanism **855***a*; (2) the motion mechanism **855***a* rotates the enclosure device **843** to the first end-position so that the food or food ingredient in the cooking container **190** is heated by the corresponding magnetron **842**; (3) the motion mechanism **855***a* rotates the enclosure device **843** to the second end-position so that the food dispensing apparatus **620** can dispense a semi-cooked food from the cooking container **190** to the cookware **11** of the cooking apparatus **103** and then the food dispensing apparatus moves the cooking container **190** to the holder on the rotatable component **845**; and (4) the rotatable component **845** is rotated to a position so that the container moving apparatus **620***b* can move the cooking container **190** to the food container cleaning mechanism **670** to get cleaned and then returned to the holder on the rotatable component **845**.

The cooking system **909** may cook a food by applying the same steps as the cooking sub-system **800** except that a semi-cooked food produced by the cooking apparatus **600** can be dispensed into the cookware **11** wherein the semi-cooked food may be used as an ingredient for the cooking apparatus **103**.

The computer system **99** is connected to the mechanisms and devices **16**; **412**; **416**; **419**; **466**; **344***a*; **344***b*; **348**; **616**; **851***a*; **412***b*; **416***b*; **419***b*; **653***a*; **653***b*; **654**; **661**; **673***a*; **673***b*; **675***m*; **679**; **501**; and **502**. The computer system **99** is also connected to the transport systems **302** and **303** to control the movements of the vehicles of the transport systems **302** and **303**.

As shown in FIG. **27**, the following tasks are performed by the computer system **99** prior to the operation of the cooking system **909**.

In Step **961**, the computer system **99** stores (in the computer system's memory) a program, configured to send or receive signals to and from the motors, actuators, inductive stoves, and pumps of the cooking system **909**.

In Step **962**, a database is installed in the computer system **99**.

In Step **963**, each of the cooking apparatuses, liquid dispensing apparatus, ingredient dispensing apparatuses, cookware cleaning mechanism, food container cleaning mechanism, food dispensing apparatus, transfer apparatus, transport system is assigned a unique ID. The computer system **99** stores the IDs of these apparatuses and mechanisms.

In Step **964**, the computer system **99** stores the information of the structure of each vehicle of the transport system **302**, including the ingredient container types that can be placed on the holders of the vehicle **790**.

In Step **965**, the computer system **99** stores a program for controlling the transport systems **302** and **303**. The program may be used to control a vehicle **790** of the transport system **302** so that the vehicle **790** may move and stop at a pre-scheduled time at a position near one of the cooking apparatuses of the cooking system **909**, where an ingredient container on a given holder of the vehicle **790** is at a dispensing position relative to the cooking apparatus. The program may be used to control a vehicle **640** of the transport system **303** so that the vehicle **640** may move and stop at a pre-scheduled time at a position near a transfer apparatus of the cooking system **909**, where the container holders of the vehicle **640** are at a transferring position near the transfer apparatus.

In Step **966**, the computer system **99** stores a list of food items which may be cooked by the cooking system **909**.

In Step **967**, for each food item in the list of Step **966**, the computer system **99** stores a cooking program configured to control the motors, actuators, inductive stoves, pumps and devices in the cooking system **909**.

In Step **968**, for each food item in the list of Step **966**, the computer system **99** stores types and quantities of the foods or food ingredients, relative timing of their dispensing and the respective ID of the destination cooking apparatus for each ingredient to be dispensed into; wherein the relative timing refers to the timing relative to the timing of the program of Step **967** corresponding to the food item. The food or food ingredient contained in an ingredient container is to be dispensed into a cookware or a cooking container.

Referring to FIG. **28**, the following tasks are routinely performed by the computer system **99** during the operation of the cooking system **909**.

In Step **971**, the computer system **99** takes an order of a food item. The order may be placed by a human either at the computer system **99**, or at a computer which sends the order to the computer system **99**.

In Step **972**, for the ordered food item of Step **971**, the computer system **99** finds the information on the types and quantities of the ingredients needed for cooking the ordered food item. Such information was stored by the computer system **99** in Step **968**.

In Step **973**, the computer system **99** locates the ingredient containers that contain the food or food ingredient found in Step **972**. The food or food ingredient may be dispensed from some larger containers into the ingredient containers. Alternatively, the food or food ingredient may already be in the ingredient containers, and their locations already stored in the memories of the computer system **99**.

In Step **974**, the computer system **99** schedules the cooking of the ordered food item by the cooking system **909**. The schedule includes the timing for running the program of Step **967** corresponding to the ordered food item. The schedule also includes the timing of dispensing of the food or food ingredient from each ingredient container into a respective cooking container or a cookware of the cooking system **909**, in accordance with the stored information by the computer system **99** in Step **968**.

In Step **975**, the computer system **99** controls the transport system **302** so that each ingredient container of Step **972** may be moved and stopped per the schedule of Step **974**.

In Step **976**, the computer system **99** runs the program of Step **967** corresponding to the ordered food item, according to the schedule of Step **974**, to send or receive signals to or from the motors, actuators, inductive stoves, temperature sensors and pumps of the cooking system **909**.

After all these steps, the cooking of the food item, including dispensing of the cooked food to a food container, is complete.

It should be noted that in the cooking system **909**, the cooking apparatus **690***a* may be substituted by the cooking apparatus **690***b*, **680***a*, **680***b* or **680***c*.

Referring to FIGS. **29**A-**29**B, a cooking apparatus **690***d* comprises: a plurality of rotatable components **846** which are respectively positioned at different levels (the figures show two rotatable components **846** but there can be more in some applications), wherein the rotatable components **846** are fixedly or rigidly connected via connectors **846***a*; and a rotational motion mechanism **601** (as in FIG. **17**A). The rotatable components **846** are directly or indirectly fixedly connected to the shaft **611** of the rotational motion mechanism **601**, so that the rotational motion mechanism **601** may produce a rotation of the rotatable components **846** around the axis of the shaft **611** wherein the axis of the shaft **611** is vertical. The rotatable components **846** may comprise a turntable but this is not a requirement.

A plurality of wheels **634** are mounted on a plurality of support components **632** which are rigidly or fixedly connected to the support component **836**; and the support component **836** is optionally rigidly or fixedly connected to the floor of the building or the ground (the connection is not shown in figures but is easy to construct). The wheels **634** are used to touch and provide support to the lowest one of the rotatable components **846**.

The cooking apparatus **690***d* further comprises a plurality of microwave ovens **850***b* each of which is mounted on one of the rotatable components **846** wherein each individual microwave oven is described as in FIGS. **18**C-**18**D. The microwave ovens **850***b* which are mounted on a same rotatable component **846** may optionally be cyclically and symmetrically positioned around the axis of the shaft **611**. A container holder **898** is fixedly connected to a first enclosure component **831** of a microwave oven **850***b* and is located inside a corresponding cooking chamber. The container holder **898** is configured to hold a cooking container **190** in the cooking chamber so that the movement of the cooking container **190** relative to the rotatable component **846** may be restricted or limited when the rotatable component **846** is moved.

As explained earlier, the electromagnetic waves produced by the magnetron **842** of the microwave generator **849***x* of the microwave oven **850***b*, can pass through (with help of the stirrer **899**) the interior of the waveguide **824** and then transfer into the cooking chamber enclosed by the enclosure device **843** of the microwave ovens **850***b* and the first enclosure component **831** of the microwave ovens **850***b*. The electromagnetic waves in the cooking chamber can heat the food or food ingredient contained in a cooking container **190** positioned inside the cooking chamber.

In some applications, some microwave ovens can be bigger than the others. It is also possible that some microwave ovens are substituted by other ovens. It is not a requirement that the number of microwaves mounted on each rotatable component **846** is the same for all.

In some embodiments, referring to FIG. **30**, a cooking system **909***d* comprises: a cooking sub-system **800** (as in FIG. **22**); and a cooking apparatus **600***d* (explained below). The cooking apparatus **600***d* comprises: a cooking apparatus **690***d* (as in FIGS. **29**A-**29**B) referred to as a first cooking apparatus; a food dispensing apparatus **620** (as in FIG. **14**); a liquid dispensing apparatus **660** (as in FIG. **15**); and an ingredient dispensing apparatus **301***d* (as in FIG. **6**D). The ingredient dispensing apparatus **301***d* is positioned next the cooking apparatus **690***d*. The ingredient dispensing apparatus **301***d* can grip and move an ingredient container from a vehicle **790** of the transport system **302** of the cooking sub-system **800** to dispense the food or food ingredient from the ingredient container into one of the cooking containers **190** of the cooking apparatus **690***d*, and the ingredient container is returned to the vehicle after the food or food ingredient are dispensed. The food dispensing apparatus **620** is positioned between the cooking apparatus **103** of the cooking sub-system **800** and the cooking apparatus **690***d* to grip and move a cooking container **190** which is positioned or otherwise held by a container holder **898** (of the cooking apparatus **690***d*) to dispense semi-cooked food from the cooking container **190** into the cookware **11** of the cooking apparatus **103**. The liquid dispensing apparatus **660** is positioned next to the cooking apparatus **690***d* to dispense liquid ingredients into a cooking container **190** of the cooking apparatus **690***d*.

The cooking system **909***d* further comprises: a container moving apparatus **620***b*; a food container cleaning mechanism **670** (described below); and a computer system **99** (as in FIG. **1**). The container moving apparatus **620***b* is a copy of the food dispensing apparatus **620** but is positioned next to the food container cleaning mechanism **670**. The part numbers in the container moving apparatus **620***b* are the same as the corresponding parts of the food dispensing apparatus **620**. The container moving apparatus **620***b* can grip and move a cooking container **190** from the cooking apparatus **690***d* and turn the cooking container **190** (optionally by 180 degrees) to a certain cleaning position so that the cooking container **190** can get cleaned by the food container cleaning mechanism **670**. Then the cleaned cooking container **190** can be moved back to the rotatable component **846** of the cooking apparatus **690***d*.

In the cooking system **909***d*, when the rotatable components **846** of the cooking apparatus **690***d* is rotated by the rotational motion mechanism **601** to a certain position, one or more of the following processes may be completed: (1) a motion mechanism **855***b* rotates a corresponding enclosure device **843** to the second end-position so that the ingredient dispensing apparatus **301***d* may dispense the food or food ingredient into a cooking container **190** which is held by the container holder **898** corresponding to the enclosure device **843**; (2) the motion mechanism **855***b* rotates the enclosure device **843** to the first end-position so that the food or food ingredient in the cooking container **190** can be heated by the corresponding magnetron **842**; (3) the motion mechanism **855***b* rotates the enclosure device **843** to the second end-position so that the food dispensing apparatus **620** can dispense a semi-cooked food from the cooking container **190** to the cookware **11** of the cooking apparatus **103** and then the food dispensing apparatus moves the cooking container **190** to the container holder **898** on the rotatable component **846**; and (4) the rotatable component **846** is rotated to a position so that the container moving apparatus **620***b* can move the cooking container **190** to the food container cleaning mechanism **670** to get cleaned and then returned to the holder on the rotatable component **846**.

The cooking system **909***d* may cook a food by applying the same steps as the cooking sub-system **800** except that a semi-cooked food produced by the cooking apparatus **600***d* can be dispensed into the cookware **11** wherein the semi-cooked food may be used as an ingredient for the cooking apparatus **103**.

Referring to FIG. **31**A, a vehicle **740** is the same as the vehicle **640** (as in FIG. **16**A) except that the container holders **662** are substituted by container holders **662***b*. The other part numbers in the vehicle **740** are the same as the corresponding part numbers in the vehicle **640**. The container holders **662***b* are configured to position or otherwise hold a cooking container **190** so that the movement of the cooking container **190** may be restricted or limited when the vehicle is moving. The computer **904***b* may control the operations of the electrical or electronic devices of the vehicle **740** by sending signals to the electrical or electronic device. The computer **904***b* may communicate with the computer system **99** via a wireless communication device.

Referring to FIG. **31**B, a transport system **308** further comprises a plurality of vehicles **740** and mini-rails **331**. The vehicles **740** and the cooking containers **190** held by the container holders **662***b* on the vehicles **740** may move along the mini-rails **331**. The transport system **308** can transfer cooking containers **190**. The computer **904***b* is connected to the computer system **99** via wireless means, so that the computer system **99** may control the timing and speed of the vehicles **740**.

It should be noted that the vehicles **740** may comprise additional components for the purpose of staying on the track.

In some embodiments, referring to FIG. **32**, a cooking system **909***e* is similarly configured to the cooking system **909***d*. The cooking system **909***e* comprises: a cooking sub-system **800** (as in FIG. **22**); a cooking apparatus **690***d* (as in FIGS. **29**A-**29**B); a transport system **308** (as in FIG. **31**B); a transfer apparatus **650***b*; and a computer system **99** (as in FIG. **1**). The transfer apparatus **650***b* is a copy of the transfer apparatus **650** but is positioned between the cooking apparatus **690***d* and the transport system **308**. The transfer apparatus **650***b* can grip a cooking container **190** (which contains or holds food or food ingredient(s)) from a vehicle **740** of the transport system **308** and move it to a container holder **898** of the cooking apparatus **690***d*. The food dispensing apparatus **620** is positioned between the cooking apparatus **103** and the cooking apparatus **690***d*. The food dispensing apparatus **620** can grip and move a cooking container **190** on one of the container holders **898** of the cooking apparatus **690***d* to dispense semi-cooked food from the cooking container **190** into the cookware **11** of the cooking apparatus **103**. The computer system **99** is connected to the electric or electronic components of the apparatuses or mechanisms in the cooking system **909***e* so that the computer system **99** may communicate with and/or control the components by known techniques.

In the cooking system **909***e*, when the rotatable component **846** of the cooking apparatus **690***d* is rotated by the rotational motion mechanism **601** to a certain position, one or more of the following processes may be completed: (1) a motion mechanism **855***b* rotates a corresponding enclosure device **843** to the second end-position so that the transfer apparatus **650***b* can grip and move a cooking container **190** containing food or food ingredient(s) from a vehicle **740** of the transport system **308** to the corresponding container holder **898** of the cooking apparatus **690***d*; (2) the motion mechanism **855***b* rotates the enclosure device **843** to the first end-position so that the food or food ingredient in the cooking container **190** can be heated by the corresponding magnetron **842**; (3) the rotatable component **846** is rotated to a position and the motion mechanism **855***b* rotates the enclosure device **843** to the second end-position so that the food dispensing apparatus **620** can dispense a semi-cooked food from the cooking container **190** to the cookware **11** of the cooking apparatus **103** and then the food dispensing apparatus moves the cooking container **190** to the container holder **898** on the rotatable component **846**; and (4) the rotatable component **846** is rotated to a position so that the transfer apparatus **650***b* can grip and move the cooking container **190** to a container holder **662***b* of a vehicle **740**.

The cooking system **909***e* may cook a food by applying the same steps as the cooking sub-system **800** except that a semi-cooked food produced by the cooking apparatus **690***d* can be dispensed into the cookware **11** wherein the semi-cooked food may be used as an ingredient for the cooking apparatus **103**.

It should be noted that a first enclosure component (**844**, **831**, or **817**) and the corresponding enclosure device **843** (as the second enclosure component) in the above-described cooking systems may be substituted by other shapes of solids, as long as they are structured to enclose a cooking chamber. For example, the first enclosure component may be substituted by a solid **848** (in the shape of a cap facing upward) comprising a flat bottom and a cylindrical wall, and the second enclosure component **843** may be substituted by a solid **847** (also in the shape of a cap, facing downward) comprising a flat bottom and a cylindrical wall, which has a same radius as the cylindrical wall of the solid **848**; see FIG. **17**H. As another example, the first enclosure component may be substituted by a solid (in the shape of a cap, facing upward) comprising a flat bottom and a cylindrical wall (which is higher than the cylindrical wall in the solid **848**), and the second enclosure component may be substituted by a flat or curved lid.

It should be noted that the transport systems **302**, **303** and **308** in the above cooking systems may comprise a single system or a plurality of disconnected sub-systems. The transport system may comprise different types of vehicles. The ingredient containers may be configured differently for different types or quantities of ingredients. It should be noted that the transport systems **302**, **303** and **308** may be combined into one transfer system.

It should be noted that the transport systems **302**, **303** and **308** may be substituted by another transport system to move ingredient containers, e.g., a transport system comprising a cyclic motion mechanism, a turntable, a chain fixed to chair wheels, a robot arm, or a conveyor mechanism.

The transport system may be substituted by any transport system disclosed in U.S. patent application Ser. Nos. 16/517,705 and 16/997,933. The entire contents of the applications are incorporated herein by reference.

In the cooking systems, the ingredient dispensing apparatuses **301** and **301***b* may be configured differently. The container holders of the transport system **302** which are next to different ingredient dispensing apparatuses may be configured to have different sizes. The ingredient containers **81** on different holders may be configured to have different sizes. The transport system **302** may comprise two or more sub-systems which are not connected with each other, and the vehicles may be configured differently on different sub-systems.

The ingredient dispensing apparatuses **301** and **301***b* may be combined into one. The ingredient dispensing apparatuses may be substituted by mechanisms each comprising a robot arm.

It should be noted that the ingredient dispensing apparatuses **301**, **301***b* and **301***d*, the food dispensing apparatus **620**, and the container moving apparatus **620***b* may be substituted by another type of dispensing apparatus, such as the robotic apparatus **222** (of FIG. **4**F), which is a combination of robot arm and robot fingers. Similarly, the motion mechanism **104** of the cooking apparatus **103** may be substituted by the robot arm **218** (of FIG. **2**I) where the moving member **217***b* is fixedly connected to the cookware **11**. The transfer apparatus **650** may be substituted by the robotic apparatus **222**.

It should be noted that the microwave generator **849***a* or **849***x* may further comprise a capacitor, a transformer, and/or other electrical devices such as in a common microwave oven. A control device is configured to control the timing and power of a magnetron **842** and the timing and speed of a stirrer **899**. The control devices are connected to the computer system **99** via wires or by wireless means, so that the computer system **99** may control the timing and power of the magnetrons **842** and the timing and speed of the stirrers **899**.

It should be noted that each microwave generator **849***a* or **849***x* may be substituted by another type of generator of electromagnetic waves. Generators of electromagnetic waves may be constructed by known techniques.

It should be noted that the drawings in the present patent application are schematic and may not be well scaled. The distances between various mechanisms and apparatuses may not be drawn to scale. The 3-dimensional positioning of various mechanisms and apparatuses in a cooking system may be done in various other ways.

A motor may be an AC or DC motor, stepper motor, servo motor, inverter motor, pneumatic or hydraulic motor, etc. A motor may optionally further comprise a speed reducer, encoder, and/or proximity sensor.

A motor is said to drive a rotation of a shaft, if the rotation produced by the motor can induce the rotation of the shaft directly or indirectly via connection of shafts (e.g., by coupling), via mechanical transmission, and/or via other means.

While this document contains many specifics, these should not be construed as limitations on the scope of an invention that is claimed or of what may be claimed, but rather as descriptions of features specific to particular embodiments. Certain features that are described in this document in the context of separate embodiments can also be implemented in combination in a single embodiment. Conversely, various features that are described in the context of a single embodiment can also be implemented in multiple embodiments separately or in any suitable sub-combination. Moreover, although features may be described above as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination can in some cases be excised from the combination, and the claimed combination may be directed to a sub-combination or a variation of a sub-combination.

A rigid component described in the present patent application can be any type of solid component which has some degree of rigidity in an application, and there is no strict or quantitative requirement for the degree of rigidity. It should be noted that there is no perfect rigid component in our world, as there are always elastic, thermal, and other deformations in any physical object. A rigid component may comprise one or more of the following: a bar, a tube, a beam, a plate, a board, a frame, a structure, a bearing housing, a shaft. A rigid component can be made of metal such as steel or aluminum, or a mixture of metals, an alloy, a reasonably rigid plastic, wood, or other materials, or a combination of different types of materials.

Similarly, a rigid connection of two or more components can be a connection which has some degree of rigidity in an application, and there is no strict quantitative requirement for the degree of rigidity. A rigid connection may be a welding of two or more metal components. A rigid connection may be a bolting of two or more components; and so on. Clearly, a typical connection of a shaft and a bearing housing by a bearing (and accessories), for example, is not a rigid connection, since the shaft can rotate relative to the bearing housing.

Most common bearings are ball bearings and roller bearings. However, a bearing in the present patent application can be of any type.

A support component described in the present patent application can be any type of rigid component. A support component may be moved or fixed relative to the floor of the building or the ground.

Only a few examples and implementations are described. Other implementations, variations, modifications and enhancements to the described examples and implementations may be made without deviating from the spirit of the present invention. For example, the term cookware is used to generally refer to a device for containing or holding a food or food ingredient during cooking.

For the purpose of present patent application, a cookware can be a wok, a pot, a pan, a basket, a bowl, a dish, a container, a board, a rack, a net, a mesh, or any object used to contain or otherwise hold a food or a food ingredient during a cooking process. The cooking also is not limited to any particular ethnic styles. The cooking may include but is not limited to: frying (including stir frying), steaming, boiling, roasting, baking, smoking, microwaving etc. The cooking apparatus may or may not use a heater.

Similarly, a food container, ingredient container, or container, can be a bowl, a plate, a cup, a jar, a bottle, a flat board, a basket, a net, a wok, a pan, or any object used to contain or otherwise hold a food or a food ingredient. A container can have a rather arbitrary geometric shape. It is possible that different ingredient containers may have different shapes. It is possible that different food containers may have different shapes. It is possible that different cooking containers may have different shapes. It is possible that different cookware may have different shapes.

An enclosure component is meant to be a solid. A plurality of solids may be shaped to enclose a space; and in this case the solids may be referred to as enclosure components. A container and a lid can be examples of enclosure components as the lid and the container may enclose a space, often referred to as the interior of the container. The lid may optionally have the shape of a cap, but this is not a requirement in any sense. The caps described above may be substituted by other shapes. A flat board and a cap may enclose a space, often referred to as the interior of the cap. Two caps with similarly shaped edges (wherein the caps are positioned against each other along the edges) may also enclose a space. The walls, the floor, the ceiling, the windows, and the doors of a typical room (in a building) are also enclosing components which may together enclose a space, often referred to as the inside of the room.

A gripper is a device used to touch and grip an object such as a container. A gripper can be a rigid or elastic object as in FIGS. **4**A-**4**E. In this patent application, a gripper may be pneumatic gripper, which is an actuating device that uses compressed air as power to pinch or grip an object. A gripper may be a vacuum chuck.

A gripping mechanism can be any mechanism that can be used to grip an object. A gripping mechanism may optionally comprise a gripper such as a vacuum chuck. A gripping mechanism may optionally comprise a plurality of rigid or elastic grippers which are moved to grip an object. A gripping mechanism may optionally comprise a robot hand. In fact, a robot hand may be used as a gripping mechanism for our purposes.

A motion mechanism can be any mechanism that can be used to produce a movement of an object, which may be a component of the motion mechanism or an object that is rigidly or fixedly connected to a component of the motion mechanism. A motion mechanism may produce a linear motion of a component. A motion mechanism may produce a rotation of a component. A motion mechanism may comprise a robot arm. A motion mechanism may be a combination motion mechanism comprising a plurality of motion sub-mechanisms. A motion mechanism may comprise: a crank rod mechanism; an eccentric motion mechanism; etc. A motion mechanism may comprise one or more the following parts: motor; encoder; shaft; coupling; bearing housing; bearings and accessories; gear and rack; screw rod and screw nut; cylinder; hydraulic cylinder; electromagnet; cam; eccentric shaft; Geneva mechanism, etc. Motion mechanisms can be more complex, and the motions produced by a motion mechanism can be a planar motion, a spherical motion, an oscillatory or vibratory motion, see e.g., U.S. patent application Ser. Nos. 16/997,196, 15/706,136 (in this application a motion mechanism may be referred to as a transport mechanism), Ser. Nos. 15/801,923, and 15/798,357. The entire contents of the above applications are hereby incorporated herein by reference.

It should be noted that the linear motion produced by the linear motion mechanism may be a linear motion between two end-positions or a linear motion with multiple stop positions. Any robot arm may be used as a motion mechanism for our purposes.

A transfer apparatus can be any apparatus that can be used to transfer an object (such as a container) from one position to another. A transfer apparatus may comprise: a gripping mechanism comprising a support component and one or more grippers; and a combination motion mechanism which is a combination of a plurality of motion sub-mechanisms, said combination motion mechanism being configured to move the support component of the gripping mechanism. A transfer apparatus may comprise a robot arm and a gripping mechanism. A robotic apparatus comprising a combination of a robot arm and a robot hand may be used as a transfer apparatus for our purposes.

An ingredient dispensing apparatus can be any apparatus that can be used to dispense a food or a food ingredient from an ingredient container into a cookware or a cooking container. A typical dispensing apparatus of the food or food ingredient may comprise: a gripping mechanism configured to grip an ingredient container, and a motion mechanism configured to move a (support) component of gripping mechanism. There are more examples in U.S. Pat. No. 10,455,987 and U.S. patent application Ser. No. 15/798,357. In particular, a robotic apparatus comprising a robot hand and robot arm may be used as an ingredient dispensing apparatus. This is often used in prior art.

A food dispensing apparatus can be any apparatus that can be used to dispense a cooked (or semi-cooked) food from a cookware into another container. A food dispensing apparatus may comprise a motion mechanism which moves the cookware. A food dispensing apparatus may alternatively comprise a robotic apparatus comprising a robot arm and a robot hand that moves the cookware, and this is often the case when the cookware is not fixedly attached to another (relatively heavy) mechanism.

There is a difference between transfer apparatus and ingredient (or food) dispensing apparatus, as follows. A dispensing apparatus needs to turn (or rotate) a gripped container upside down or by some angle of say, 90 to 180 degrees, to dispense the food or food ingredient contained in the container to another container. In comparison, a transfer apparatus does not need to turn (or rotate) a gripped container, since the food or food ingredient is not to be dispensed from the container. Indeed, it is advantageous (though not always a strict requirement) for the transfer apparatus to keep the gripped container in an upright or nearly upright position, so as to not let the food or food ingredient drop out. Even if the container is sealed by a cover or lid, there is no need for the food or food ingredient to touch the cover or lid.

Each vertical motion mechanism as described above may be substituted by a motion mechanism which can produce a linear or non-linear motion in an upward or downward direction, where an upward direction needs not to be exactly vertical. It can have an inclination angle between 0 and 90 degrees. The same applies to each horizontal motion mechanism described above.

A liquid dispensing apparatus can be any apparatus that can be used to dispense a liquid ingredient from a container into a cookware. A liquid dispensing apparatus may comprise liquid pipes, a liquid pump, a valve, and/or flow sensors, etc. There are more examples in U.S. Pat. No. 10,455,987.

A cooking apparatus can be any apparatus comprising a cookware. A cooking apparatus may optionally further comprise a motion mechanism configured to move the cookware. The motion mechanism may optionally comprise a motion sub-mechanism configured to move the cookware to stir the food or food ingredient(s) in the cookware. The motion mechanism may optionally comprise a motion sub-mechanism configured to move the cookware to dispense a cooked (or semi-cooked) food from the cookware. A cooking apparatus may optionally comprise a transfer apparatus configured to move the cookware. Said transfer apparatus may optionally grip and turn the cookware to dispense a cooked (or semi-cooked) food from the cookware. Examples of cooking apparatuses are given in U.S. Pat. No. 10,455,987 and U.S. patent application Ser. Nos. 16/997,196, 15/706,136, 16/155,895, 15/801,923, and 15/869,805, the entire disclosures of which are hereby incorporated herein by reference.

A cleaning apparatus can be any apparatus that can be used to clean an object, e.g., a funnel, or a container such as cookware, food container, or ingredient container. A cleaning apparatus comprises a liquid source (e.g., tap water, or a water tank) and a liquid pipe to transfer the liquid from the source to the object; wherein the liquid flow may be controlled by a valve, a liquid pump, and/or by other known techniques; wherein the liquid may be referred to as a cleaning liquid, such as hot water, for the purpose of cleaning the object. In some applications, the liquid may be sprayed onto the object at high speed, but this is not a requirement. A cleaning apparatus may optionally further comprise a stirrer which is rotated to stir the cleaning liquid in the object, e.g., a container, which is cleaned by the cleaning apparatus. A cleaning apparatus may optionally comprise a motion mechanism configured to move the water pipes and stirrers away from or towards the object, which is cleaned or to be cleaned by the cleaning apparatus.

A transport system can be any system that can be used to transfer a container (such as, an ingredient container, a food container, a cookware, or a cooking container). In some applications (but not always), a transport system can move a container after the container is placed on a member of the transport system. For example, a transport system may include a plurality of vehicles each configured to carry and transport a container; wherein the vehicles may optionally move on rail tracks. A transport system may optionally comprise a rotating turntable, or a cyclic motion mechanism, a chain, and/or a belt. Examples of transport systems are given in U.S. Pat. No. 10,455,987 and U.S. patent application Ser. Nos. 15/798,357, 16/997,933, and 16/155,895, the entire disclosures of which are hereby incorporated herein by reference. A transport system may only comprise a transfer apparatus.

A container holder is a solid which has an adequate shape to position or hold a container of a certain shape.

A "container holder configured to position or hold a container" may be any solid which has the shape to (steadily) position the container or to (steadily) hold the container. For example, if the container is a bowl, then a table, a horizontally placed net, or a ring of a matching shape, a platform, a device consisting of two properly placed parallel sticks, can be a container holder configured to position or hold the bowl. In other applications, a "container holder configured to position or hold a container" may be a device or mechanism that can be moved to position or hold the container. Thus, a gripping mechanism which can grip the container may also be considered a container holder configured to hold the container.

A container transfer apparatus can be any transfer apparatus used to move a container to a (different) member of a transport system. The container transfer apparatus can optionally be a part of the transport system.

A heater for the purpose of cooking in the known technique may substitute any stove and heater disclosed in the present application.

In our patent application, a computer system may or may not comprise a network. A computer system may be a single computer in some simpler applications.

A loading apparatus can be any transfer apparatus used to move a container to a (different) member of a transport system. The loading apparatus can optionally be a part of the transport system.

Control by a computer or computer system of a motor, an actuator, a heater, or electrical or electronic devices is by known techniques.

It should be noted that a cooking system may comprise several different motion mechanisms. For the purpose of our patent application, if a first apparatus (or equipment) is different from a second apparatus (or equipment), then a motion mechanism of the first apparatus is implicitly different from a motion mechanism of the second apparatus. For example, we may say that "a first cooking apparatus comprises a motion mechanism and a second cooking apparatus comprises a motion mechanism." Such words should be compared with: "a first vehicle (e.g., a truck) comprises a motor and a second vehicle (e.g., a car) comprises a motor." Implicitly, the motor of the second vehicle is a different motor than the motor of the first vehicle; and the two motors may or may not have the same configuration. Thus, in plain meanings, the motion mechanism of the second cooking apparatus is a different motion mechanism than the motion mechanisms of the first cooking apparatus. Different motion mechanisms may or may not have the same design and configuration. The same comment applies to various gripping mechanisms, various support components, various motors, and/or various grippers, etc.

Generally speaking, components belonging to different equipment are implicitly not the same mechanism, even if they have the same design and configuration and/or they are referred to by the same words (e.g., gripping mechanism).

What is claimed is:

1. An automated cooking system comprising:

a first cooking apparatus comprising:

a plurality of microwave ovens, each said microwave oven comprising:

a first enclosure component comprising a first solid;

a second enclosure component comprising a second solid; and

a first motion mechanism configured to move the second enclosure component between a first position and a second position, said first motion mechanism comprising a first motor or other driving mechanism; and

a magnetron or microwave generator;

wherein the first enclosure component and the second enclosure component are configured to enclose a cooking chamber when the second enclosure component is moved to the first position by the first motion mechanism, wherein the cooking chamber is configured to position a cooking container, wherein the cooking container is a container used to contain or hold a food or one or more food ingredients;

wherein the magnetron or microwave generator is configured to generate electromagnetic waves of microwave frequencies in the cooking chamber to heat the food or food ingredients contained or held in the cooking container positioned in the cooking chamber; and

a second motion mechanism comprising a second motor, said second motion mechanism being configured to move the first enclosure components of the microwave ovens among a plurality of positions; and

a second cooking apparatus comprising a cookware, said cookware being configured to contain or hold a food or one or more food ingredients;

wherein the second cooking apparatus is configured to produce a cooked food from the food or food ingredients;

wherein the food or food ingredients heated by one of the plurality of microwave ovens of the first cooking apparatus may be used as a food ingredient for the second cooking apparatus.

2. The cooking system of claim 1, further comprising a plurality of ingredient containers each configured to contain or hold food ingredients.

3. The cooking system of claim 1, further comprising a storage configured to position or store a number of ingredient containers, wherein each ingredient container is used to contain or hold one or more food ingredients.

4. The cooking system of claim 1, wherein the first cooking apparatus further comprises an ingredient dispensing apparatus configured to move an ingredient container to dispense one or more food ingredients from the ingredient container to the cooking container that is positioned in one of the cooking chambers of the first cooking apparatus, wherein the ingredient container is used to contain or hold the food ingredients, said ingredient dispensing apparatus comprising:

a gripping mechanism comprising a support component and one or more grippers, said gripping mechanism being configured to grip the ingredient container; and

a third motion mechanism configured to move the support component of the gripping mechanism, said third motion mechanism comprising a third motor or other driving mechanism.

5. The cooking system of claim 1, wherein the second cooking apparatus further comprises an ingredient dispensing apparatus configured to move an ingredient container to dispense one or more food ingredients from the ingredient container to the cookware of second cooking apparatus,

wherein the ingredient container is used to contain or hold the food ingredients, said ingredient dispensing apparatus comprising:

a gripping mechanism comprising a support component and one or more grippers, said gripping mechanism being configured to grip the ingredient container; and

a fourth motion mechanism configured to move the support component of the gripping mechanism, said fourth motion mechanism comprising a fourth motor or other driving mechanism.

6. The cooking system of claim **1**, further comprising a transport system configured to transport an ingredient container, wherein the ingredient container is used to contain or hold one or more food ingredients.

7. The cooking system of claim **6**, wherein the transport system comprises a plurality of vehicles, said cooking system further comprising a container transfer apparatus configured to load an ingredient container to one of the plurality of vehicles, wherein the ingredient container is used to contain or hold one or more food ingredients, said container transfer apparatus comprising:

a gripping mechanism comprising a support component and one or more grippers, said gripping mechanism being configured to grip the ingredient container; and

a fifth motion mechanism configured to move the support component of the gripping mechanism, said fifth motion mechanism comprising a fifth motor or other driving mechanism.

8. The cooking system of claim **1**, wherein the second cooking apparatus further comprises a liquid dispensing apparatus configured to dispense a liquid ingredient to the cookware of the second cooking apparatus, said liquid dispensing apparatus comprising a liquid pipe, a liquid container, and/or a pump.

9. The cooking system of claim **1**, further comprising a dispensing apparatus configured to dispense a heated food or food ingredients from the cooking container that is positioned in one of the cooking chambers of the first cooking apparatus to the cookware of the second cooking apparatus, said dispensing apparatus comprising:

a gripping mechanism comprising a first support component and one or more grippers, said gripping mechanism being configured to grip the cooking container; and

a sixth motion mechanism configured to move the first support component of the gripping mechanism, said sixth motion mechanism comprising:

a first motion sub-mechanism comprising a sixth motor and a second support component, said first motion sub-mechanism being configured to produce a motion of the first support component of the gripping mechanism;

a second motion sub-mechanism comprising a seventh motor and a third support component, said second motion sub-mechanism being configured to produce a motion of the second support component of the first motion sub-mechanism; and

a third motion sub-mechanism comprising an eighth motor and a fourth support component, said third motion sub-mechanism being configured to produce a motion of the third support component of the second motion sub-mechanism.

10. The cooking system of claim **9**, wherein the motion produced by the first motion sub-mechanism in the dispensing apparatus is a rotation, wherein the motion produced by the second motion sub-mechanism in the dispensing apparatus is a vertical linear motion, wherein the motion produced by the third motion sub-mechanism in the dispensing apparatus is a rotation.

11. The cooking system of claim **1**, further comprising a dispensing apparatus comprising a robotic apparatus comprising a robot hand and one or more robot fingers, said dispensing apparatus being configured to move the cooking container to dispense a cooked food from the cooking container to the cookware of the second cooking apparatus, wherein the cooking container is used to contain the cooked food and is positioned in the cooking chamber of one of the plurality of microwave ovens of the first cooking apparatus prior to the dispensing.

12. The cooking system of claim **1**, wherein the second cooking apparatus further comprises a motion mechanism configured to move the cookware to dispense a cooked food from the cookware to a food container, wherein the food container is used to contain or hold the cooked food, said motion mechanism comprising a ninth motor.

13. The cooking system of claim **1**, wherein the second enclosure component of each microwave oven comprises a cap or a lid.

14. A cooking system comprising

a first cooking apparatus, said first cooking apparatus comprising:

a plurality of microwave ovens, each said microwave oven comprising:

a first enclosure component comprising a first solid;

a second enclosure component comprising a second solid;

a first motion mechanism configured to move the second enclosure component between a first position and a second position, said first motion mechanism comprising a first motor or other driving mechanism; and

a magnetron or microwave generator;

wherein the first enclosure component and the second enclosure component are configured to enclose a cooking chamber when the second enclosure component is moved to the first position by the first motion mechanism, wherein the cooking chamber is configured to position a cooking container, wherein the cooking container is a container used to contain or hold a food or one or more food ingredients;

wherein the magnetron or microwave generator is configured to generate electromagnetic waves of microwave frequencies in the cooking chamber to heat the food or food ingredients contained or held in the cooking container positioned in the cooking chamber; and

a second motion mechanism comprising a second motor, said second motion mechanism being configured to move the first enclosure components of the microwave ovens among a plurality of positions.

15. A cooking system comprising:

a first cooking apparatus comprising:

a plurality of microwave ovens, each said microwave oven comprising:

a first enclosure component comprising a first solid;

a second enclosure component comprising a second solid;

a first motion mechanism configured to move the second enclosure component between a first position and a second position, said first motion mechanism comprising a first motor or other driving mechanism; and a magnetron or microwave generator;

wherein the first enclosure component and the second enclosure component are configured to enclose a cooking chamber when the second enclosure component is moved to the first position by the first motion mechanism, wherein the cooking chamber is configured to position a cooking container, wherein the cooking container is a container used to hold a food or one or more food ingredients;

wherein the magnetron or microwave generator is configured to generate electromagnetic waves of microwave frequencies in the cooking chamber to heat the food or food ingredients contained or held in the cooking container positioned in the cooking chamber;

a second motion mechanism comprising a second motor, said second motion mechanism being configured to move the first enclosure components of the microwave ovens among a plurality of positions; and

a transfer apparatus configured to grip and move the cooking container to and from one of the cooking chambers of the first cooking apparatus, wherein the cooking container is used to contain or hold a food or one or more food ingredients, said transfer apparatus comprising:

a gripping mechanism comprising a support component and one or more grippers, said gripping mechanism being configured to grip the cooking container; and

a third motion mechanism configured to move the support component of the gripping mechanism, said third motion mechanism comprising a third motor or other driving mechanism.

16. The cooking system of claim **15**, wherein the motion produced by the second motion mechanism is configured to be an intermittent rotation around a vertical axis.

17. The cooking system of claim **15**, further comprising an ingredient dispensing apparatus configured to move an ingredient container to dispense one or more food ingredients from the ingredient container to the cooking container that is positioned in one of the cooking chambers of the first cooking apparatus, wherein the ingredient container is used to contain or hold the food ingredients, said ingredient dispensing apparatus comprising:

a gripping mechanism comprising a support component and one or more grippers, said gripping mechanism being configured to grip the ingredient container; and

a fourth motion mechanism configured to move the support component of the gripping mechanism, said fourth motion mechanism comprising a fourth motor or other driving mechanism.

18. The cooking system of claim **15**, wherein each said microwave oven further comprises a cover configured to be rigidly, fixedly, or elastically connected to the second enclosure component of the microwave oven, said cover being configured to cover the cooking container when the second enclosure component is moved to the first position and when the cooking container is positioned in the cooking chamber of the microwave oven, wherein the cooking container is used to hold a food or one or more food ingredients.

19. The cooking system of claim **15**, wherein the motion produced by the first motion mechanism of one of the plurality of microwave ovens of the first cooking apparatus is configured to be a rotation or a linear motion.

20. The cooking system of claim **15**, wherein the gripping mechanism of the transfer apparatus comprises one or more robot fingers, wherein the third motion mechanism of the transfer apparatus comprises a robot arm.

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